

Accounting for Beginners

Learn Accounting Like Never Before

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Chapter 1: Accounting Fundamentals

What is Accounting?

Accounting is a *system* that allows businesses to **identify, record, and communicate** financial information. Basically, a transaction happens, and the accounting system needs to identify what that transaction is, record it in the accounting records (journal and ledger), and communicate that information through the financial statements.

Example (Practical Application)

For example, let's say that we own a coffee shop. Our business is to sell coffee. A customer walks in and buys a cup of coffee for \$4. Our accounting system must **identify** this transaction. We know that we sold coffee and received cash (\$4) for it. Once we identify that transaction, we need to **record** it. If we want others to invest in our coffee shop, we must show them how well we are doing. Therefore, we need to **communicate** these transactions through our financial statements (later about these).

WATCH VIDEO

Accounting Principles and Assumptions

Accounting Principles

Cost Principle (this principle tells us that we record everything at cost. If you buy a building for \$100,000, you record it at \$100,000. Simple as that.)

Revenue Recognition Principle (this principle tells us that we only record revenues, when we earn them. We earn revenues when we provide a service or sell a product. If we sell a product for \$100, we just earned revenue of \$100.)

Matching Principle (this principle tells us that expenses associated with specific revenues, should be matched in the same period. For example, If I provide a service in

2017 that is a revenue to the company. But if they pay me salaries in 2018, that is an expense to the company, directly connected to the revenue earned in 2017. Therefore, we must match the expense to the revenue in 2017.)

Full-Disclosure Principle (this principle tells us that we must inform the users of our financial records about everything.)

Accounting Assumptions

Business Entity Assumption (this assumption tells us that we assume that we as owners are separate from the business entity. If you own a coffee shop, you are you own entity, and the coffee shop is a separate entity.)

Going-Concern Assumption (this assumption tells us that we assume the business will continue running and will not go out of business.)

Periodicity Assumption (this assumption tells us that we use artificial periods of time for reporting purposes, such as monthly, quarterly, or yearly.)

Unit of Measurement Assumption (this assumption tells us that we use dollars for measurement.)

WATCH VIDEO

The Accounting Equation

$$\text{ASSETS} = \text{LIABILITIES} + \text{STOCKHOLDERS' EQUITY}$$

This is the accounting equation. All of accounting comes to this. Stockholders' Equity represents what the owners own from the company's assets. Liabilities represents what the creditors own in the company's assets.

Assets are something of value that the company has. **Liabilities** are what the company owes to creditors. **Stockholders' Equity** is what the owners own.

Asset Accounts: Cash, Supplies, Equipment, Building, Land, Accounts Receivable, etc.

Liabilities Accounts: Accounts Payable, Notes Payable, Utilities Payable, Salaries Payable.

Stockholders' Equity Accounts: Common Stock, Dividends, Revenues (Service Revenue, Sales Revenues, Interest Revenues), and Expenses (Rent Expense, Insurance Expense, Utilities Expense, Salaries Expense).

WATCH VIDEO

Effects on the Accounting Equation

Let's see how various transactions affect the accounting equation.

WATCH VIDEO

Financial Statements

Income Statement

The income statement contains revenues and expenses. Revenues minus expenses are either Net Income or Net Loss. If revenues are larger than expenses, then we will have net income. If expenses are larger than revenues, it will result in a net loss. In practice, large amount of **people refers to the income statement as a profit and loss statement**. Take a look below at our sample Income Statement issued by Best Coffee Shop Company.

Best Coffee Shop Company
INCOME STATEMENT
 For the year ended Dec. 31, 2021

Revenues:		
Coffee Sales Revenues	35,000	
Pastries Sales Revenues	20,000	
Total Revenues		55,000
Expenses:		
Cost of Goods Sold	12,000	
Salaries Expense	10,000	
Rent Expense	5,000	
Utilities Expense	2,500	
Total Expenses		29,500
Net Income (Loss)		25,500

This sample shows what we earned and what we expensed for the past year.

Revenues shows us how much we earned, and expenses shows us how much in expenses we incurred.

In this income statement we can see that at the end of the year we have a Net Income of \$25,500.

The Income Statement always shows the results of a company for the entire period (for example, past year).

Statement of Retained Earnings

The statement of Retained Earnings is consisted of the beginning balance, plus Net Income/-Net Loss, minus dividends. This statement shows us the ending balance of retained earnings, which shows up on the Balance Sheet. Take a look at the Statement of Retained Earnings sample below:

Best Coffee Shop Company
Statement of Retained Earnings
 For the year ended Dec. 31, 2021

Beginning Retained Earnings, Jan. 1	\$ 20,000
Net Income	25,500
	\$ 45,500
Less: Dividends on Common Stock	5,500
Ending retained Earnings, Dec. 31	\$ 40,000

The statement of retained earnings shows us the earnings the company retained. We add the previous balance to net income and we subtract the earnings that were distributed to the owners through dividends.

The Statement of Retained Earnings always shows the results of a company for the entire period.

Balance Sheet

The balance sheet is practically the same as the accounting equation. We are listing all assets, liabilities, and stockholders' equity accounts.

Best Coffee Shop Company
BALANCE SHEET
 December 31, 2021

ASSETS		LIABILITIES	
Cash	10,000	Accounts Payable	15,000
Accounts Receivable	2,000	Notes Payable	12,000
Supplies	5,000	Total Liabilities	<u>27,000</u>
Prepaid Rent	10,000	STOCKHOLDERS' Equity	
Equipment	10,000	Common Stock	20,000
Land	20,000	Retained Earnings	40,000
Building	30,000	Total Stockholders' Equity	<u>60,000</u>
Total Assets	<u>87,000</u>	Total Liabilities & Stockholders' Equity	<u>87,000</u>

As you can notice, the Balance Sheet is the same as the Accounting Equation. Assets must equal Liabilities plus Stockholders' Equity. Unlike other financial statements, the *Balance Sheet reports the standing of the company at a specific point in time* (for example, on December 31, 2021).

Statement of Cash Flows

The statement of cash flows is consisted of three sections: Cash flows from Operating Activities, Cash flows from Investing activities, and Cash flows from Financing activities. We will learn about this section by the end of the course.

Best Coffee Shop Company
STATEMENT OF CASH FLOWS
 For the Period Ended Dec. 31, 2021

Operating Activities	
Net Income	25,500
Increase in Accounts Receivable	(2,000)
Decrease in Inventory	1,500
Increase in Accounts Payable	<u>1,000</u>
Cash Provided by (Used in) Operating Activities	<u>26,000</u>
Investing Activities	
Purchase of Equipment	<u>(30,000)</u>
Cash Provided by (Used in) Investing Activities	<u>(30,000)</u>
Financing Activities	
Issuance of Common Stock	<u>5,000</u>
Cash Provided by (Used in) Financing Activities	<u>5,000</u>
Net Increase (Decrease) in Cash	1,000
Cash Balance, January 1	<u>9,000</u>
Cash Balance, December 31	<u>10,000</u>

As you can see in the sample, the Statement of Cash flows explains what happens to the cash in our company. It explains where it comes from and where it goes. The balance at the end of the statement is the same amount as the on the Balance Sheet.

The Statement of Cash Flows always shows the results of a company for the entire period.

Video Explanation

See the following video for better understanding and preparation of these financial statements. We will not prepare the Statement of Cash Flows until the end of this book.

WATCH VIDEO

Time to Exercise - Problem Solving

In the following video we will see how some transactions affect the accounting equation.

WATCH VIDEO

In the next video we will see some additional examples on how transactions affect the accounting equation.

WATCH VIDEO

Chapter 2: Business Transactions

Debits and Credits

As you saw in the videos, it is too bothersome to record every increase and decrease of an account. We must come up with a better way to make our lives easier. Well, we don't have to because somebody has already come up with a great way. All we must do is to learn it.

Debits and Credits Video Lecture

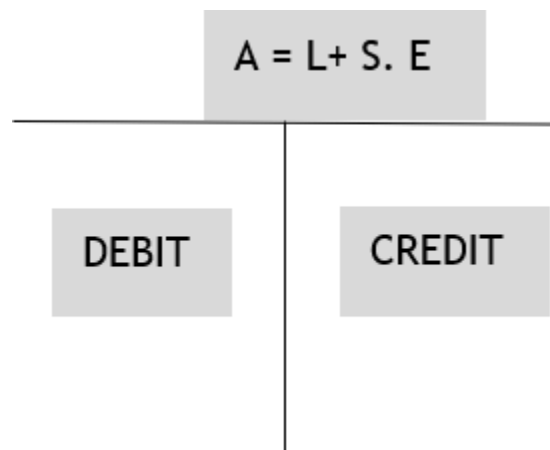
[WATCH VIDEO](#)

Let's see another comprehensive explanation of debits and credits.

Debits and Credits Second Video Lecture

[WATCH VIDEO](#)

There is something that we call **normal balance**. Normal balance can be either a debit or a credit, depending on the account. *The account's normal balance tells us that with that balance we increase the account.* Look at the figure below.

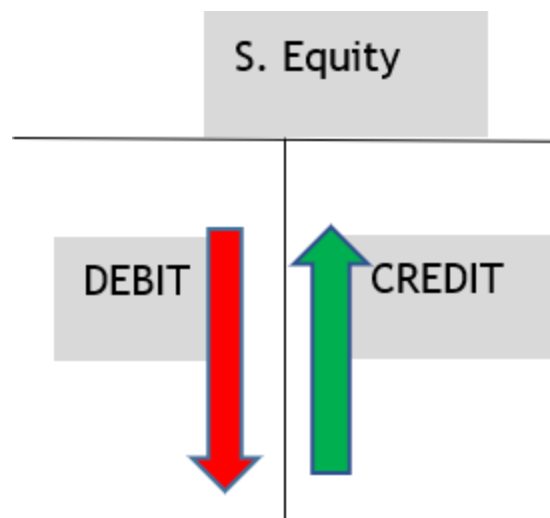


When I insert the accounting equation on top of the account (only for instructional purposes), you can see the normal balances. The normal balance of assets is debit. The normal balance of liabilities and stockholders' equity is credit. This means that **we increase all assets with a debit** (and decrease them with a credit), and **we increase all liabilities and stockholders' equity accounts with a credit** (and decrease them with a debit).

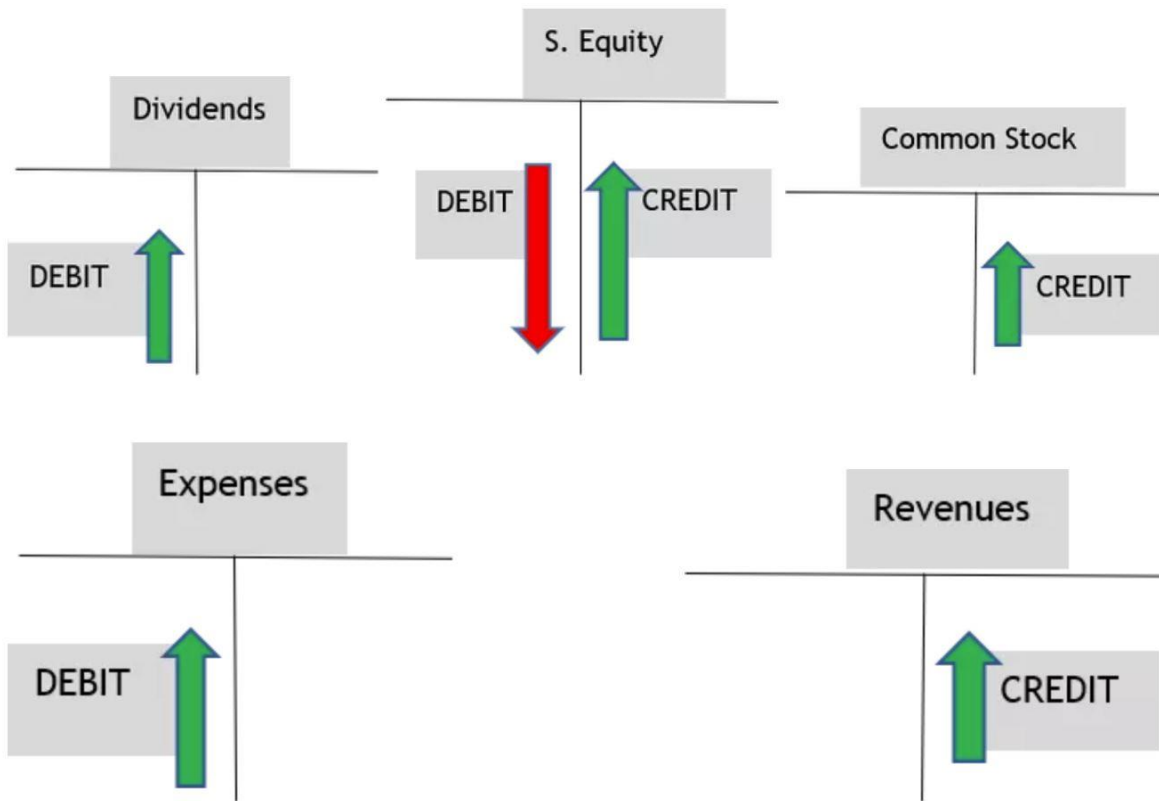
In the following lectures, we will learn more about normal balances of specific accounts, the process of journalizing (recording journal entries) and posting to the general ledger.

Journalizing and Posting

Before we learn how to journalize and post, we need to learn more about the Stockholders' Equity account.



We know that **stockholders' equity normal balance is a credit**. That means we *increase it with a credit and decrease it with a debit*. However, there are 4 accounts that affect equity. Common Stock and Revenues increase the equity, and Expenses and Dividends decrease the equity.



If common stock increase equity (and equity is increased with a credit), the *normal balance of **common stock** must also be **credit***. If revenues increase equity (and equity is increased with a credit), the *normal balance of **revenues** must also be **credit***.

If dividends decrease equity (and equity is decreased with a debit), the *normal balance of **dividends** must also be **debit***. If expenses decrease equity (and equity is decreased with a debit), the *normal balance of **expenses** must also be **debit***.

Think about this analogy and watch the videos. It is very important that you understand and know these concepts.

Journal entries are recorded in the journal and then all transactions are posted to the general ledger. This is a very simple process. Follow the entire journalizing and posting process in the following video.

Journalizing and Posting Video Lecture

[WATCH VIDEO](#)

Account Balances (Normal Balance)

There is something that we call normal balance. Normal balance can be either a debit or a credit, depending on the account (as we saw in previous lessons). The account's normal balance tells us that with that balance we increase the account. To calculate the balance of every account, we must follow the steps in the video below.

Account Balances Lecture Video

[WATCH VIDEO](#)

If we want to calculate the balances of all accounts from the "Journalizing and Posting Video Lecture", we need to do the following:

Asset Accounts Balances

The asset accounts in the "Journalizing and Posting Video Lecture" are: Cash, Accounts Receivable, Supplies, Prepaid Insurance, Prepaid Rent, and Equipment.

- For the **Cash account**, we add all debits and subtract all credits (since the normal balance of Cash is debit). Debits (\$90,000+\$5,000+\$5,000) minus credits (\$8,000+\$11,000+\$4,800+\$4,000+\$500) is \$71,700 Debit Balance.
- For the **Accounts Receivable account**, we add all debits and subtract all credits (since the normal balance of A/R is debit). Debits (\$7,000+\$3,000) minus credits (\$5,000) is \$5,000 Debit Balance.
- For the **Supplies account**, we add all debits and subtract all credits (since the normal balance of Supplies is debit). Debits (\$4,000+\$700) minus credits (\$0) is \$4,700 Debit Balance.

- For the **Prepaid Insurance account**, we add all debits and subtract all credits (since the normal balance of Prepaid Insurance is debit). Debits (\$4,800) minus credits (\$0) is \$4,800 Debit Balance.
- For the **Prepaid Rent account**, we add all debits and subtract all credits (since the normal balance of Prepaid Rent is debit). Debits (\$8,000) minus credits (\$0) is \$8,000 Debit Balance.
- For the **Equipment account**, we add all debits and subtract all credits (since the normal balance of Equipment is debit). Debits (\$20,000+\$7,000) minus credits (\$0) is \$27,000 Debit Balance.

Important Note: Anything Prepaid is actually an asset account. It means that we bought something of value.

To insert the balance of an account in the T-Account, we simply put a line on the normal balance side and write the balance. Look at the figure below:

Cash		A/R		Supplies	
90,000	8,000	7,000	5,000	4,000	
5,000	11,000	3,000		700	
5,000	4,800				
	4,000				
	500				
Balance	71,700	Balance	5,000	Balance	4,700

Prepaid Insurance		Prepaid Rent		Equipment	
4,800		8,000		20,000	
				7,000	
Balance	4,800	Balance	8,000	Balance	27,000

Asset Accounts Balances

Liability Accounts Balances

The only liability account we have in the "Journalizing and Posting Video Lecture" is Accounts Payable.

- For the **Accounts Payable account**, we add all credits and subtract all debits (since the normal balance of Accounts Payable is credit). Credits (\$11,000+\$700) minus debits (\$11,000) is \$700 Credit Balance.

Finding the balance for the liability accounts in the T-Account will look like the figure below.

Accounts Payable	
11,000	11,000
	700
	700
	Balance

Liability Account Balance

Stockholders' Equity Accounts Balances

The stockholders' equity accounts we have in the "Journalizing and Posting Video Lecture" are: Common Stock, Dividends, Revenues, and Utilities Expense.

- For the **Common Stock account**, we add all credits and subtract all debits (since the normal balance of Common Stock is credit). Credits (\$110,000) minus debits (\$0) is \$110,000 Credit Balance.
- For the **Dividends account**, we add all debits and subtract all credits (since the normal balance of Dividends is debit). Debits (\$4,000) minus credits (\$0) is \$4,000 Debit Balance.
- For the **Revenue account**, we add all credits and subtract all debits (since the normal balance of Revenue is credit). Credits (\$5,000+\$7,000+\$3,000) minus debits (\$0) is \$15,000 Credit Balance.
- For the **Utilities Expense account**, we add all debits and subtract all credits (since the normal balance of Expenses is debit). Debits (\$500) minus credits (\$0) is \$4,000 Debit Balance.

Finding the balance for the stockholders' equity accounts in the T-Accounts will look like the figure below.

Common Stock			
	110,000		
	110,000	Balance	

Revenues			
	5,000		
	7,000		
	3,000		
	15,000	Balance	

Dividends			
4,000			
Balance	4,000		

Utilities Expense			
	500		
Balance	500		

Trial Balance

The next step in the accounting cycle, after finding the account balances, is to prepare the Trial Balance. The Trial Balance is simply a list of all accounts showing their balances. It looks like the figure below.

Best Coffee Shop Company
TRIAL BALANCE
December 31, 2021

	Name of the Account	Debits	Credits
asset accounts	Cash	71,700	
	Accounts Receivable	5,000	
	Supplies	4,700	
	Prepaid Insurance	4,800	
	Prepaid Rent	8,000	
liability accounts	Equipment	27,000	
	Accounts Payable		700
equity accounts	Common Stock		110,000
	Dividends	4,000	
	Revenues		15,000
	Utilities Expense	500	
	Totals	125,700	125,700

Trial Balance

As you can notice, the trial balance lists all the accounts we have. The order in which we list the accounts resembles the accounting equation. First, we list all the asset accounts, then the liability accounts, and finally, the stockholders' equity accounts.

The total debits equal the total of the credits. This is a great assurance that we have done a good job recording the journal entries. However, it is not a guarantee. Mistakes could still be present. For example, if you have failed to record an entry, that will not disturb the debit and credit balance. Additionally, you might have recorded either the debit or the credit to a wrong account. That would be another mistake that the trial balance would fail to detect.

Chapter 3: Period-End Procedures

End of Period Procedures

At the end of the period (whether it is a year, a quarter, or a month), we must follow some specific rules. There are transactions in the accounting cycle that expand beyond one period. For example, when we pay rent every 10th of the month, we live in an apartment from December 10th, 2021 to January 10th, 2022. We pay in 2021 but live in the apartment until 2022. This transaction expands in two periods. In the next video we will examine these transactions.

The **end of the period is important** because that is when **we prepare the financial statements** of the company. Everything we say in there must be true. If something happened, but we didn't record it and we didn't show it to our investors, we will be in trouble.

Adjusting Entries Video Lecture

WATCH VIDEO

As you can note from the video above, we have different types of these transactions that expand beyond two periods. Here, we are discussing about **ACCRUALS** and **DEFERRALS**. Let's first understand what these words mean.

Deferrals

To defer means to *postpone* or to put off until a later time. In our examples in the video above, the Cash transaction happened, and then, the *earning of the revenue was deferred*, or the *occurrence of the expense was deferred*. Examples of these are any **Prepaid Expenses** or any **Unearned Revenues**.

Prepaid Expenses

We call them Prepaid Expenses because *they were paid for, but we haven't used them yet*. An example is supplies (e.g. paper, pens). When we buy supplies, we paid

for it. It is something of value that we own (an asset). However, when we use these supplies (i.e. paper, pens, pencils), they lose their value and usefulness and we throw them away (they become an expense).

To sum up, **we buy Supplies (asset)**, but we have not used it yet (not yet an expense). **Once, we use it, it becomes an expense.** Prepaid Expenses are not expenses. **Prepaid Expenses are assets.** We call them Prepaid Expenses because ultimately, they will become expenses.

Unearned Revenues

We call them Unearned Revenues because we have not earned our revenue, even though we were paid. The best example is a singer. Imagine that a singer schedules a concert. All fans will buy tickets in advance. The singer will receive all the cash from the fans, but, however, the singer did not perform the concert yet. Therefore, the singer did not earn the revenue (did not have the concert yet). It is true, however, that **the singer OWES his/her fans to perform at a concert**. Therefore, the **Unearned Revenues are classified as a Liability**. When the singer performs at the concert, he/she will earn the revenue by singing all songs at the concert. When the concert finishes, the singer would have earned all revenue.

To sum up, **the singer collects all the cash for a concert**. This is an Unearned Revenue (a liability). Once he/she performs at the concert, he/she earned the revenue and does not owe anything to his/her fans. Unearned Revenue is not a Revenue account, it is a liability account. We call it Unearned Revenue because **ultimately the singer will earn the revenue**.

Deferrals Video Lecture

WATCH VIDEO

Accruals

To accrue means to accumulate or to pile up over time. In our examples in the first video lecture, we noted that something happens first, and then, we are either being paid, or we pay someone else. Or in accounting terms, we earn (accrue) the revenue first, and later we are paid for it. Or, we use some service first, and then we pay for it. Examples of these are Accrued Revenues and Accrued Expenses.

Accrued Expenses

Accrued Expenses are simple. They truly are Expenses. Think of your workers in your company. They work for you for 2 weeks, let's say, and at the end of the 2 weeks you pay them. This means that you accrued the Salary Expense for 2 weeks and paid them at the end. Another example of accrued expenses is electricity. You use the electricity in your office for a month, but you pay at the end of the month, when the bill arrives.

Accrued Expenses Video Lecture

[**WATCH VIDEO**](#)

Accrued Revenues

Accrued Revenues are also very simple. They truly are Revenues. Think that you are a contractor. You agreed to install a new kitchen for one of your customers. The customer will pay you once you complete the project. So, first you work and install the kitchen (earning your revenue), but once you are done, you will get paid.

Accrued Revenues Video Lecture

[**WATCH VIDEO**](#)

Adjusted Trial Balance

Once we complete the process of adjusting (updating) the accounts, we are ready to prepare the Adjusted Trial Balance. The Adjusted Trial Balance is essentially an updated Trial Balance. Since with the adjusting entries we update our accounts, the adjusted trial balance should reflect the new balance of those accounts.

The Accounting Cycle

The accounting cycle is what we do in accounting, step by step. First, a business transaction will take place. Once that happens, we record a journal entry in the journal. Then, we post the amounts to the general ledger. Before we do anything else, we need to prepare the Trial Balance.

Accounting Cycle Video Lecture

[WATCH VIDEO](#)

Closing entries

The closing entries are completed right after the financial statements are prepared. We only close the *temporary (nominal) accounts*. *Temporary accounts are all Revenues, Expenses, and Dividends*. All other accounts (Assets, Liabilities, Retained Earnings) are *permanent (real) accounts* and are never closed. Instead, we roll over their balances to the next period.

To close an account means that we need to bring it to zero (0). For example, a Revenue account might have a credit balance of \$100. To close it, we need to debit it for \$100. The new balance would be zero. Or, an Expense account might have a debit balance of \$50. To close it, we need to credit it for \$50. The new balance will be zero. Or, a Dividends account might have a debit balance of \$20. To close it, we need to credit it for \$20. The new balance will be zero.

Closing entries is best explained and learned through a visual explanation. Therefore, please watch the video below.

Closing Entries Video Lecture

[WATCH VIDEO](#)

Chapter 4: Merchandising Businesses

We have discussed many things so far. Mainly, we learned about the accounting fundamentals and the accounting procedures during the period and at period-end. When we think about general types of businesses, we can conclude that there are 3 types: service businesses, merchandising businesses, and manufacturing businesses.

Service businesses are businesses that provide services to their customers. Examples include attorneys, CPAs, tutoring, consulting, hair salons, taxi, etc. These businesses provide a valuable service to their customers and get paid for it. **Merchandising (or retail) businesses** are such businesses that sell products to customers. Basically, retailers buy products and then resell them for a higher price, yielding a profit. Examples include supermarkets, clothing stores, gas stations, etc. Finally, **manufacturing businesses** produce their own products and usually sell those products to retailers to reach their customers. This lesson will discuss merchandising (retail) businesses.

How do Merchandising Businesses Work?

Merchandising businesses **purchase** products (merchandise; inventory) and **sell** those products to customers (either for cash or on account). If the products were sold on account, they later collect the payment. This process is called the **operating cycle** of a merchandising business.

New Accounts

Merchandisers (Retailers) have some additional accounts that we have not mentioned before. These accounts include: Sales, Sales Returns and Allowances, Sales Discounts, Cost of Goods Sold, and Inventory.

New Revenue Accounts

Instead of Service Revenues, retailers have an account called **Sales Revenues, or simply Sales**. This is a revenue account and its normal balance is credit. Other, very important accounts are the **Sales Returns and Allowances** account and the **Sales Discounts** account. When a customer returns a product, we record it as a Sales Returns and Allowances. When a customer gets a discount, we record it as a Sales

Discount. Allowances on the other hand, happen when a customer received damaged product or some of the specifications are mistaken. For example, if you ordered a custom blue Mercedes, but instead, received a red one, would mean you received the wrong product. Of course, you call customer service and request that they pick up their red car and bring you the blue one. Customer service tells you that instead of paying so much for shipping, they are willing to reduce the price of the car for \$6,000 if you were willing to keep the red car. If you accept, it would be recorded as a Sales Returns and Allowances. Both Sales Returns and Allowances and Sales Discounts accounts are contra-revenue accounts. Contra means opposite. As contra-revenue accounts, both have a normal balance of debit, which means they reduce stockholders' equity.

New Expense Accounts

The new expense account is **Cost of Goods Sold**. As its name implies, Cost of Goods Sold is the cost of the goods we sold. Think about the operations of a clothing store. The clothing store buys clothes first, and then resells them to customers. If they bought a jacket for \$40 and resold it to a customer for \$90, the \$40 is the Cost of Goods Sold (the \$90 is Sales Revenue). The Cost of Goods Sold is an expense account. Its normal balance is a debit.

New Asset Accounts

The new asset account is **(Merchandise) Inventory**. Inventory is an asset account. It is something of value that we own. All the products that a retailer holds to resell, should be recorded as inventory. The clothes that a clothing store has on their shelves are inventory. All the products in a supermarket are an inventory. The normal balance of Inventory is debit. It is very important to note that **inventory is always recorded at its cost**. You will see exactly what this means when we discuss discounts from a buyer's perspective.

Retailers Video Lecture

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Perspective

All retailers buy products and then resell them for a higher price. This means that they are both buying and selling. The accounting for merchandising businesses, therefore, has 2 perspectives. We have the **buyer's perspective** and the **seller's perspective**. We will look at both perspectives. Additionally, we will look at 5 possible transactions (from both perspectives) that are specific to retailers. These transactions include: **(1)** purchasing and selling inventory; **(2)** taking and giving a discount; **(3)** returning purchased inventory or accepting returns from sold inventory; **(4)** receiving or giving an allowance; and **(5)** paying for shipping charges as a buyer and as a seller.

Perpetual versus Periodic Inventory Systems

Seller's Perspective

In this section we will take a look at all 5 transactions from the seller's perspective of a retailer. We will learn the rules for selling inventory, giving a discount to customers, accepting returns from customers, and giving an allowance to our customers.

Selling Inventory

When a retailer sells inventory on account, they need to Debit Accounts Receivable and Credit Sales Revenue. In the same entry, they need to update their inventory and the expense account (Debit Cost of Goods Sold and Credit Inventory). Let's say that we have sold inventory for \$1,000. We bought this inventory for \$800. The journal entry will look like this:

		Debit	Credit
Nov. 2	Accounts Receivable	1,000	
	Sales Revenue		1,000
	COGS	800	
	Inventory		800

Recording Sale of Inventory on Account

In a perpetual inventory system, we need to always update the inventory account. As you can see from the entry above, we debit A/R because we sell on account, so customers owe us. We credit Sales Revenue because we sold the product, and, therefore, earned our revenue. We debit the Cost of Goods Sold (COGS) because we need to record our cost of the inventory we are selling. Finally, we credit Inventory because we have less inventory since we sold it.

It is also possible that we sell inventory for Cash. Let's see it in an example. We sold inventory for \$4,000. The cost of the inventory was \$3,200. The journal entry for this transaction is shown below:

		Debit	Credit
Nov. 5	Cash	4,000	
	Sales Revenue		4,000
	COGS	3,200	
	Inventory		3,200

Recording Sale of Inventory for Cash

The only difference in the entry above is that we debit Cash instead of Accounts Receivable. This is the case because we sold the inventory for cash. Let's see what happens in other situations.

Sales Discounts

It is very common for businesses to offer discounts to their customers. When that happens, there are specific accounting procedures that we need to follow. However, whenever we offer a discount as a seller, we record that discount in our newly learned **Sales Discounts** account. When we sell inventory on account, it is a good practice to add selling terms about discounts to accelerate collections. We can add the terms: **2/10, net 30**. These terms mean that we are going to give the customer 2% discount if they paid us within 10 days. If they do not pay within 10 days, they need to pay the full amount in 30 days. Other terms include **1/15, net 45** (1% discount if paid within 15 days, full amount in 45 days); or **1/10, net EOM** (1% discount if paid within 10 days, full amount by the End of Month).

Let's take a look at a practical example. On November 1st, we have sold inventory for \$2,000 on account, with credit terms *2/10, net 30*. The cost of inventory was \$1,400. This is the journal entry:

Nov. 1	Accounts Receivable.....	2000	
	Sales Revenue.....		2000
	COGS.....	1400	
	Inventory.....		1400

Recording Sales Entry

Independent Scenario: Customer paid within 10 days, on November 5th, and **took the discount**. This is the journal entry:

Nov. 5	Cash	1,960	
	Sales Discounts	40	
	Accounts Receivable		2,000

Journal Entry to Record Discount Taken

Since the customer took the discount of 2%, they need to pay us only \$1,960 (\$2,000*98%). We debit the Sales Discounts for the amount of the discount, \$40 (\$2,000*2%). Finally, we credit the full amount of the Accounts Receivable because the customer does not owe us anything anymore.

Independent Scenario: Customer paid in 20 days, on November 21st. The **customer did not take the discount**. This is the journal entry:

Nov. 21	Cash	2,000	
	Accounts Receivable		2,000

Payment Entry - Discount not Taken

As you can see from the entry above, the customer did not take the discount. Therefore, we received the full amount of \$2,000. We credit Accounts Receivable because the customer paid the full amount.

Sales Returns

When a company sells products (inventory), it is not unusual that the customers return those products. Customers can return all the products they purchased, or simply, some of the products. The reasons are numerous: possible defective products, wrong specifications, etc. When a customer returns a product, we cannot debit the Sales Revenue account. Instead, we debit the Sales Returns and Allowances account. The best way to learn this is by seeing an example.

On November 1st, we have sold inventory for \$2,000 on account. The cost of inventory was \$1,400. This is the journal entry:

Nov. 1	Accounts Receivable.....	2000	
	Sales Revenue.....		2000
	COGS.....	1400	
	Inventory.....		1400

Sales Entry

On November 15th, the customer returned half of the products because they did not meet the specifications ordered. To record the entry for the return, we record the following entry:

Nov. 15	Sales Returns and Allowances.....	1,000
	Accounts Receivable.....	1,000
	Inventory.....	700
	COGS.....	700

Sales Returns

As you can see from the entry above, we just reversed the sales entry. The only difference from a complete reversal is that we must debit the **Sales Returns and Allowances** account. Since the customer returned half of the products, they owe us 1/2 less, and they returned half of the inventory (1,400/2).

Sales Allowances

Sales allowances are different than sales returns. The difference is that the customer does not return anything. For example, imagine that you have ordered a blue car for \$20,000. When the car arrived at your house, you noticed that car was actually red. Probably, you immediately call the factory and tell them about the mistake they made. You demanded your blue car. However, the factory tells you: "If you accept the red car, we will give you an allowance of \$8,000". Many customers will accept this deal. They have a new car for \$12,000, instead of \$20,000. The only caveat is that it is red and not blue as you ordered. Let's see how a sales allowance transaction is recorded.

On November 1st, we have sold inventory for \$2,000 on account. The cost of inventory was \$1,400. This is the journal entry:

Nov. 1	Accounts Receivable.....	2000
	Sales Revenue.....	2000
	COGS.....	1400
	Inventory.....	1400

On November 4th, we received a call from our customer, telling us that some of the inventory did not meet the specifications in the purchase order. We agreed with the

customer to provide \$600 allowance instead of exchanging the inventory. This is the journal entry:

Nov. 4	Sales Returns and Allowances.....	600
	Accounts Receivable.....	600

Sales Allowance

As you can see in the entry above, we reduced the Accounts Receivable of our customer for \$600. We also recorded a **Sales Allowance** of \$600. We do not record any changes to the Inventory and COGS accounts because remember, the customer did not return anything, they just agreed to accept the wrong inventory for a lower price.

Buyer's Perspective

In this section we will take a look at all 5 transactions from the buyer's perspective of a retailer. We will learn the rules for buying inventory, getting a discount from our suppliers, handling returns to suppliers, and getting an allowance from our suppliers.

Purchasing Inventory

If we want to sell products for profit, we first need to buy those products from somewhere. This is exactly the buyer's perspective. Let's take a look at an example of buying inventory.

On November 1st, we have bought inventory for \$1,400 on account. This is the journal entry:

Nov. 1	Inventory.....	1,400
	Accounts Payable.....	1,400

Purchasing Inventory

The entry above is simple. We purchase inventory, so we debit inventory. Since the purchase is on account, we credit Accounts Payable for the amount we owe to the supplier.

Discounts as a Buyer

It is not only us as retailers to provide discounts for accelerated collections. Our suppliers too, offer discounts if we pay faster. The journal entry to record discounts as a buyer is a bit different than the entry to record discount as a seller. This is the moment to remind you that **inventory is always recorded at its cost**. Let's see an example of a discount as a buyer.

On November 1st, we have bought inventory for \$1,400 on account, credit terms *1/15, net 30*. This is the journal entry:

Nov. 1	Inventory.....	1,400	
	Accounts Payable.....		1,400

Purchasing Inventory

Independent Scenario: We paid our supplier on November 5th. **We took the discount** of 1%. This is the journal entry:

Nov. 5	Accounts Payable.....	1,400	
	Inventory.....		14
	Cash.....		1,386

Purchase Discount

Now, the entry above might look a little bit tricky. But let's go step by step and explain it. It is true that we paid on November 5th. But we paid only \$1,386 ($99\% \times 1,400$) because we took a discount for paying faster. Since we paid, it means we do not owe anymore. Therefore, we debit Accounts Payable for the full amount. *The tricky part is the credit to the inventory account.* We cannot record sales discounts because we are not the seller, we are the buyer in this situation. Let

me remind you again: **Inventory is always recorded at its cost.** We paid \$1,386 for the inventory, which means that is its cost. If you take a look at the November 1st entry, we recorded inventory at \$1,400. This is not correct. Therefore, we need to correct the balance in the inventory account by crediting it for \$14, the amount of the discount. The new balance in the inventory account is now \$1,386, its cost (exactly what we paid for it).

Inventory			
Nov. 1	1,400	14	Nov. 5
Balance	1,386		

Inventory Account Balance

Now, the inventory accounts are recorded at cost in our books, as you can see in the figure above.

Independent Scenario: We paid our supplier on November 25th. **We did not take the discount.** Instead, we paid the full amount. This is the journal entry:

Nov. 5	Accounts Payable.....	1,400
	Cash.....	1,400

Purchase Discount not Taken

As you can see in the entry above, we did not take the discount. We just paid the full amount. Now, the cost of inventory is \$1,400.

Inventory Returns as a Buyer

It is possible that the inventory we get from our suppliers is defective or it did not meet our specifications in the purchase order. In that case, we are going to return some or all of the inventory we purchased. let's see the treatment of these situations through an example.

On November 1st, we have bought inventory for \$1,400 on account. This is the journal entry:

Nov. 1	Inventory.....	1,400	
	Accounts Payable.....		1,400

Purchasing Inventory

Once we received the inventory, we realized that one half of the inventory was defective. Therefore, we returned half of the inventory on November 5th. This is the journal entry:

Nov. 5	Accounts Payable.....	700	
	Inventory.....		700

Inventory Purchase Return

In the entry above, we debit accounts payable to reduce it for \$700, and credit Inventory to reduce it for \$700. This is the case because we returned that inventory and we do not owe that amount anymore.

Receiving Allowance as a Buyer

Just as the sales allowance, we can also get a purchase allowance. The supplier might have sent us wrong products and is willing to give us an allowance, rather than paying for all the shipping back and forth. Let's take a look at an example.

On November 1st, we have bought inventory for \$1,400 on account. This is the journal entry:

Nov. 1	Inventory.....	1,400	
	Accounts Payable.....		1,400

Purchasing Inventory

Once we received the inventory, we noticed that some of the inventory did not meet the specifications in the purchase order. On November 4th, the supplier offered use a \$400 allowance instead of exchanging the inventory, and we agreed. This is the journal entry:

Nov. 4	Accounts Payable.....	400	
	Inventory.....		400

Purchase Allowance

We debit Accounts Payable because we owe \$400 less, since the supplier gave us the allowance. We credit Inventory for \$400, not because we return any of the inventory, but because the Inventory costs less. If on November 1st, we order inventory for \$1,400, and now we need to pay \$400 less for it, that means that the cost of inventory is really \$1,000 (1,400-400). Therefore, we need to reduce the cost of inventory by \$400. Remember, **inventory is always recorded at its cost**.

Shipping (Freight)

Whenever a company buys and sells inventory, it is very likely that shipping will be involved. There are 2 important questions to be asked about shipping. (1) Who owns the inventory while it is being shipped; and (2) Who will pay for the shipment of inventory?



As you can notice in the figure above, Seattle Company is buying inventory from Chicago Company. When the inventory is in the warehouse of Chicago Company, it belongs to Chicago Company. When the inventory is in the warehouse of Seattle Company, it belongs to Seattle Company. However, who owns the inventory when it is in the truck, on its way to Seattle? That would depend on the terms. We have two options:

FOB (Free on Board) Shipping Point

When the terms are FOB Shipping Point, it means that the title (the ownership) of the inventory exchanges hands at the shipping point. Since the shipping point is Chicago, as soon as the inventory leaves the Chicago Company and is loaded on the truck, Seattle Company owns the inventory. Generally, when the terms are FOB Shipping Point, the buyer pays for the inventory. This is the journal entry that Seattle Company (the buyer) will make:

Inventory	50	
Cash		50

Seattle Company will debit inventory because it is part of the cost of getting the inventory to their warehouse. Remember, inventory is recorded at cost. The cost of getting the inventory to their warehouse is capitalized in the inventory account.

FOB (Free on Board) Destination

When the terms are FOB Shipping Point, it means that the title (the ownership) of the inventory exchanges hands at the destination. Since the destination is Seattle, the title will exchange hands as soon as the inventory is unloaded from the truck in

Seattle. This means that Chicago Company owns the inventory until it arrives in Seattle. Generally, when the terms are FOB Destination, the seller pays for the inventory. This is the journal entry that Chicago Company (the seller) will make:

Shipping Expense	50	
Cash		50

The seller debits shipping expense because that is the cost of doing business. As you can see, recording the payment for shipping will depend on few factors. Namely, it will depend on whether you are the buyer or seller and the terms of shipping, FOB Shipping Point or FOB Destination.

Chapter 5: Inventory

We have explained about what inventory is in Chapter 4. However, that is simply not enough. *Tracking inventory is very important for every business.* Specifically, how are you going to track the inventory in your business?

Imagine that you have a clothing store. On the shelves and in the storage, you have 2,000 T-shirts. It is very easy to track these if you bought them all at the same price. However, in business, we always buy small pieces of inventory over time as our customers demand. In the case of the 2,000 T-shirts, we probably bought them on 5 occasions over 12 months. On each occasion we bought 400 T-shirts. Even that is not a problem. The biggest problem is that the prices change over time. We bought the first batch of T-shirts (first 400) for one price (i.e. \$2.00 each). We bought the second batch (second 400) for another price (i.e. \$2.50 each). The third batch (third 400) for \$3.00 each, fourth batch (fourth 400) for \$3.50 each, and the last batch (last 400) for \$4.00 each. As you can see there has been an inflation (an increase in prices) over time.

If all T-shirts are exactly the same but they all cost differently, which cost are you going to use as your Cost of Goods Sold (COGS) when you sell them? Are you going to use \$2, \$2.50, \$3, \$3.50, or \$4? Are you going to tell your customers to wait until you check which T-shirt they picked and then sell it to them? This chapter will provide answers to all of these questions.

Inventory Assumptions

Imagine that you are going to a supermarket to do your weekly grocery shopping. You pick milk, eggs, cereal, bread, and other items. Once you are ready to check out, you stop at the register. The employee at the register tells you to wait. They pick up your milk, go back to the shelves, check which carton of milk you picked, note the cost of that carton and come back to the register. Then, they do the same for the eggs, the cereal, the bread, and all other items you have. Once the employee is done checking, she/he comes back to the register and tells you how much to pay. Right? Absolutely not. If that was the case, we would all spend days just for grocery shopping. Then, if that is not the case, how does the supermarket know which carton of milk they sold to you? How do they know what the amount of their Cost of Goods Sold? They track their inventory by using inventory assumptions.

The three inventory assumptions we are going to learn are **First-In-First-Out (FIFO)**, **Last-In-First-Out (LIFO)**, and the **Weighted Average Cost** Methods. Let's look at the video lecture for an explanation on all three.

Inventory Assumptions Video Lecture

WATCH VIDEO

First-In-First-Out (FIFO) Method

As you learned in the video lecture, the FIFO method assumes that we first sell the first item we bought. Keep in mind the keyword in the last sentence, "assumes". When we assume something, we think that is what happened. Did it really happen? We do not really care. We only assume what happened. So, if we bought 3 T-shirts: the first one for \$1, the second one for \$2, and the third one for \$3, and we sell one, we only assume it is the first one we bought (cost of \$1). We do not care, and we do not have time to check which T-shirt was actually picked by our customer. We just assume that the customer picked the first one. Therefore, our Cost of Goods Sold will be \$1. Let's solve a more complicated problem using the FIFO method.

Problem-Solving Video FIFO

WATCH VIDEO

Last-In-First-Out (LIFO) Method

As you learned in the video lecture, the LIFO method assumes that we first sell the last item we bought. Keep in mind the keyword in the last sentence, "assumes". When we assume something, we think that is what happened. Did it really happen? We do not really care. We only assume what happened. So, if we bought 3 T-shirts: the first one for \$1, the second one for \$2, and the third one for \$3, and we sell one, we only assume it is the last one we bought (cost of \$3). We do not care, and we do not have time to check which T-shirt was actually picked by our customer. We just

assume that the customer picked the first one. Therefore, our Cost of Goods Sold will be \$3. Let's solve a more complicated problem using the LIFO method.

Problem-Solving Video LIFO

[**WATCH VIDEO**](#)

Weighted Average Cost Method

As you learned in the video lecture, when we use the Weighted Average Cost Method, we need to find the average of all products, and that will be our COGS per item. So, if we bought 3 T-shirts: the first one for \$1, the second one for \$2, and the third one for \$3, and we sell one, we first need to find the average cost per T-shirt. To do that we add all costs together ($1+2+3 = 6$). Then, we divide with the number of T-shirts we have (3). Therefore, $\$6/3 \text{ T-shirts} = \2 per T-shirt . The average cost per T-shirt is \$2. If we sell one T-shirt, do we care which one our customer picked? We do not care, and we do not have time to check which T-shirt was actually picked by our customer. We just know that the average cost per T-shirt is \$2. Therefore, our Cost of Goods Sold will be \$2. Let's solve a more complicated problem using the Weighted Average method.

Problem-Solving Video Weighted Average Method

[**WATCH VIDEO**](#)

Lower of Cost or Market (LCM)

One last concept for this chapter is the Lower of Cost or Market rule. The rule states that a company needs to evaluate the value of its inventory against the market. Market is defined as the cost of replacing the inventory. Basically, you look at the balance of your inventory and the replacement cost on the market, and then you choose the lower. The lower value is then reported on the Balance Sheet.

Let's look at an example.

We have the following inventory items:

Inventory Items	# of Items	per unit		Total Cost	Total Market	LCM
		Cost	Market			
Jackets						
Cotton Jackets	10	\$ 100	\$ 80	\$ 1,000	\$ 800	\$ 800
Leather Jackets	8	\$ 200	\$ 220	\$ 1,600	\$ 1,760	\$ 1,600
Shirts						
Dress Shirts	15	\$ 70	\$ 75	\$ 1,050	\$ 1,125	\$ 1,050
Casual Shirts	10	\$ 40	\$ 30	\$ 400	\$ 300	\$ 300
				\$ 4,050		\$ 3,750

We can see that we sell cotton and leather jackets, and dress and casual shirts. The cost of the inventory is what we have recorded in the Balance Sheet. The market is the current replacement cost. Our job here is to choose the lower of the two.

For example, the total cost for cotton jackets is \$1,000, but the market is \$800. Therefore, the market is lower, and we need to make a correction, or record our inventory at \$800. As far as leather jackets, the total cost is \$1,600 and the market is \$1,760. The lower is what we already have recorded on the Balance Sheet. Therefore, no change is needed for this item.

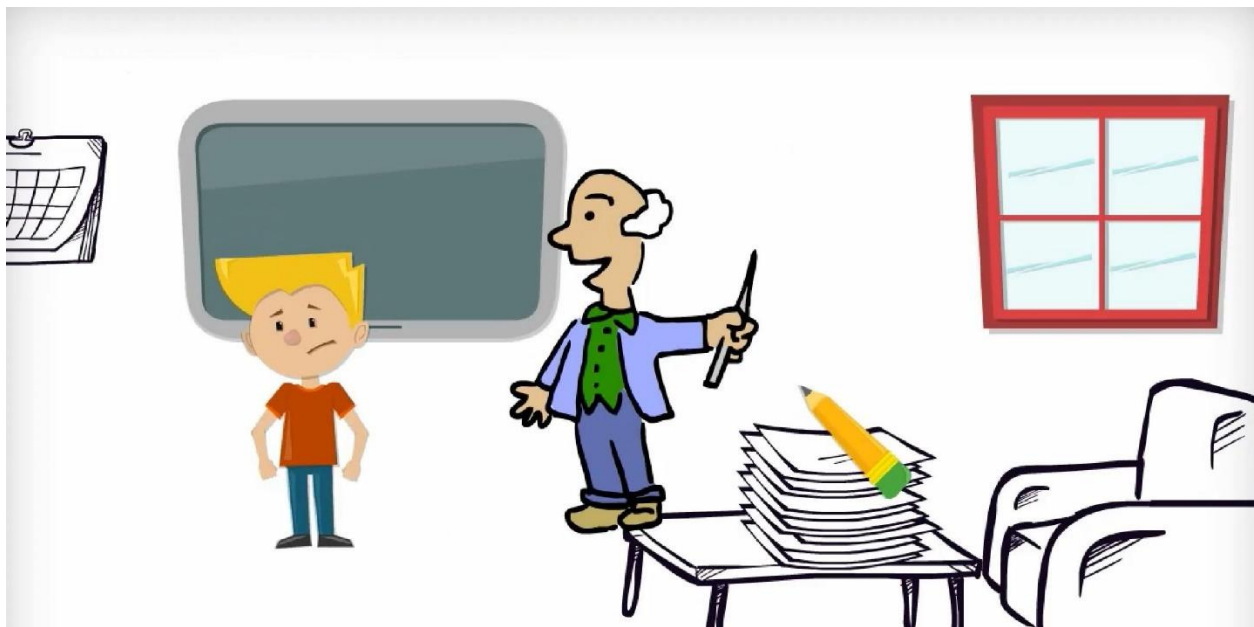
Cumulatively, the total cost of our inventory is \$4,050 (recorded on the balance sheet), and when we apply the LCM rule, the inventory recorded should be \$3,750. This means that we need to reduce the value of our inventory for \$300 (\$4,050 - \$3,750). To do this, we record the following journal entry:

Cost of Goods Sold	300	
Inventory		300

This is how we adjust the inventory cost to market value (current replacement cost).

Chapter 6: Internal Controls, Cash, and Bank Reconciliation

Imagine that we are the owners of a tutoring business. We only have one tutoring office. Since we are working for the business, we handle everything in our business. We pay the rent for the office, we buy and pay for our supplies, we handle the payroll for our employees, we collect the payments from our customers, we deposit those collections in the bank, etc. Looking at our example, we are in control of the entire tutoring business.



However, what if our business did so well, we decided to open tutoring offices all around our state? Will one person be able to drive to all tutoring centers and handle everything? What if decided to open tutoring centers all around the country? It would be impossible to control everything. That is why we would have to establish an internal control system.

[WATCH VIDEO](#)

The Purpose an Internal Control System

The purpose of the internal control system is to enable the decision-makers in a business to control and monitor all business activities and transactions. The internal control system will enable owners of the business to *protect their assets*, make sure that the *accounting is reliable*, and the *operations of the business are efficient*, and finally make sure that *all employees follow the policies of the business*.

Sarbanes-Oxley Act (SOX)

The Sarbanes-Oxley Act, or SOX, was enacted by Congress in 2002 after the reveal of many accounting scandals, such as Enron, WorldCom, etc. The Sarbanes-Oxley Act contains sections specifically for the internal controls of companies traded publicly. Section 404 of this Act is specific to this chapter because it requires the company's management 'to assess the effectiveness of internal controls for financial reporting'. The auditors, on the other hand, are required 'to attest and report on the management's assessment of the company's internal controls'.

Principles of Internal Control

In this section we will review the most important principle of an Internal Control System that is applicable to most companies.

Establishing Clear Responsibilities

Companies should have established clear responsibilities of its employees. When something goes wrong in the business, we will know exactly who was at fault. Then, depending on the mistake made, we can take appropriate action. If responsibilities were not clearly established, it would be very difficult to determine the fault and assigning it to a specific individual.

Maintain Proper and Adequate Records

Keeping and maintaining records for all business activities and transactions helps the business protect its assets and makes sure that its employees are following the company's policies and procedures. For example, inventory will be harder to be stolen if proper records are kept for receiving inventory and inventory sales.

Employees can be monitored better if there is a record of them in which they sign when they show up for work and when they leave.

Insure Company's Assets and Bond the Key Employees

Insuring the company's assets for casualty and theft protects the company from such losses. When the company bonds employees who handle a lot of the company's cash, it is basically an insurance in the case that the employee steals cash from the company.

Record-Keeping Should be Separated from Asset Custody

Basically, it is a good principle to separate an employee who handles an asset from access to the accounting records. For example, if we give the accountant, who is responsible for our accounting records, to make the cash deposits to the bank, we increase the risk for the accountant to be able to steal the cash deposits. It would be difficult to detect this theft, because the accountant might omit the cash receipts from the accounting records.

Segregation of Duties

Segregating, or dividing responsibilities, is crucial for the company's internal control system. Imagine what could happen if one employee is responsible for purchasing inventory, receiving that inventory, paying for the inventory, and updating the accounting records. The risk of theft is extremely high.

Example Scenario. One possible scheme is when one employee establishes a 'shell company' (a company that does not do anything). This employee might order (purchase) the inventory from the 'shell company', claim the inventory was received (when it never happened), pay the 'shell company', and record in the accounting records that inventory was received, and the payment was properly processed. In this case, not only our company did not receive any inventory, but we also paid the 'shell company' for nothing.

A system for segregation of duties will put the risk explained in the above scenario to minimum. If we have different employees with different responsibilities (one to order the inventory, one to receive the inventory and verify it meets the purchase order, one to authorize the payment, and one to update the accounting records), it would be very unlikely for such a scenario to occur.

Integrate Internal Controls System with Technology

Integrating technology in the Internal Control System should improve the internal controls significantly. For example, instead of having manual time clocks for employees, a digital time clock will enable a unique employee login and password or PIN, to monitor when employees arrive and leave work. Additionally, a cash register that keeps an electronic log file (recording all transactions) can also improve internal controls.

Perform regular and independent reviews

Having regular internal audits and hiring an independent audit firm to review all financial statements will ensure the reliability of accounting, the existence of assets, the efficiency of operations, and whether company's policies and procedures are being followed.

Limitations of Internal Control

Generally, an effective Internal Control System will significantly reduce the risks a company might face. However, an Internal Control System is not a guarantee that fraud, waste, or inefficiency will not occur in a company. There are two general limitations of internal control: (1) cost-benefit and (2) human factor.

Cost-Benefit

The cost-benefit principle prescribes that a company should only implement those internal controls in which the benefits will be greater or at least equal to the costs of its implementation. Let's say that every day our business receives around \$1,000 cash. At the end of the day, we deposit the money in the local bank. Is it beneficial to the business to buy \$30,000 safe to keep these \$1,000 from being stolen? Every decision that we make in the company should take the cost-benefit principle in consideration.

Human Factor

Another limitation to the Internal Control System in a company is the human factor. Do you remember the scenario in which we had one employee handle many responsibilities, such as ordering, receiving, and paying for inventory? Even if our

company has hired 4 employees to handle the different responsibilities, it is still possible that all 4 employees will collude and all 4 will decide to steal from the business together. However, it is much more unlikely.

Cash and Cash Equivalents

It easy to recognize cash because it includes currency and coins. However, it is important to note that cash are also considered all deposits made in the bank's checking and savings accounts, a cashier's check, certified checks, and money orders.

Additionally, we have something called Cash Equivalents. Cash Equivalents represent all short-term investments that can be converted to cash and the maturity date is within three months or less. Examples include U.S. Treasury bills, Commercial Paper, Money Market Funds, Short-term Government Bonds, Marketable Securities, etc., that mature within three months or less.

[*WATCH VIDEO*](#)

Cash Management

Managing the cash in a company is an extremely important task because cash is essential for the company's operations. Imagine a company that runs out of cash in the middle of the month and cannot pay for its current liabilities (utilities, rent, payroll, etc.). Such a scenario will be devastating to the business. The principles for cash management follow the general principles of an effective Internal Control System with some specifications to cash. To manage the cash properly, a company's treasurer should attempt to achieve the following goals:

Collect Cash from Credit Customers (Receivables) As Soon As Possible

Collecting cash as fast as possible is important because the company can have enough cash available to meet the current liability payments. In order to improve collections, a company might hire an effective collections team, or simply offer discounts to customers who pay faster.

Pay Liabilities as Late As Possible

I know this sounds a little hypocritical, but it seems to be the best practice for cash management. If a supplier is offering you a discount to pay faster, you will be encouraged to pay within the discount period. Otherwise, you would pay by the regular due date. For example, if an invoice is due in 45 days, and no discounts are offered for paying early, you could invest the cash in a short-term investment and earn money on your cash, instead of paying the liability early.

Invest the Excess Cash on Hand

This goal aligns with the example mentioned above. If have cash that is excess, and it is not used for current liability payments, it is wise to invest that excess to earn interest or make capital investments.

Plan Appropriate for Spending

Planning the spending can ensure that shortage of cash will not occur, and excess cash will be invested. Appropriate planning is a good cash management practice.

Internal Controls for Cash Receipts

When the business receives cash, it must have a procedure to record the transaction and deposit the cash to the bank. Cash is probably the most attractive asset to be stolen.

Cash Received from Sales (Cash Register)

When our company sells products or provides services, we might receive a payment from our customers in cash. We must ensure that the customer receives a receipt to check the correctness of the employee that receives the cash. For the employee to print a receipt, we must have a register that requires the employee to enter the cash received, records the transaction, and does not allow the employee to erase that transaction. Even when a transaction needs to be deleted, it must be done by an authorized employee and the register should record the logs of such deletions.

For example, let's look at one working shift at our company. The register clerk receives cash all day from customers and issues them a receipt by entering each amount of cash receipts in the register. At the end of the shift, the supervisor compares the balance entered in the register with the cash on hand. Finally, a third employee deposits the cash and the accountant records the entry in the accounting records.

Cash Received by Mail

Not always do the employees receive cash on the register. Many times, companies receive cash by mail. In this case, an employee opens the mail. Preferably there are two employees that open the mail to prevent theft. The employee who opens the mail prepares 2 records. One record for the register and another for the accountant. Finally, a third person compares what was entered in the books and the cash deposited in the bank.

Internal Controls for Cash Disbursements

Having proper internal controls for cash disbursements is salient because it is the company's cash that is being spent and it is an area of abuse. Who will pay, who will write the checks? How do we know these are accurate? In a small business that might be easy to track, but not so easy in a multinational corporation.

Cash Budget

Cash budget is a schedule of planned cash receipts and cash disbursements. When we plan the cash receipts and disbursements ahead of time, the chances for cash shortages will be reduced.

Voucher System

A voucher system is a system that many companies use to control the disbursements of cash. This system only allows specific and authorized departments and employees to incur liabilities on behalf of the company.

For example, let's say that our company wants to buy inventory. We need to prepare a purchase order and send it to our supplier, the accounting department, and the employee that will be receiving the inventory. When the supplier ships the inventory to us, the employee that receives the inventory will check if the inventory is as on the purchase order. This employee will confirm the receipt of the inventory and send the invoice from the supplier to the accounting department. The accounting department will approve the payment of the invoice and authorize the cashier to send a check to the supplier.

Petty Cash Fund

A petty cash fund is a very small cash fund intended for accidental and unusual purchases. For example, if we have tax preparation business, once we prepared the

tax return of our client, we need to print the return, so they can sign it. What happens if we ran out of paper? Do we go through the voucher system we have in place to order paper, and tell our client to come back tomorrow or next week? That would not be very convenient for our client. For such occurrences, we establish a petty cash fund, to skip the voucher process in certain situations.

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Establishing a Petty Cash Fund

To establish a petty cash fund of \$500, we record the following entry:

Jan. 1, 2021	Petty Cash	500	
	Cash		500

We debit Petty Cash because the designated employee who holds the funds for the petty cash fund received \$500. We credit cash because we withdrew cash from our cash account.

Reimbursing a Petty Cash Fund

Now, let's say that we paid out of the petty cash fund for various accidental expenses. We need to pay \$100 for the driver to unload the inventory we ordered. We also had to pay \$200 for some supplies because we ran out unexpectedly. Because all of these transactions are small and accidental, we do not bother to record every single small transaction. Instead, we only record an entry when we reimburse the petty cash account. Let's say that at the end of the month, the employee who holds the petty cash fund says there is only \$200 in the petty cash fund. S/He shows receipts for \$100 inventory and \$200 supplies. The journal entry is as follows:

Jan. 31, 2021	Inventory	100	
	Supplies	200	
	Cash		300

We credit cash for \$300 because this is an entry for reimbursement of the petty cash fund. This means we give another \$300 to the employee who holds the petty cash fund. We debit supplies and inventory to record what the cash was spent on.

The example above explained a perfect situation in which cash was not missing. Cash balance was not over either. However, very often, that can happen too. Let's take a look at 2 examples:

Example 1. Our employee who was holding a petty cash fund of \$400, at the end of the month showed a balance in the petty cash fund of \$100. When asked for receipts, s/he showed \$150 inventory, \$100 supplies, and \$40 freight charges. To reimburse the petty cash fund, we make the following entry:

Jan. 31, 2021	Inventory	150	
	Supplies	100	
	Freight-In	40	
	Cash Over and Short	10	
	Cash		300

In the journal entry we credit cash for \$300 because that is how much is needed to reimburse the fund to be \$400 again. The cash was spent on supplies, \$100, inventory, \$150, and freight, \$40. The missing \$10 will go to the debit side Cash Over and Short account. Of course, we should have a discussion with our employee about why this happened, but things like this, do happen.

Example 2. Our employee who was holding a petty cash fund of \$300, at the end of the month showed a balance in the petty cash fund of \$50. When asked for receipts, s/he showed \$100 inventory, \$100 supplies, \$20 freight charges, and \$40 for other miscellaneous expenses. To reimburse the petty cash fund, we make the following entry:

Jan. 31, 2021	Inventory	100	
	Supplies	100	
	Freight-In	20	
	Misc. Expenses	40	
	Cash		250
	Cash Over and Short		10

Again, we credit cash for \$250 because to reimburse the fund from \$50 to \$300. The money was spent on inventory, supplies, freight, and miscellaneous expenses. The total receipts were for \$260. The cash account is now over (employee probably found the missing \$10 from before). Therefore, we credit the Cash Over and Short account.

Increasing/Decreasing a Petty Cash Fund

Sometimes, we might realize that our petty cash fund is too small. Let's have a petty cash fund of \$200. To increase it to \$1,000, we make the following journal entry:

Feb. 1, 2021	Petty Cash	800	
	Cash		800

Sometimes, we realize the petty cash fund is too large. Let's have a petty cash fund of \$1,200. To decrease it to \$500, we make the following journal entry:

Feb. 1, 2021	Cash	700	
	Petty Cash		700

Bank Reconciliation

A good way to check if everything is correct within our cash account is to involve a neutral third party. That would be the bank. The bank holds and records all money going in and out of our cash account. While we do the same in our books, not

always do these two equal. Many times, businesses find out that their cash balance in the books is different than the cash balance in their checking account (even though it is the same account). The process of explaining the differences between the balance of cash according to the company's books and the balance of cash according to bank's records (bank statement) is called bank reconciliation.

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There are about 5 general reasons why there would be a difference between the book's cash balance and bank's cash balance. In some of these cases our records have not been updated, and in others the bank might have made an error.

Outstanding Checks

Outstanding checks is a term we use about the checks that we as a company wrote to pay for some service that we used. While we know about these checks, and we updated our records (company's books), the bank is not yet aware because our supplier did not deposit our check yet. In this case, the amount written on the check has already been subtracted from the cash balance in our books, but it hasn't yet been subtracted from our balance in the bank.

Deposits in Transit

Deposits in transit is a term that we use for in a situation in which we have made deposits to the bank and we updated our company's books, but the bank still hasn't processed the deposit. Therefore, the deposit does not yet show up on the bank's balance as it does in our books.

NSF Checks

NSF, or non-sufficient checks is a term we use when a customer paid us with a check, we updated our books, and we deposited the check in our checking account in the bank. However, the customer did not have sufficient funds in their account, and therefore, the check bounced. The bank informed us that the customer did not really pay us. In this case, our book's balance shows a higher balance than it is supposed to. We need to correct it by subtracting it from our books.

Collections by the bank on our behalf

In some cases, the bank acts as a collector of our receivables. When the bank collects money on our behalf, the bank's balance has been updated, but, however, our book's balance has not. Therefore, we need to update the balance of cash in our books.

Errors

For cases such as outstanding checks and deposits in transit, we know that the bank's balance of cash needs to be updated. For cases such as NSF checks and collections by the bank on our behalf, we know that our books need to be updated. However, errors can be made on either side. In the case of errors, the correction should be done by whoever made the error.

Now that we know the most common reasons, we can discuss some other reasons as well. It is possible that the bank has some annual fees for keeping our account open that we forgot to account for. It is also possible that some of our employees stole the cash. This is fraud and we should try and identify the employee that did it.

Example of Bank Reconciliation

The best possible way of learning the process of bank reconciliation is to see how it is done first, and then to do it yourself. For that reason, we will take a look at a comprehensive example.

On March 31, 2022, our company, Best Coffee Shop, found out that the cash balance in our books was \$10,500 and the cash balance on the bank statement was \$10,850. In order to explain this difference, we need to perform a bank reconciliation. In performing the bank reconciliation for the month, we identified the following transactions that could explain the difference in the balances.

We wrote a check to our supplier for inventory we purchased this month (outstanding checks) for \$3,000.

We have also deposited \$4,500 in the bank on Friday. The bank teller said it would take about a week for these funds to show up in our account (deposits in transit).

A check we received from our customer Anthony M., for \$500, was recorded in our books, and the check was deposited in the bank. the bank now says that the

check bounced since Anthony M. did not have enough funds (NSF check). In addition, they charged us \$40 because Anthony's check bounced.

Our bank informed us that they have collected \$2,500 on our behalf from our customer Taylor (collections by bank on our behalf).

Further, we have found that \$200 were subtracted from our bank's balance for a check written by Best Auto Body Shop to someone else. Our company is named Best Coffee Shop, and clearly, the bank has made an error they need to correct (bank error).

Finally, we found out that our company wrote a check for \$450 to the utility company and the check was correctly withdrawn by the bank. However, in our books, we have recorded that we paid the utilities company \$540 (book error).

The bank reconciliation is presented below:

	Book's Balance				Bank's Balance		
Mar. 31, 2022	10,500				10,850		
	-	540	NSF Check	-	3,000	Outstanding Checks	
	+	2,500	Collections by bank on our behalf	+	4,500	Deposits in Transit	
	+	90	Book Error	+	200	Bank Error	
Adjusted Balance	12,550				12,550		

Let's explain how we arrived at the adjusted balance.

For **transaction 1)**, we know that our books have already been updated, but the bank's balance need to be updated. Therefore, we deduct \$3,000 from the bank's balance.

For **transaction 2)**, we know that our books have been updated, but the bank's balance need to be updated. therefore, we add \$4,500 to the bank's balance.

For **transaction 3)**, the bank found out that the Anthony gave as an NSF check, but we did not know until we got the bank statement. Therefore, we need to subtract the amount from the balance on our books along with the bank charge of \$40, because Anthony did not really pay us, and the bank charged us for the NSF check.

For **transaction 4)**, the bank knew they collected money on our behalf, but we did not know until we received the bank statement. Therefore, we add the amount to our book's balance.

For **transaction 5)**, the bank made a mistake, so we need to call them and tell them to correct their mistake. therefore, they will need to add the \$200 back.

Finally, **for transaction 6)**, we made an error by deducting too much cash from our books (\$540 instead of \$450). Therefore, we need to correct this error by adding back the extra \$90 we subtracted earlier.

As you can see above, some adjustments were made to our book's cash balance, and some to the bank's cash balance. We cannot make the adjustments on the bank's side, but we can ask them to make corrections if needed. However, the adjustments made to the books, need to be followed by a journal entry. In the example above, the transactions that required adjustments to the book's balance were 3, 4, and 6.

Transaction 3 Journal entry

This is the journal entry to record the journal entry for the adjustment that arose from transaction 3):

Mar. 31, 2022	A/R - Anthony M.	500	
	Bank Charge	40	
	Cash		540

In the journal entry above, we debit the Accounts Receivable of Anthony M., because his check bounced, and, therefore, still owes us money. We also debit Bank Charge Expense because the bank chard our account for the NSF check. We credit the cash account to adjust the balance appropriately.

Transaction 4 Journal Entry

This is the journal entry to record the journal entry for the adjustment that arose from transaction 4):

Mar. 31, 2022	Cash	2,500	
	A/R - Taylor		2,500

We debited cash because we cash was collected on our behalf from the bank. We credit the Accounts Receivable of Taylor, because she does not owe us anymore.

Transaction 6 Journal Entry

This is the journal entry to record the journal entry for the adjustment that arose from transaction 6):

Mar. 31, 2022	Cash	90	
	Utilities Expense		90

We debit cash not because we received it, but to correct the previous error in which we deducted \$90 too much. We debit Utilities Expense for \$90, because previously we debited it for \$540 instead of \$450.

Chapter 7: Receivables

Many businesses provide services or sell products to customers by extending credit to them. In accounting, we call that **selling on account**. In that case, we debit Accounts Receivable for that particular customer. As we discussed in previous chapters, Accounts Receivable is an account that represents the amount a customer owes us. Let's say that on November 1st, we provided a tutoring service to a customer in the amount of \$500, on account. The journal entry will look like this:

Nov. 1	Accounts Receivable.....	500	
	Service Revenue.....		500

In this case we are expecting for the customer to pay us in the near future. If we provided the service for cash, the journal entry will look like this:

Nov. 1	Cash.....	500	
	Service Revenue.....		500

In the case in which we received cash, we do not have to wait for the customer to pay us, because we already got paid - we received cash.

Valuation of Accounts Receivable

If our credit customers pay, our business will not have any problems.

Unfortunately, not all customers pay. Some customers might not have any money to pay, others might have gone bankrupt. Yet others, might not want to pay because of various reasons. This is a problem for our business. As we know by now, Accounts Receivable is an asset and it shows on the Balance Sheet. If we **provided a service for \$500 on account**, we will **record the Accounts Receivable for**

\$500 on the Balance Sheet. When investors in our company look at the Balance Sheet, they will think that **our company expects to collect these \$500 in the near future**. But what if our customer that owes us \$500 went bankrupt? What if we determined that this specific customer is not going to pay us? Does that mean that we lie to our investors? Certainly. We show our investors an Accounts Receivable account of \$500 on the Balance Sheet, deceiving them into believing that we will collect the \$500. This practice is not acceptable. To correct this, we must **write-off** this account because we know we will not collect it. There are two methods to account for such uncollectible accounts: **(1)** the direct write-off method and **(2)** the allowance method.

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Direct Write-Off Method

The direct write-off method does exactly what its name suggests. It writes off an account directly. Let's see it through an example. On November 1st, 2021 we provided a tutoring service to a customer in the amount of \$500, on account. This is the journal entry:

Nov. 1	Accounts Receivable.....	500	
	Service Revenue.....		500

We sent an invoice to the customer on November 30th, 2021 but the customer did not pay. We sent an invoice to the customer on December 31st, 2021 but the customer did not pay. We sent an invoice to the customer on January 31st, 2022 but the customer did not pay. Finally, we realized that this customer is not going to pay. But we do not want to lie to our investors and let this Accounts Receivable account remain on the Balance Sheet. We need to write it off. This is the journal entry to write off the account:

Feb. 1	Bad Debt Expense.....	500
	Accounts Receivable.....	500

As you can see from the entry above, we debit a new account, called **Bad Debt Expense**. The Bad Debt Expense account is an expense account, representing the amount we are not able to collect. We also credit the Accounts Receivable account because we cannot collect it, and we do not want to lie to our investors that we will. This method is relatively easy. When we determine that a customer will not pay, we simply record the bad debt expense and write-off the accounts receivable account. However, there is one major flaw with this method. *It violates the matching principle.*

Direct Write-Off and the Matching Principle

As we learned in Chapter 1, the matching principle meant that the revenues and the expenses associated with those revenues must be matched in the same time period. However, look at our previous example. We provided the service on November 1st, 2021, recognizing revenue on November 1st, 2021. We wrote-off the account and recognized the bad debt expense on February 1st, 2022, the next year (next period). This means that revenues and expenses were not recognized in the same period, which violates the matching principle. This is not acceptable.

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Allowance Method

In order to solve the problem with the violation of the matching principle, we must use the allowance method instead. Doing business and selling on account means that there will always be customers who will not pay for some reason. Therefore, instead of waiting to determine whether a customer pays or not, at the end of each year, we can estimate how much will be uncollectible from our Accounts

Receivable. This is how this method works. On December 31, 2021 we estimated that \$700 will be uncollectible. This is the journal entry:

Dec. 31	Bad Debt Expense.....	700	
	Allowance for Doubtful Accounts.....		700

Allowance Method Estimate

As you can see from the entry above, we debit Bad Debt Expense because that is the amount, we expect to be uncollectible. However, **we do not credit Accounts Receivable because by December 31st, we do not who is not going to pay.** Since we do not know exactly which customer is not going to pay, we credit the **Allowance for Doubtful Accounts** account. The Allowance for Doubtful Accounts is a contra-asset account, meaning its normal balance is credit. The purpose of this account is to reduce the amount of our accounts receivable. This is how it will look like on the Balance Sheet.

Best Coffee Shop Company BALANCE SHEET December 31, 2021			
ASSETS		LIABILITIES	
Cash	10,000	Accounts Payable	12,000
Accounts Receivable	2,000	Notes Payable	15,000
Less: Allowance for Doubtful Accounts	700	Total Liabilities	27,000
Supplies	4,700		
Prepaid Rent	10,000	STOCKHOLDERS' EQUITY	
Equipment	10,000	Common Stock	20,000
Land	27,000	Retained Earnings	16,000
		Total Stockholders' Equity	36,000
Total Assets	<u>63,000</u>	Total Liabilities and Stockholders' Equity	<u>63,000</u>

Pay careful attention to the Balance Sheet presented above. When we calculate total assets, we add all assets, but deduct the amount of the Allowance for Doubtful Accounts because this account is a contra-asset account. When the investors see our Balance Sheet, they know we are probably not going to be able to collect \$700 from

our Accounts Receivable. The accounts receivable amount is \$2,000 and the allowance for doubtful accounts amount is \$700. The difference is \$1,300 (\$2,000 - \$700). This difference is called **net realizable value** - the net amount we are probably going to be able to collect (realize) from our accounts receivable.

Let's say that on February 1st, 2022 we determined that one particular customer will not pay us the \$500 he owes us. We need to write-off his account. This is the journal entry to record the write-off of this account:

Feb. 1	Allowance for Doubtful Accounts.....	500
	Accounts Receivable.....	500

Allowance Method - Account Write-Off

As you can notice from the entry above, we write-off the uncollectible account by crediting the accounts receivable. However, in this case, we debit the Allowance for Doubtful Accounts. We do not debit Bad Debt Expense because we already did on December 31, 2021. Every write-off after we made the estimate on December 31, will be debited against the Allowance for Doubtful Accounts account. Let's take a look at the new balance of the Allowance for Doubtful Accounts:

Allowance for Doubtful Accounts			
		\$700	Dec. 31, 2021
Feb. 1, 2022	\$500		
		\$200	Adjusted Balance

Allowance for Doubtful Accounts - Adjusted Balance

If we have one more customer who will not pay \$200, the adjusted balance in the Allowance for Doubtful Accounts will be 0 (zero). That means that we have made an accurate estimate of the uncollectible accounts. However, making a 100% accurate estimate is very rare. If we make a mistake in our estimate, there will be a balance in the Allowance for Doubtful Accounts. *If we overestimate the amount to be uncollectible*, like in our example above, there will be a *credit balance*. *If we underestimate*, there will be a *debit balance*.

Now that we know how the allowance method works, including the journal entries to record the estimate of uncollectible accounts and writing-off an account, it is time to learn how to make the estimate. There are two ways of making an estimate: **(1)** focus on Sales and **(2)** focus on Receivables.

Allowance Method based on Sales

Let's remind ourselves of the journal entry used to record the estimated uncollectible accounts:

Dec. 31	Bad Debt Expense.....	700	
	Allowance for Doubtful Accounts.....		700

When we base our estimate of uncollectible accounts on Sales, we focus on the Income Statement. Therefore, we estimate the balance of the Bad Debt Expense account. The beginning balance of Bad Debt Expense will always be zero (0). This is because all expenses are temporary accounts and are closed (brought to zero) at the end of the year. Let's take a look at an example.

On December 31, 2021, we estimated that uncollectible accounts will be 2% from gross Sales. Sales were \$3,000,000. Since our estimate is 2% of Sales, and Sales is \$3,000,000, we estimate that the balance of Bad Debt Expense will be \$60,000. Remember, the beginning balance of the Bad Debt Expense account is always zero (0).

	Bad Debt Expense
Beg. Balance	0
Adjusting Entry	\$60,000
Adj. Balance	60,000

The journal entry to record the estimate of uncollectible accounts is the adjusting entry, which looks like this:

Dec. 31, 2021	Bad Debt Expense.....	60,000
	Allowance for Doubtful Accounts.....	60,000

$$\$3,000,000 * 2\% = \$60,000$$

Further, let's say that on March 1, 2022, we determined that we cannot collect \$56,000. The journal entry to write-off the uncollectible accounts will look like this:

Mar. 1, 2022	Allowance for Doubtful Accounts.....	56,000
	Accounts Receivable.....	56,000

As you can notice, we made a mistake in our estimate. We overestimated the amount of uncollectible accounts by \$4,000. Therefore, the Allowance for Doubtful Accounts will carry a balance of \$4,000.

Allowance for Doubtful Accounts			
		\$60,000	Dec. 31, 2021
Mar. 1, 2022	\$56,000		
		\$4,000	Adjusted Balance

Basing our estimate as a percentage of sales is not a very accurate way of estimating uncollectible accounts. The reason for this, is that not all sales are made on account. Some of the sales were cash sales. Maybe we can improve our estimate by basing the estimate only on credit sales. Or maybe, we can base the estimate on the balance in our accounts receivable account, or precisely the customers who did not pay yet. Let's see how we make the estimates based on Receivables in the following section.

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Allowance Method based on Receivables

As we discussed previously, the estimate of uncollectible accounts is more accurate if it were based on Receivables. We can either base the estimate as a **percentage of the receivables balance**, or we can create an **aging receivable schedule** (the most accurate method).

The important thing to note in this section is that when we base our estimate on receivables, we focus on the Balance Sheet and calculate the balance of the Allowance for Doubtful Accounts. This is different from the estimate that was

based on sales, in which we focused on the Income Statement and calculated the balance of the Bad Debt Expense. Let's continue our example from before.

We already have a credit balance of \$4,000 in our Allowance for Doubtful Accounts account. The balance of Accounts Receivable is \$1,600,000. On December 31, 2022, we estimated that 4% of our accounts receivable balance will be uncollectible, or \$64,000 ($\$1,600,000 \times 4\%$). Remember, when we base the estimate on receivables, we calculate the balance of Allowance for Doubtful Accounts.

Allowance for Doubtful Accounts	
\$4,000	Beg. Balance
60,000	Adjusting Entry
\$64,000	Adj. Balance

We already had a balance of \$4,000 in the Allowance for Doubtful Accounts. Our estimate of \$64,000 is the new adjusted balance of this account. To get from a \$4,000 beginning balance to \$64,000 ending balance, we need to credit the account for \$60,000. This is our adjusting entry:

Dec. 31, 2022	Bad Debt Expense.....	60,000
	Allowance for Doubtful Accounts.....	60,000

As you can notice, the beginning balance in the Allowance for Doubtful Accounts makes all the difference when we base our estimate on receivables.

When we write-off an account, we follow the same principle as we did before. Let's see an example.

Let's say that on March 1, 2022, we determined that we cannot collect \$56,000. The journal entry to write-off the uncollectible accounts will look like this:

Mar. 1, 2022	Allowance for Doubtful Accounts.....	56,000
	Accounts Receivable.....	56,000

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Aging Receivables Schedule

The most accurate and the most widely used method is the Aging Receivables Schedule. This is how an aging receivables schedule looks like:

Customer	Balances	Not Due Yet	1-30 Days Past Due	31-60 Days Past Due	61-90 Days Past Due	90+ Days Past Due
John Smith	\$ 6,000		\$ 6,000			
ABC Company	\$ 1,000	\$ 1,000				
Christina Baez	\$ 12,000			\$ 12,000		
Jane Doe	\$ 3,000					\$ 3,000
Timothy Kang	\$ 5,000	\$ 5,000				
Stigna, LLC	\$ 14,000				\$ 14,000	
Total Receivables	\$ 41,000	\$ 6,000	\$ 6,000	\$ 12,000	\$ 14,000	\$ 3,000
Uncollectible %		1%	3%	15%	20%	80%
Uncollectible Estimate	\$ 7,240	\$ 60	\$ 180	\$ 1,800	\$ 2,800	\$ 2,400

The aging receivable schedule shown above presents how the estimate for uncollectible accounts is calculated, using this method.

Specifically, we have probabilities (**Uncollectible %**) about how much of each balance will not be collected. The longer a customer's balance is past due, the less

probability we will collect. The balance of ABC Company is not due yet. Therefore, the probability of it to be uncollected is very low, 1%. However, the balance of Jane Doe is past due for over 90 days, meaning the probability that we will not collect is very high, 80%.

To make the estimate, we multiply each of the balances (by their past due dates) with the estimated probability of not collecting those balances. Finally, we add those estimates to get \$7,240, the estimated uncollectible amount.

The accounting when using this method is the same as the “Allowance Method based on Receivables”, described above.

Notes Receivable

Notes Receivable are different than Accounts Receivable. The Note Receivable is a written promissory note that specifies the exact amount owed, interest, and the due date for payment.

There are few situations when our company would issue Notes Receivables. We can issue a Notes Receivable when (1) we sell inventory to a new customer, for whom we are not sure about his/her payment habits, or a customer who has a bad credit; (2) to allow a customer who hasn't paid the Accounts Receivable balance, to have more time to pay us.

Selling Inventory by Issuing a Note

On September 1st, 2022, our company sold inventory for \$1,000 to our new customer, John Smith. We issued a note for \$1,000, 10% interest, due within 90 days. The cost of the inventory was \$700. This is the journal entry:

Sep. 1, 2022	N/R - J. Smith	1,000	
	Revenue		1,000
	Cost of Goods Sold	700	
	Inventory		700

In 90 days, on November 29th, 2022, John Smith paid the full balance plus interest. This is the journal entry:

Nov. 29, 2022	Cash	1,025	
	N/R - J. Smith		1,000
	Interest Revenue		25

The interest is calculated to be \$25 ($\$1,000 \times 10\% \times 90/365$). Cash received is the principal of the note, \$1,000 plus the interest of \$25. We credit the N/R because John Smith does not owe us anymore.

Issuing a Note for Overdue Customer

On January 2nd, 2023, our company sold \$2,000 inventory to Timothy Kang, on account. The cost of the inventory was \$1,300. This is the journal entry:

Jan. 2, 2023	A/R - T. Kang	2,000	
	Revenue		2,000
	Cost of Goods Sold	1,300	
	Inventory		1,300

The above journal entry is the entry we record when we sell inventory on account.

Timothy Kang could not pay his balance by the due date and he asked our company for time extension. On May 1, 2023, we agreed to issue a note to Timothy for \$2,000, but now we added 7% interest because he is a risky customer. The note is due within 60 days. This is the journal entry to record the issuance of the note:

May 1, 2023	N/R - T. Kang	2,000	
	A/R - T. Kang		2,000

Basically, we eliminate Timothy's accounts receivable and transfer that amount to Notes Receivable. In 60 days, two things can happen: (1) Timothy pays the full balance plus interest (honoring the note); or (2) Timothy does not pay anything (dishonoring the note).

Honoring the Note

Timothy paid the full balance plus interest within 60 days. This is the journal entry:

Jun. 29, 2023	Cash	2,023	
	N/R - T. Kang		2,000
	Interest Revenue		23

The interest revenue is calculated to be \$23 ($\$2,000 \times 7\% \times 90/365$). We received both the principal, \$2,000 and the interest, \$23. We credit the N/R because Timothy Kang does not owe us anymore.

Dishonoring the Note

Timothy did not pay anything in 60 days. This is the journal entry to record a dishonored note:

Jun. 29, 2023	A/R - T. Kang	2,023	
	N/R - T. Kang		2,000
	Interest Revenue		23

When a customer dishonors a note receivable, we reinstate their A/R account and eliminate the N/R balance. The reason for this is because we want to continue to collect the \$2,023 from Timothy. We also want to make sure that whenever Timothy applies for credit in our company, the dishonored note will show up and we will reject that request.

Chapter 8: Fixed Assets

Fixed assets, or plant assets, are assets that we intend to use for a longer period of time. These assets help us carry out our operations. Examples of fixed assets are: building, land, equipment, vehicles, etc. Imagine that we have a tutoring business. In the business we have desks, blackboard, and a projector. All of these assets are fixed assets. We intend to use them for a longer period and they help us carry out our operations - tutoring students. Another example is a business that manufactures bottles. The fixed assets are the factory building, the machine to produce the bottles, the forklifts, etc. Again, these assets are helping us carry out our operations - produce bottles.

There is a big difference between current assets (i.e. cash, accounts receivable, supplies) and fixed assets. For example, when we use our supplies, we record an entry recording the reduction in supplies. Here is an example: On October 31, we have used \$200 worth of supplies. This is the journal entry:

Oct. 31	Supplies Expense.....	200
	Supplies.....	200

As you can see in the entry above, we immediately reduce Supplies by crediting them, and debit Supplies Expense to recognize the use of supplies. This is not the case with fixed assets. We cannot reduce the amount of a building, for example. The fixed assets have some specific rules when we account for them.

In this chapter, we will learn about the accounting for (1) purchasing (acquiring) a fixed asset, (2) using a fixed asset, and (3) disposing of a fixed asset.

Acquisition (purchase) of Fixed Assets

Recording the acquisition of fixed assets should be fairly simple. Every asset that we acquire we will record it at cost. However, the cost might be a little trickier to identify. When we are determining the cost of a fixed asset, we have to consider all

reasonable costs to get the asset in our business and get it ready to use. The perfect example is the machine we needed for our bottle production business. Let's say that we bought the machine for \$750,000. However, we also paid \$15,000 for the machine to be shipped to our factory, and another \$20,000 for experts to assemble the machine and get it ready for use. Therefore, the cost of this machine is \$785,000 (\$750,000+\$15,000+\$20,000). If we did not pay for the shipping cost and the assembly cost, we could not have used the machine. This would be the journal entry:

Sept. 1	Machine Equipment.....	785,000	
	Cash.....		785,000

Another example is Buildings. When we purchase a building, there are many costs associated with purchasing that building. Some examples include: the price of the building, title fees, attorney fees, taxes, repairs or remodeling to prepare it for its intended use, etc. On the other hand, we might construct our own building instead of buying it. The costs would then include construction costs, architecture fees, building permits fees, etc.

Lump Sum Purchase

Many times, when we purchase a building, we purchase the building, the land, and the land improvements. This is called a lump sum purchase. The appraised values of these three fixed assets might be different than the purchase price. For example, our business bought a building, land, and land improvements for \$80,000. The appraised value of the building was \$70,000, land - \$20,000, and land improvements - \$10,000. The appraised value of this lump sum purchase was \$100,000. However, we only paid \$80,000. Then, how are we going to record these three fixed assets in our books? Since we need to record them at cost, we need to record them at a total of \$80,000. So, how much of the \$80,000 goes to the building, how much to the land, and how much to the land improvements? The

answer is, we have to allocate the cost based on the appraised values. To do that, we first calculate the percentage of each asset of the total appraised value:

	Appraised Value	Percentage	Calculation
Building	70,000	70%	70,000/100,000
Land	20,000	20%	20,000/100,000
Land Improvements	10,000	10%	10,000/100,000
	<u>\$100,000</u>	<u>100%</u>	

Once we find the percentages, each of the fixed assets will be assigned the appropriate percentage of the purchase price. This is the calculation:

	Purchase Price	Percentage	Allocated Cost
Building	80,000	70%	56,000
Land	80,000	20%	16,000
Land Improvements	80,000	10%	8,000
			<u>80,000</u>

We multiply the purchase price with the appropriate percentage to get the allocated cost. Therefore, the building will have a cost of \$56,000 ($\$80,000 \times 70\%$), the land will have a cost of \$16,000 ($\$80,000 \times 20\%$), and the land improvements will have a cost of \$8,000 ($\$80,000 \times 10\%$). The journal entry to record this purchase will look like this:

Building.....	56,000	
Land.....	16,000	
Land Improvements.....	8,000	
Cash.....		80,000

We simply debit the fixed assets for the amount of the allocated cost, and we credit the amount we paid.

Use of Fixed Assets

As we mentioned in the introduction of this chapter, when we use a fixed asset, we cannot reduce its cost. Instead, we depreciate fixed assets. Depreciation is an allocation of its cost in the period in which we use that particular asset. Please keep in mind that depreciation is not a decline in value (devaluation), and it is not the deterioration of that asset. We depreciate a fixed asset by allocating its cost over the period when we use it. To depreciate a fixed asset, we need to follow the allowed methods for depreciation: (1) Straight-Line Method, (2) Units-of-Production Method, and (2) Double Declining Balance Method. There is also a 4th allowed method, called Sum-of-the-Years'-Digits Method, but it is not part of this book. In this book, we will focus on the three methods mentioned above - the methods that are used most commonly in business.

Adjusting Entry and New Accounts

Before we start the discussion about the different depreciation methods, let's take a look at the journal entry appropriate to record depreciation, regardless of the method used.

Dec. 31	Depreciation Expense - Building.....	5,000
	Accumulated Depreciation - Building.....	5,000

As you can see in the entry above, we debit Depreciation Expense for the particular asset, and we also credit the Accumulated Depreciation for the same asset. Depreciation Expense is simply an expense account, which shows the use of the fixed asset. However, the Accumulated Depreciation account is a contra-asset account. Its normal balance is Credit. Since it is a contra-asset account, the Accumulated Depreciation account shows up on the balance sheet, right below the associated fixed asset.

Best Coffee Shop Company
BALANCE SHEET
December 31, 2021

ASSETS

Current Assets	
Cash	10,000
Accounts Receivable	45,000
Supplies	20,000
Prepaid Insurance	24,000
Total Current Assets	99,000
 Property, Plant, and Equipment	
Building	56,000
Less: Accumulated Depreciation - Building	5,000
Equipment	40,000
Less: Accumulated Depreciation - Equipment	1,000
Land Improvements	8,000
Less: Accumulated Depreciation - Land Improvements	700
Land	16,000
Total PP&E	113,300

As you can notice from the Balance Sheet segment above, the Accumulated Depreciation accounts are subtracted from the original cost of the fixed assets. Take for example, the Building account. The original cost is \$56,000. However, the Accumulated Depreciation for the Building is \$5,000. Therefore, the book value of the building is \$51,000 (\$56,000 - \$5,000). Book value is the difference between the cost of the fixed asset and its accumulated depreciation.

In the following sections, we will learn the methods used to calculate the depreciation of a fixed asset. Regardless of the method used, the adjusting journal entry and the presentation in the balance sheet will be the same as described in this section.

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Straight-Line Method

The straight-line method is the most widely used method by businesses. The reason for such a wide use of this method is its simplicity. The formula used to calculate the depreciation expense, using the straight-line method looks like this:

$$\text{Depreciation Expense} = \frac{\text{Cost} - \text{Salvage Value}}{\text{Useful Life (in periods)}}$$

The cost is the original cost of the fixed asset. Salvage value is an estimate of the value of the asset when we dispose of it after the useful life has expired. Finally, the useful life of a fixed asset is the amount of time the fixed asset will be useful for our operations, in terms of periods (months, years, etc.).

Let's take a look at an example. On January 1st, 2021, our company bought equipment at a cost of \$120,000. The useful life of the equipment was estimated to be 4 years, or 100,000 units. The salvage value at the end of the equipment's useful

life was estimated to be \$20,000. Calculate the depreciation expense for each of the 4 years, using the straight-line method.

If we follow the formula from above, the depreciation expense would be \$25,000 per year. The calculation of the \$25,000 per year is shown below.

$$\text{Depreciation Expense} = \frac{\$120,000 - \$20,000}{4 \text{ years}}$$

The adjusting journal entry for each year will be:

Dec. 31	Depreciation Expense - Equipment.....	25,000
	Accumulated Depreciation - Equipment.....	25,000

This same adjusting entry will be recorded at the end of each of the 4 years. This simplicity is the reason why the straight-line method is the most widely used method across businesses. Let's take a look at all the details during those 4 years.

Year	Depreciation Expense	Accumulated Depreciation
2021	25,000	25,000
2022	25,000	50,000
2023	25,000	75,000
2024	25,000	100,000
	<u>100,000</u>	

The depreciation recorded each year for 4 years is \$25,000. The total depreciation expense over the useful life of the asset is \$100,000, or the depreciable cost. Depreciable cost is the difference between the cost of the asset and its salvage value. The accumulated depreciation account is a permanent account located on the balance sheet. Therefore, it accumulates over the years.

Units-of-Production Method

The units-of-production method (or units-of-output method) is probably the most accurate method to measure depreciation. However, using a more accurate method, usually means more expensive method. The formula to calculate the depreciation expense, using the units-of-production method looks like this:

$$\text{Depreciation Expense} = \frac{\text{Cost} - \text{Salvage Value}}{\text{Useful Life}} \\ (\text{in units of measurement})$$

The cost is the original cost of the fixed asset. Salvage value is an estimate of the value of the asset when we dispose of it after the useful life has expired. Finally, the useful life of a fixed asset is the amount the fixed asset will be useful for our operations, in terms of units of measurement (units, miles, etc.).

Let's take a look at an example. On January 1st, 2021, our company bought equipment at a cost of \$120,000. The useful life of the equipment was estimated to be 4 years, or 100,000 units. The salvage value at the end of the equipment's useful life was estimated to be \$20,000. Our company produced 20,000 units during 2021, 25,000 units during 2022, 30,000 units during 2023, and 25,000 units during 2024. Calculate the depreciation expense for each of the 4 years, using the units of production method.

If we follow the formula from above, the depreciation expense would be \$1 per unit produced. The calculation of the \$1 per unit produced is shown below.

$$\text{Depreciation Expense} = \frac{\$120,000 - \$20,000}{100,000 \text{ units}}$$

To find the actual depreciation expense for each year, we need to multiply the units produced that year with \$1, as shown below.

Year	Depreciation per unit	Units Produced	Depreciation Expense	Accumulated Depreciation
2021	\$1	20,000	20,000	20,000
2022	\$1	25,000	25,000	45,000
2023	\$1	30,000	30,000	75,000
2024	\$1	25,000	25,000	100,000
			100,000	

As you can see from the table above, the depreciation expense recorded each year will depend on the number of units produced each year.

Double Declining Balance Method

The Double Declining Method for depreciation is an accelerated method allowing the company to allocate most of its cost in the early years. To calculate the depreciation under the double declining balance method, we need to follow three steps.

We need to find the Straight-Line rate (1/Useful Life [in periods])

Double Declining Rate = Double the Straight-Line Rate (multiply by 2)

Multiply the doubled straight-line rate with the Net Book Value (NBV)

Let's take a look at an example. On January 1st, 2021, our company bought equipment at a cost of \$120,000. The useful life of the equipment was estimated to be 5 years, or 100,000 units. The salvage value at the end of the equipment's useful

life was estimated to be \$20,000. Calculate the depreciation expense for each of the 5 years, using the double declining balance method.

The Straight-Line Rate is 1 divided by the useful life in periods. In our case, that is 1/5, or 20%. Thus, the double declining rate is 40% (20%*2). Before we see the results, I want to remind you that the Net Book Value is the Original Cost of the Asset minus the Accumulated Depreciation Balance.

Year	Beg. NBV	DDB Rate	Depreciaton Expense	Accumulated Depreciation	End. NBV
2021	120,000	40%	48,000	48,000	72,000
2022	72,000	40%	28,800	76,800	43,200
2023	43,200	40%	17,280	94,080	25,920
2024	25,920	40%	5,920	100,000	20,000
2025	20,000	40%	-	-	-
			100,000		

As we mentioned earlier, we can depreciate an asset for more than the depreciable cost. In our case, the depreciable cost is \$100,000 (Cost - Salvage Value). Let's go through the calculations together.

In 2021, we just bought the equipment. We have not depreciated anything. Therefore, the balance in the Accumulated Depreciation account was 0. Book Value is the original cost minus the balance in the Accumulated Depreciation account. In our case, for year 2021, it is 120,000 - 0 = 120,000. Thus, the beginning Net Book Value is 120,000. We multiply 120,000 with 40% to get the depreciation expense for year 2021, \$48,000. The adjusting journal entry on December 31, 2021, looks like this:

Dec. 31, 2021	Depreciation Expense.....	48,000
	Accumulated Depreciation.....	48,000

Since now we have a balance of 48,000 in our Accumulated Depreciation account, the Net Book Value changes. The new Net Book Value is \$72,000 (120,000 - 48,000). Therefore, to find the depreciation expense for year 2022, we multiply

72,000 with the DDB rate of 40%, to get \$28,800. The adjusting journal entry on December 31, 2022, looks like this:

Dec. 31, 2022	Depreciation Expense.....	28,800	
	Accumulated Depreciation.....		28,800

By the end of 2022, the accumulated depreciation account has accumulated to \$76,800 (48,000+28,800). This means that the new Net Book Value will be \$43,200 (120,000 - 76,800). Therefore, to find the depreciation expense for year 2023, we multiply 43,200 with the DDB rate of 40%, to get \$17,280. The adjusting journal entry on December 31, 2023, looks like this:

Dec. 31, 2023	Depreciation Expense.....	17,280	
	Accumulated Depreciation.....		17,280

The new balance in the accumulated depreciation account is now 94,080. The Net Book Value at the end year 2023 will be \$25,920 (120,000 - 94,080). If we multiply \$25,920 with 40% to find the depreciation expense for year 2024, we will get \$10,368. This is wrong. Remember, we cannot depreciate more than the depreciable cost - in our case \$100,000. The Ending Net Book Value must equal the salvage value. Therefore, the most we can depreciate during the year 2024 is \$5,920. We cannot depreciate anything during 2025.

Look again at the amortization table below. When we do this, the total depreciation expense is \$100,000. The balance in the accumulated depreciation account is \$100,000. The Ending Net Book Value is \$20,000, exactly our salvage value.

Year	Beg. NBV	DDB Rate	Depreciaton Expense	Accumulated Depreciation	End. NBV
2021	120,000	40%	48,000	48,000	72,000
2022	72,000	40%	28,800	76,800	43,200
2023	43,200	40%	17,280	94,080	25,920
2024	25,920	40%	5,920	100,000	20,000
2025	20,000	40%	-	-	-
			100,000		

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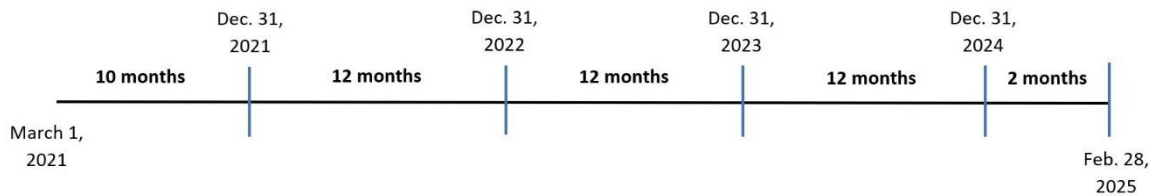
Partial Year Depreciation

In all of our examples so far, we assumed that we bought the fixed asset on January 1st. This means that we got to use it for an entire year. We all know that we do not buy fixed assets only at the beginning of the year. Sometimes, we buy assets on April 1st. If we buy an asset on April 1st, we only use it for 9 months that year, from April 1st, until December 31st. If we buy the asset on October 1st, we only use it for 3 months that year, from October 1st to December 31st. Calculating depreciation expense for an asset that we used it for less than 12 months, is called partial year depreciation.

When calculating partial year depreciation, all the calculations are the same as we discussed previously, except the calculation of the first and the last year of the asset's useful life. Let's look at an example.

On March 1, 2021 we bought machinery equipment for \$22,000. The estimated useful life of the machine is 4 years. The estimated salvage value at the end of the useful life is \$2,000. Calculate the depreciation expense for each of the 4 years, using the straight-line method.

The useful life is 4 years, or 48 months (4 years*12 months). The time line of the asset use is shown below:



You can notice from the figure above that we use the asset only for 10 months out of 12 for the 1st year (from March 1, 2021 to December 31, 2021). We use the asset for the full 12 months in year 2022, 2023, and 2024, and only 2 months in year 2025. In other words, we use for a total of 48 months (4 years).

When we use the formula to calculate depreciation expense using the straight-line method, we get a yearly depreciation of \$5,000.

$$\text{Depreciation Expense} = \frac{\$22,000 - \$2,000}{4 \text{ years}}$$

The \$5,000 is a yearly depreciation, for a full year (12 months). To find the depreciation for the 10 months, we simply multiply the yearly depreciation (\$5,000) with 10/12, to get \$4,167. The depreciation expense for the years 2022, 2023, and 2024 will be \$5,000 because we are using the machinery equipment for a full year. In 2025, the depreciation expense will be \$833 (\$5,000 * 2/10), because we only use it for 2 months.

The example above shows calculation of partial year depreciation using the straight-line method. The calculations for double-declining balance method are the same for the first and last year. All we need to change from the regular methods is to account only for the months we use the asset.

Change in Depreciation Estimates

When we calculate the depreciation expense, we use 2 estimates - useful life and salvage value. Many times, making estimates means making mistakes. Sometimes the useful life might be longer or shorter than we expected. Sometimes the salvage value might be higher or lower than expected. If that is the case, we can change the estimate when we realize that we made a mistake. However, we can only make a change in estimate prospectively. That is, what happened in the past (the amount we depreciated) cannot be changed, we can only change how much we depreciate in the future. The formula used to change the estimate looks like this:

$$\text{Revised Depreciation Expense} = \frac{\text{Book Value} - \text{Revised Salvage Value}}{\text{Remaining Useful Life}}$$

On January 1, 2021 we bought a vehicle for \$25,000. The estimated useful life of the vehicle was 4 years. The estimate salvage value was \$5,000. At the end of the second year, we realized that the vehicle's useful life will be a total of 6 years and at the end of the 6 years, the salvage value will be \$3,000.

Using the regular formula for calculating depreciation expense under the straight-line method, the depreciation expense will be \$5,000.

$$\text{Depreciation Expense} = \frac{\$25,000 - \$5,000}{4 \text{ years}}$$

We will depreciate the asset for the first 2 years for \$5,000 each year. However, at the end of the second year we changed the estimate. Since we already depreciated 2 years, or \$10,000 (\$5,000+\$5,000), the balance in the accumulated depreciation account is \$10,000. Our book value at the end of year 2 is \$15,000 (\$25,000 -

\$10,000). The original useful life was 4 years and we changed it to 6 years. Since we already depreciated 2 years, we have 4 years remaining. The revised salvage value was \$3,000. Using the formula for revised depreciation we get a depreciation expense of \$3,000 for the remaining 4 years of useful life.

$$\text{Revised Depreciation Expense} = \frac{\$15,000 - \$3,000}{4 \text{ years remaining}}$$

To sum up, the depreciation expense for the years 2021 and 2022 will be \$5,000 each year. The depreciation expense after our revision, for years 2023, 2024, 2025, and 2026 will be \$3,000 per year.

Disposing of Fixed Assets

So far, we covered the acquisition of fixed assets and the use of those assets. It is now time to learn the accounting for disposing of fixed assets. To dispose of fixed assets means that we are getting rid of it. To get rid of it, we can sell it, throw it away, donate it, or trade it for something else.

Whatever we decide to do, we need to take the fixed asset off our balance sheet because when we get rid of it, we do not want our investors to believe that we still have it.

Throwing Away a Fixed Asset

We have an equipment for bottle production on our books recorded at a cost of \$50,000. The accumulated depreciation for this equipment is \$40,000. Record the journal entry to dispose of this equipment.

Loss on Asset Disposal.....	10,000
Accumulated Depreciation - Equipment.....	40,000
Equipment.....	50,000

As you can see from the entry above, we credit the Equipment account to bring it to 0, since we do not have it anymore. We debit the accumulated depreciation account to bring it to 0 since we do not have the asset anymore. We know that in every journal entry, debits and credits must equal. Therefore, we debit Loss on Asset Disposal for \$10,000. This amount, \$10,000 is the actual Book Value of the Equipment. When we threw the asset away, we need to record a loss of \$10,000.

The Loss on Asset Disposal Account is a non-operating expense account. Our business' primary operations are not buying and selling equipment. Our operation is producing bottles. Therefore, this account will be a non-operating expense.

Selling a Fixed Asset

Selling a fixed asset is different than throwing it away or donating it, because we receive money from the sale. Let's take a look at 3 different examples.

Example 1 - Sale at book value

We have an equipment for bottle production on our books recorded at a cost of \$50,000. The accumulated depreciation for this equipment is \$40,000. Record the journal entry to sell the equipment for \$10,000.

Cash.....	10,000
Accumulated Depreciation - Equipment.....	40,000
Equipment.....	50,000

As you can notice in the entry above, we received Cash for \$10,000 and wrote-off the equipment and its associated accumulated depreciation account. Since we sold the asset for book value, we have no losses or gains.

Example 2 - Sale for less than book value

We have an equipment for bottle production on our books recorded at a cost of \$50,000. The accumulated depreciation for this equipment is \$40,000. Record the journal entry to sell the equipment for \$8,000.

Cash.....	8,000
Loss on Sale of Asset.....	2,000
Accumulated Depreciation - Equipment.....	40,000
Equipment.....	50,000

As you can notice in the entry above, we received Cash for \$8,000 and wrote-off the equipment and its associated accumulated depreciation account. Debits and credits must equal, so we debit Loss on Sale of Asset for \$2,000. This is the case because we sold the asset for \$8,000, or \$2,000 less than the book value (which is \$10,000).

The Loss on Sale of Asset Account is a non-operating expense account. Our business' primary operations are not buying and selling equipment. Our operation is producing bottles. Therefore, this account will be a non-operating expense.

Example 3 - Sale for more than book value

We have an equipment for bottle production on our books recorded at a cost of \$50,000. The accumulated depreciation for this equipment is \$40,000. Record the journal entry to sell the equipment for \$14,000.

Cash.....	14,000
Accumulated Depreciation - Equipment.....	40,000
Equipment.....	50,000
Gain on Sale of Asset.....	4,000

As you can notice in the entry above, we received Cash for \$14,000 and wrote-off the equipment and its associated accumulated depreciation account. Debits and credits must equal, so we credit Gain on Sale of Asset for \$4,000. This is the case because we sold the asset for \$14,000, or \$4,000 more than the book value (which is \$10,000).

The Gain on Sale of Asset Account is a non-operating revenue account. Our business' primary operations are not buying and selling equipment. Our operation is producing bottles. Therefore, this account will be a non-operating revenue.

Chapter 8 Appendix: Intangible Assets

Intangible assets cannot be seen or touched, but still have a value. Examples of intangible assets are patents, copyright, franchises, trademarks, goodwill, etc.

When a company purchases an intangible asset, it records it at its cost. If the intangible asset's **life is limited**, we need to allocate its cost over the useful life through **amortization** (just like depreciation). If the intangible asset's life is **unlimited, then, there is no amortization**.

Patents

A patent is a right to the person or company that invented it to sell and use for 20 years. An example of a patent is the patent for a treadmill. Someone invented it and they have the right to use it or sell it to others.

As you can notice, patents have a limited life. That means they are subject to amortization. For example, our company might buy a patent for \$30,000 and a useful life of 20 years. The annual amortization will be \$1,500 ($\$30,000/20$ years). The journal entry will be as following:

Amortization Expense - Patents	\$1,500	
Accumulated Amortizaiton - Patents		\$1,500

Copyright

Copyright is a right of the person who made it to publish and sell. Copyrighted material includes music, movies, art work, literature, etc. The life of a copyright is for the length of the creator's life plus 70 years.

Since Copyrights have a limited life, they can be amortized as well.

Trademarks

Trademark is a brand name that might include the name of a company, a phrase, a symbol, etc. Examples are Coca Cola, Google, Jeep, Ziploc and others. Trademarks can be renewed indefinitely. If trademarks are renewed indefinitely, they do not need to be amortized.

Goodwill

Goodwill is an asset. This issue is still debated in accounting research. Generally, goodwill is recorded when one company acquires another company. Let's say that Company A buys Company B for \$100,000. The net assets (assets minus liabilities) of Company B are \$20,000. When Company A pays \$100,000, it only receives \$20,000 worth of assets. The difference, \$80,000 is recorded as Goodwill, which is an asset.

So, Goodwill is the excess amount paid of the cost of acquiring Company B. We cannot amortize Goodwill. However, we can test it for impairment. Goodwill is impaired if the book value exceeds its fair value, and therefore, we need record an impairment loss.

Many accounting scholars debate the treatment of Goodwill. While many say that it is an asset, just like it is now, others are saying that Goodwill should be a reduction to equity.

Chapter 9: Current, Payroll, Estimated, and Contingent Liabilities

Current Liabilities

Current Liabilities are debts (money we owe), and they are due within one year or less. As opposed to current liabilities, long-term liabilities are due beyond one year, as explained in Chapter 10. The most common current liabilities and how we account for them, are presented below:

Accounts Payable

Accounts payable are current liabilities and we record them when we purchase something on credit. For example, our company bought supplies on account, for \$500. This is the journal entry:

Supplies	500	
Accounts Payable		500

When we pay the Accounts Payable, we record the following entry:

Accounts Payable	500	
Cash		500

Sales Taxes Payable

Sales Taxes Payable is another current liability. We record this liability whenever we sell a product for which we collect sales tax. The sales tax we collect as a company needs to be remitted to the State. For example, we sold a product for \$200 cash. The sales tax in our state was 7%. This is the journal entry:

Cash	214	
Sales Revenue		200
Sales Tax Payable		14

The sales tax is calculated to be \$14 ($\$200 \times 7\%$). When we remit the sales tax to the State, we record the following entry:

Sales Tax Payable	14	
Cash		14

Unearned Revenues

We covered Unearned Revenues in Chapter 3. The unearned revenues were the adjusting entries we had to make at the end of an accounting period.

Short-Term Notes Payable

Short-Term Notes Payable are another form of current liabilities. The S-T Notes Payable are written promises to pay. Basically, we sign a promissory note which is normally due within a year or less. The difference between Accounts Payable and Short-Term Notes Payable is that we generally need to sign a promissory note and pay interest for S-T Notes Payable. Short-Term Notes Payable can be obtained directly from the bank or received by a supplier to extend the period to pay an accounts payable.

Obtained from Bank

On September 1, 2022, our company borrowed \$10,000, with 7% interest from First Bank. The note is due within 60 days. The journal entry to record this transaction is as follows:

Sep. 1, 2022	Cash	10,000	
	S-T Notes Payable		10,000

The note was due in 60 days. When we repay the note, we need to pay both the principal and the interest of the note. The interest is calculated as $\text{Principal} \times \text{Interest Rate} \times \text{Time}$. In our case the interest is 7% per year. That is why our time will be $60/365$, which stands for 60 days out of the entire year. The interest we need to pay for the 60 days is \$115 ($\$10,000 \times 7\% \times 60/365$). The journal entry to record the payment is as follows:

Oct. 30, 2022	Interest Expense	115	
	S-T Notes Payable	10,000	
	Cash		10,115

Obtained from Supplier

Our company had a balance in Accounts Payable in the amount of \$700. We failed to pay within the due date and asked the supplier to extend that payment due date. However, because the supplier does not trust us as much, they offered for us to sign a short-term notes payable. On June 1, 2022, the note payable we signed was \$700, with 12% interest, due within 30 days. This is the journal entry to record this transaction:

Jun. 1, 2022	S-T Notes Payable	700	
	A/P		700

When we pay the note, we need to pay both principal and interest. The interest is \$14 ($\$700 \times 12\% \times 30/365$). The journal entry to record the payment of the note is as follows:

Jul. 1, 2022	Interest Expense	14	
	S-T Notes Payable	700	
	Cash		714

Payroll Liabilities

Employee Earnings

If the company has employees, those employees need to be paid. We can generally pay salaries (annual payments) or wages (per hour payments). Additionally, the company might need some of the employees to work overtime, having to pay overtime pay. When paying employees, businesses must follow the Fair Labor Standards Act, which is a Federal Law.

Let's say that our employee, Jane Doe, receives \$30 per hour. If Jane Doe works overtime (in excess of 40 hours per week), we pay her 1.5 times the regular pay. Jane doe worked 50 hours last week. She earned \$1,650 last week, as shown below:

Regular Earnings	1,200
40 hours * \$30 per hour	
Overtime Earnings	450
10 hours * \$30 * 1.5	
Total Earnings	<u>1,650</u>

Employee Deductions

The \$1,650 that Jane Doe earned is the Gross Pay. This is not the amount she will take home. The amount that Jane Doe is taking home is called Net Pay. Net Pay is the amount that we get when we subtract all the deductions from the Gross Pay. The most common deductions include: federal income tax withholding, state income tax withholding, local income tax withholding, medical insurance coverage deduction, contributions to a pension plan, etc.

W-4 Form

When you get hired for a company, the company will provide you with a W-4 form (Employee's Withholding Allowance Certificate). This is how it looks:

Form W-4 Department of the Treasury Internal Revenue Service		Employee's Withholding Allowance Certificate ▶ Whether you're entitled to claim a certain number of allowances or exemption from withholding is subject to review by the IRS. Your employer may be required to send a copy of this form to the IRS.		OMB No. 1545-0074 2018	
1 Your first name and middle initial		Last name		2 Your social security number	
Home address (number and street or rural route)		3 ☐ Single ☐ Married ☐ Married, but withhold at higher Single rate. Note: If married filing separately, check "Married, but withhold at higher Single rate."			
City or town, state, and ZIP code		4 If your last name differs from that shown on your social security card, check here. You must call 800-772-1213 for a replacement card. ▶ ☐			
5 Total number of allowances you're claiming (from the applicable worksheet on the following pages)		5			
6 Additional amount, if any, you want withheld from each paycheck		6 \$			
7 I claim exemption from withholding for 2018, and I certify that I meet both of the following conditions for exemption.					
• Last year I had a right to a refund of all federal income tax withheld because I had no tax liability, and • This year I expect a refund of all federal income tax withheld because I expect to have no tax liability. If you meet both conditions, write "Exempt" here		▶ 7			
Under penalties of perjury, I declare that I have examined this certificate and, to the best of my knowledge and belief, it is true, correct, and complete.					
Employee's signature (This form is not valid unless you sign it.) ▶		Date ▶			
8 Employer's name and address (Employer: Complete boxes 8 and 10 if sending to IRS and complete boxes 8, 9, and 10 if sending to State Directory of New Hires.)		9 First date of employment		10 Employer identification number (EIN)	

For Privacy Act and Paperwork Reduction Act Notice, see page 4. Cat. No. 10220Q Form **W-4** (2018)

As you can notice from the W-4 form, the employee needs to fill in the name, address, social security number, marital status, number of allowances, and any additional amount they want withheld from their paycheck. This form helps the

employers to determine the amount that will be withheld from the employee paycheck.

Federal Insurance Contributions Act (FICA)

FICA, or Federal Insurance Contributions Act, requires companies to withhold a part of the employees' pay for Social Security (6% of earnings up to a maximum of \$128,400 earnings for 2018) and Medicare (1.5% of earnings up to a maximum of \$128,700 earnings for 2018).

Employer's Payroll Taxes

The companies are also responsible to pay payroll taxes for their employees. Namely, companies need to match the employees' FICA tax, pay Federal Unemployment Tax (FUTA), and State Unemployment Tax (SUTA).

Estimated Liabilities

Estimated liabilities are simply liabilities that we know will occur, but, however, we do not know the exact amount. Not knowing the exact amount is problematic, but we can reasonably estimate those amounts. Some of the examples of estimated liabilities that are covered in this section are health and pension benefits and warranty liabilities.

Health and Pension Benefits

All companies pay salaries or wages to its employees. However, many companies also offer health and pension benefits packages. Health benefits may include medical, dental, or vision coverage. The company also might offer life insurance or disability insurance. The company might pay part of these benefits for the employees or pay the full amount of the benefits.

Our company pays \$7,400 for yearly medical insurance for our employee John Smith. We also contribute to his retirement in the amount of 12% of his \$100,000 gross salary. The journal entry is as follows:

John Smith Benefits Expense	19,400	
	John Smith Medical Insurance Payable	7,400
	John Smith Retirement Contribution Payable	12,000

As you can notice from the journal entry, we debit Benefits Expense for the employee, along with crediting our estimated liabilities.

Warranty Liabilities

Usually, many companies provide warranty on the products they sell. For example, a car manufacturer might provide a warranty on the cars it sells. Usually, the terms of the warranty for a car manufacturer are in terms of years and miles. When you buy a new car, you might get a 5-year warranty, or 100,000 miles warranty. The warranty is generally for situations in which the product will fail to perform as it was advertised, and the company will be obligated to replace or fix the product.

We sold a car for \$25,000 on June 1, 2022. The warranty was 5 years or 100,000 miles, bumper to bumper. When we sell the car, we need to estimate the possible warranty expense we will incur. Let's say that in our experience, the warranty expense is usually 2% of the car's price. The journal entry will be as follows:

Warranty Expense	500	
Estimated Warranty Payable		500

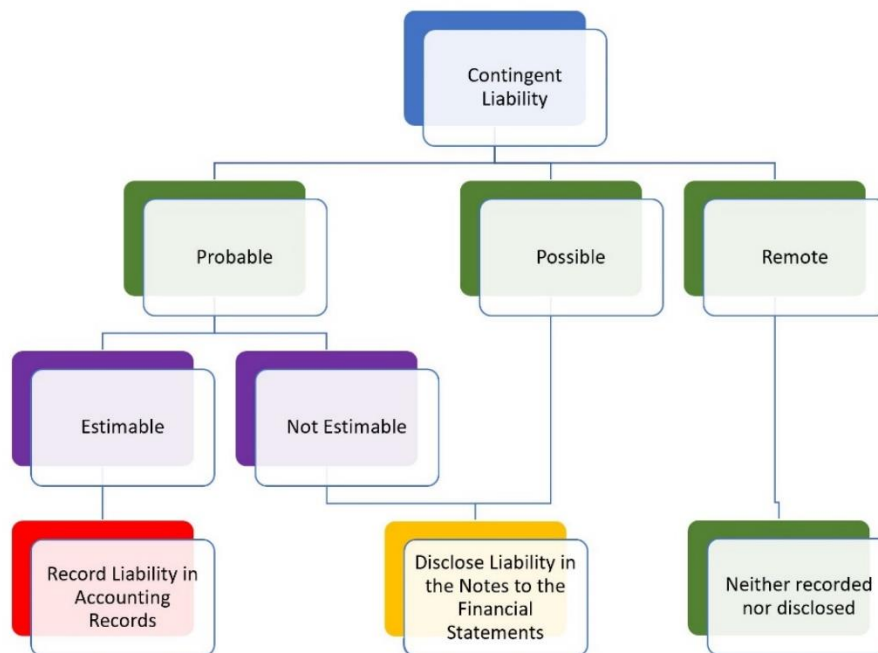
The journal entry above is only an estimate of the warranty expense. Let's say that the car broke down, the customer returned it to be repaired, and the costs to repair it were \$400. This the journal entry we need to make:

Estimated Warranty Payable	400	
Inventory - Parts		400

Whenever we make an estimate, we make mistakes. It is our job to rely on past data and forecasting analysis to make sure we are as accurate as possible.

Contingent Liabilities

The word contingent means "to depend on" something. Thus, contingent liabilities will depend on some outcome. The possible outcomes are presented below:



Take for example that we own a trucking company. One of our drivers went to a loading dock, truck was loaded with commodities, and driver was on his way to deliver those commodities to another destination. On the way, the load slipped from the trailer, straight onto the road. This event caused multiple car accidents and injuries. Now, our company is being sued by everyone involved for damages. Do we record this event in our accounting records? Do we know the amounts we will owe? This liability we have, is a contingent liability. It will depend on the outcome of the lawsuit.

Remote Scenario

The lawsuit against us was for \$10,000,000. When we consulted our attorneys and they told us that it was not our responsibility to load the truck. The truck was loaded by the employees of the company where the driver picked up the commodities. Therefore, it was a remote possibility (unlikely), that we will be responsible for the damages. In this scenario, we should neither disclose this liability in the notes to the financial statements nor record a liability in our accounting records.

Possible Scenario

The responsibility for not securing the load properly is hard to determine. When we consulted our attorneys, they told us that it is possible that we will be held accountable. However, they did not know about the amount that our company will be responsible for. In this scenario, we need to disclose this potential contingent liability in the notes to the financial statements.

Probable Scenario (Estimable and Non-Estimable)

Probable and Not Estimable

In this scenario, the attorneys advised us that we will probably lose the lawsuit and we will have to pay damages. However, they said that when it comes to damages, many of the cases are settled for a lesser amount. The amount that we could settle this lawsuit for, was not known and could not be estimated by attorney because it depended upon many factors. In a scenario where the liability is probable, but not estimable, we need to disclose the contingent liability in the notes to the financial statements.

Probable and Estimable

In this scenario, the attorneys advised us that we will probably lose the lawsuit and we will have to pay estimated damages of \$10,000,000. In a scenario where the liability is probable and estimable, we need to record the journal entry to our accounting records and disclose the contingent liability in the notes to the financial statements. The journal entry would look like this:

Lawsuit Contingent Expense	10,000,000
Lawsuit Contingent Liability	10,000,000

Recording Contingent Liability

Chapter 10: Long-Term Liabilities

In this chapter we will learn about long-term liabilities. Before we do that, let's remind ourselves of the accounting equation:

$$\text{ASSETS} = \text{LIABILITIES} + \text{STOCKHOLDERS' EQUITY}$$

The Accounting Equation

In order to grow our business, we need to increase our assets. For example, in order for a factory that produces automobiles to expand, it probably needs another factory. A retail store might want to open another store and increase its inventory. The factory, the store, and the inventory are all assets. This tells us that, to expand a company we need to increase our assets. However, increasing the assets means spending money we might not have.

Based on the accounting equation, there are two ways to increase our assets. Either issuing more stock (Chapter 11) or getting a loan and purchasing the assets we need to expand. In this chapter we are going to cover the long-term liabilities. We will specifically discuss the accounting for bonds and long-term notes payable.

Bonds

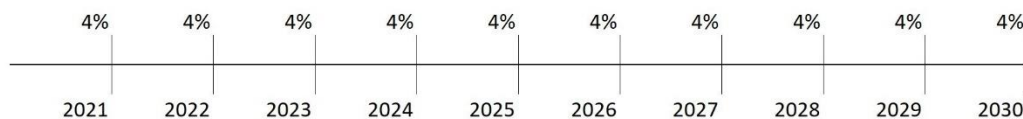
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Bond is a type of a long-term liability. Basically, we issue the bond and by that issuance, **we promise to pay back the amount on the bond plus interest**. Let's take a look at a sample bond and notice the key sections:



As you can see in the picture above, the bond certificate includes \$1,000 which is the par value (face value) of the bond. We can also see that it is a 10-year bond. Finally, the 4% interest is the stated (contract) interest rate and it is paid annually.

Our company issues this bond in order to get more cash to expand our company. We issue this certificate, and someone gives us \$1,000. We make 2 promises every time we issue a bond. The **first promise** is that we will pay interest of 4% every year, and the **second promise** is that we will pay the \$1,000 after 10 years. The 2 promises are illustrated below:



First Promise (Paying Interest)

The first promise is that our company will pay **4% interest** on the face amount of the bond (\$1,000).



Second Promise (Paying Face amount at Maturity)

The second promise is that our company will pay the **face amount of the bond** (\$1,000) at the maturity date (after 10 years).

In this chapter we will learn the accounting procedures for issuing a bond, paying the interest and face amount of the bond, and retirement of bonds. Finally, we will discuss about long-term notes payable.

Issuing Bonds

Issuing bonds means that we provide a written promise to whoever is buying our bond that we will pay the face amount of the bond at maturity and we will pay interest every year (or every quarter, or every 6 months).

Issuing Bonds at Face Value

When the stated (contract) interest rate of the bond is equal to the market (effective) interest rate, we issue the bonds at face (par) value. If our bond's stated interest rate is 5% and the market rate is also 5%, then we will issue the bond at face value. If the market interest rate is the same as ours, it means that nobody is offering to pay more, or less interest than us. Let's look at an example:

Our company issued bonds with a face amount (par value) of \$100,000 on January 1, 2021. The bonds' annual contract (stated) rate is 7%, and interest is paid annually on December 31. The bonds mature in 3 years. The annual market (effective) rate is 7%. This is how the bond looks like:



The journal entry to record the issuance of the bond, looks like this:

Jan. 1, 2021	Cash	100,000	
	Bonds Payable		100,000

Issuance of the bond

We debit Cash because we are receiving cash, and we credit Bonds Payable (long-term liability) because we need to pay the face amount after 3 years (a long-term liability). For each of the 3 years, we also need to pay interest in the amount of \$5,000 ($\$100,000 \times 5\%$). The entry to make the first interest payment will look like this:

Dec. 31, 2021	Interest Expense	5,000	
	Cash		5,000

Interest Payment

It is going to be the same interest payment entry for years 2021, 2022, and 2023. At the end of 2023 we would have fulfilled our promise to pay interest of 5% every year. However, we will also have to fulfill the other promise, to pay the face amount at maturity. This is the journal entry:

Dec. 31, 2023	Bonds Payable	100,000	
	Cash		100,000

Payment at Maturity

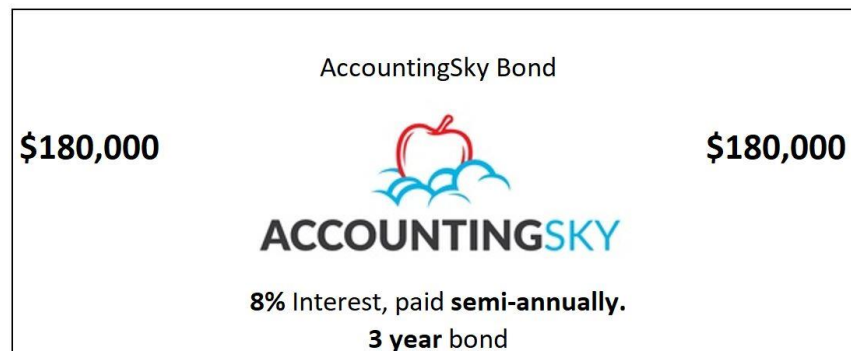
As you can notice, we paid the full face amount at the maturity date, and since we do not owe anymore, we debit the Bonds Payable account.

Issuing Bonds at a Discount

Issuing bonds at face value is straightforward. There aren't any complications. Unfortunately, the market (effective) interest rate is not always the same as our contract rate. Many times, we can issue a bond with a contract rate of 5%, only to find out that the market rate is 7%. This is a problem because *while we offer 5% interest, everyone else on the market is offering 7%*. This means that all of the investors will buy the 7% interest rate bonds because that way they will earn more interest. In this case **our bond is unattractive**, and nobody wants to buy it. For that reason, we need to **issue the bond at a discount** (for less than the face amount). Let's look at an example:

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Our company issued bonds with a face amount (par value) of \$180,000 on January 1, 2021. The bonds' annual **contract (stated) rate is 8%**, and interest is paid semi-annually on June 30 and December 31. The bonds mature in 3 years. The annual **market (effective) rate is 10%**, and the bonds were sold for \$170,862. This is how the bond looks like:



The journal entry to record the issuance of the bonds looks like this:

Jan. 1, 2021	Cash	170,862	
	Discount on Bonds Payable	9,138	
	Bonds Payable		180,000

Issuing the bond at a discount

The face amount of our bond is \$180,000, but we only received \$170,862 (debiting Cash for \$170,862). This is because our bond was unattractive because **our contract rate of 8% was lower than the market rate of 10%**. The only way to make our bond attractive was to sell it for less than the face amount (**at a discount**). We still credit Bonds Payable for \$180,000 because that is the actual bond we sold. Discount on Bonds Payable is debited because that is the discount we provided.

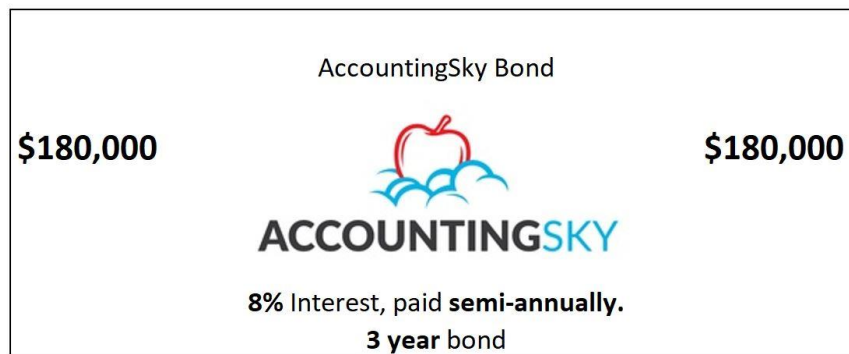
Discount on Bonds Payable is an **adjunct account** to Bonds Payable. The net effect of Bonds Payable - Discount on bonds Payable equals to the carrying amount of the bond (in our case $\$180,000 - \$9,138 = \$170,862$).

When we make the interest payments on this bond, things get a little complicated. We have 2 ways to calculate the interest payment. One is to use the **(1) straight-line method** and the other one is to use the **(2) effective interest method**.

Straight-Line Method

When we make the interest payments, we need to record the interest expense (debit Interest Expense), the cash payment (credit Cash), and the amortization of the discount. **Amortizing the discount** means eliminating that discount over the maturity of the bond. Since the original balance of the Discount on Bonds Payable was debit, we need to credit it in order to eliminate it.

Looking at our example, we promised to pay 8% semi-annually (every 6 months).



The cash payment is calculated to be \$7,200 ($\$180,000 \times 8\% \times 6/12$). The expression 6/12 means for 6 months out of 12, since it is a **semi-annual** payment.

Jun. 30, 2021	Interest Expense	8,723	
	Discount on Bonds Payable		1,523
	Cash		7,200

First Semi-Annual Interest Payment

The *credit to discount on Bonds Payable* is calculated using the **straight-line method**. the straight-line method allows us to amortize the discount equally over the interest payments. The Discount on Bonds Payable in our case has a balance of \$9,138. The bond matures in 3 years and we have 2 payments each year. This means we have a total of 6 payments. **We divide the balance in Discount on Bonds Payable (\$9,138) with 6 total payments to get \$1,523 amortization of the discount per interest payment.** The debits and credits must equal in every journal entry we make. Therefore, the amount of Interest Expense will be \$8,723 ($\$7,200 + \$1,523$).

Every six months, we will make the same exact journal entry to record the interest payments. After every payment, however, the carrying amount of the bond will change. The beginning carrying amount was \$170,862, but after every interest payment it will increase for \$1,523 (the amount of amortization of the discount). After our last interest payment, the carrying amount of the bond will equal the face amount of the bond, \$180,000.

	Carrying Amount	plus Amortized Discount
Jan. 1, 2021	170,862	
Jun. 31, 2021	172,385	1,523
Dec. 31, 2021	173,908	1,523
Jun. 31, 2022	175,431	1,523
Dec. 31, 2022	176,954	1,523
Jun. 31, 2023	178,477	1,523
Dec. 31, 2023	180,000	1,523

Therefore, at the end of the maturity, we will need to fulfill the promise to pay the face amount at maturity. The journal entry will look like this:

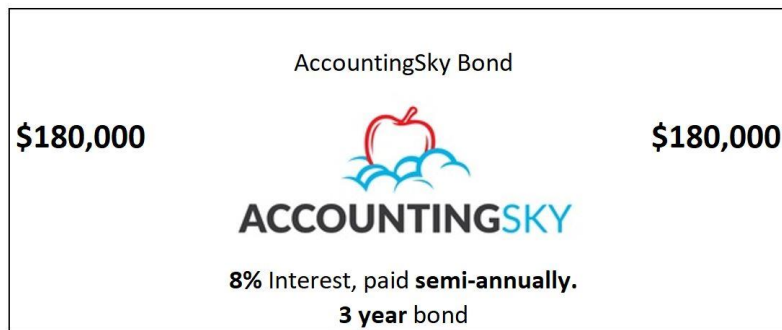
Dec. 31, 2023	Bonds Payable	180,000	
	Cash		180,000

Paying the Bonds Payable at Maturity

Effective-Interest Method

When we use the effective interest method to record the interest payments, we follow a different procedure to calculate the amount of interest expense. The journal entry will be the same as when we were using the straight-line method. We still need to record the interest expense (debit Interest Expense), the cash payment (credit Cash), and the amortization of the discount (credit Discount on Bonds Payable). However, some of the amounts will be different.

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The cash payment (Credit Cash) will be the same exact amount. however, instead of calculating the amortization of the discount first, in this method, we calculate the amount of the interest expense first. To find the amount of Interest Expense, using the effective interest method, we need to use the following formula:

$$\text{Interest Expense} = \text{Carrying Amount} * \text{Market Rate} * \text{Time Period}$$

Basically, we multiply the carrying amount (in our case \$170,862, but which changes after every interest payment) with the market rate at the date of issuance (in our case 10%). We also need to calculate the interest expense for 6 months out of 12 (6/12) because the interest stated on our bond is paid semi-annually. The journal entry would look like this:

Jun. 30, 2021	Interest Expense	8,543	
	Discount on Bonds Payable		1,343
	Cash		7,200

We credited Cash for \$7,200 because that is exactly what we promised (8% per year on \$180,000, paid every 6 months). The Interest expense was debited for \$8,543 (\$170,862*10%*6/12). Debits and credits must equal, which means the amount of amortization of the Discount on Bonds Payable account will be \$1,343.

If you remember, the carrying amount was calculated as the balance in Bonds Payable minus the Discount on Bonds Payable. The new balance in the Discount on

Bonds Payable account is \$7,795 (\$9,138 - \$1,343). That means the new carrying amount of the bond is \$172,205 (\$180,000 - \$7,795). This is a very important information because we will need it to calculate the interest expense for the second interest payment. This is how the second interest payment looks like:

Dec. 31, 2021	Interest Expense	8,610
	Discount on Bonds Payable	1,410
	Cash	7,200

We still credit cash for \$7,200 because that is what we promised when we issued the bond ($\$180,000 \times 8\% \times 6/12$). The interest expense is \$8,610 ($\$172,205 \times 10\% \times 6/12$), by using the formula for calculating the interest expense using the effective interest method. The difference is the amortization of the discount \$1,410. The new balance in the Discount on Bonds Payable account is \$6,385 (\$7,795 - \$1,410). Now, the new carrying amount of the bond is \$173,615 (\$180,000 - \$6,385). This new carrying amount will be used to calculate the interest expense for the third interest payment. Here is the schedule of the carrying amount of the bond:

	Carrying Amount	plus Amortized Discount
Jan. 1, 2021	170,862	
Jun. 31, 2021	172,205	1,343
Dec. 31, 2021	173,615	1,410
Jun. 31, 2022	175,096	1,481
Dec. 31, 2022	176,651	1,555
Jun. 31, 2023	178,284	1,633
Dec. 31, 2023	180,000	1,716

For the 4th payment, we will use \$175,096 as carrying amount, and for the 5th payment, \$176,651. For the final interest payment, we will use \$178,284 as the interest payment. (Note that there is some rounding in the calculations above). The

amounts of interest expense and amortization of the discount will change in every payment. However, the cash payment will remain the same, \$7,200.

Issuing Bonds at a Premium

Unfortunately, the market (effective) interest rate is not always the same as our contract rate. Many times, we can issue a bond with a contract rate of 7%, only to find out that the market rate is 5%. *While we offer 7% interest, everyone else on the market is offering 5%.* This means that all of the investors will buy our 7% interest rate bonds because that way they will earn more interest. In this case **our bond is very attractive**, and everyone wants to buy it. For that reason, we need to **issue the bond at a premium** (for higher price than the face amount). Let's take a look at an example:

Our company issued bonds with a face amount (par value) of \$180,000 on January 1, 2021. The bonds' annual **contract (stated) rate is 8%**, and interest is paid semi-annually on June 30 and December 31. The bonds mature in 3 years. The annual **market (effective) rate is 6%**, and the bonds were sold for \$189,750. This is how the bond looks like:



The journal entry to record the issuance of the bonds looks like this:

Jan. 1, 2021	Cash	189,750	
	Bonds Payable		180,000
	Premium on Bonds Payable		9,750

The face amount of our bond is \$180,000, but we received \$189,750 (debiting Cash for \$189,750). This is because our bond was very attractive because **our contract rate of 8% was higher than the market rate of 6%**. Since everyone wants to buy

our bond, we can sell it for more than the face amount (**at a premium**). We still credit Bonds Payable for \$180,000 because that is the actual bond we sold. Premium on Bonds Payable is credited because that is the premium we received.

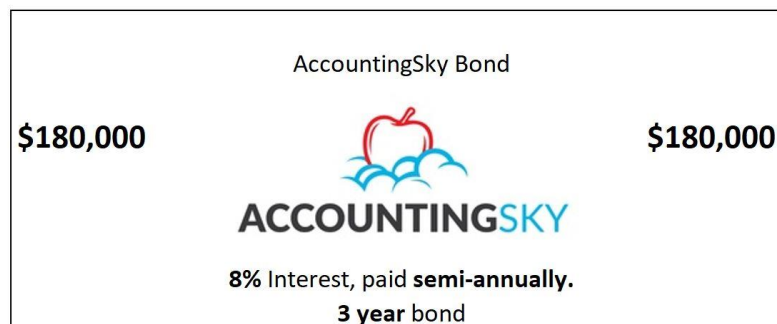
Premium on Bonds Payable is an **adjunct account** to Bonds Payable. The net effect of Bonds Payable + Premium on Bonds Payable equals to the carrying amount of the bond (in our case \$180,000 + \$9,750 = \$189,750).

When we make the interest payments on this bond, things get a little complicated. We have 2 ways to calculate the interest payment. One is to use the **(1) straight-line method** and the other one is to use the **(2) effective interest method**.

Straight-Line Method

When we make the interest payments, we need to record the interest expense (debit Interest Expense), the cash payment (credit Cash), and the amortization of the premium. **Amortizing the premium** means eliminating that premium over the maturity of the bond. Since the original balance of the Premium on Bonds Payable was credit, we need to debit it in order to eliminate it.

Looking at our example, we promised to pay 8% semi-annually (every 6 months).



The cash payment is calculated to be \$7,200 (\$180,000 * 8% * 6/12). The expression 6/12 means for 6 months out of 12, since it is a **semi-annual** payment.

Jun. 30, 2021	Interest Expense	5,575	
	Premium on Bonds Payable	1,625	
	Cash		7,200

The *debit to Premium on Bonds Payable* is calculated using the **straight-line method**. The straight-line method allows us to amortize the premium equally over the interest payments. The Premium on Bonds Payable in our case has a balance of \$9,750. The bond matures in 3 years and we have 2 payments each year. This means we have a total of 6 payments. **We divide the balance in Premium on Bonds Payable (\$9,750) with 6 total payments to get \$1,625 amortization of the premium per interest payment.** The debits and credits must equal in every journal entry we make. Therefore, the amount of Interest Expense will be \$5,575 (\$7,200-\$1,625).

Every six months, we will make the same exact journal entry to record the interest payments. After every payment, however, the carrying amount of the bond will change. The beginning carrying amount was \$189,750, but after every interest payment it will decrease for \$1,625 (the amount of amortization of the premium). After our last interest payment, the carrying amount of the bond will equal the face amount of the bond, \$180,000.

	Carrying Amount	minus Amortized Premium
Jan. 1, 2021	189,750	
Jun. 31, 2021	188,125	1,625
Dec. 31, 2021	186,500	1,625
Jun. 31, 2022	184,875	1,625
Dec. 31, 2022	183,250	1,625
Jun. 31, 2023	181,625	1,625
Dec. 31, 2023	180,000	1,625

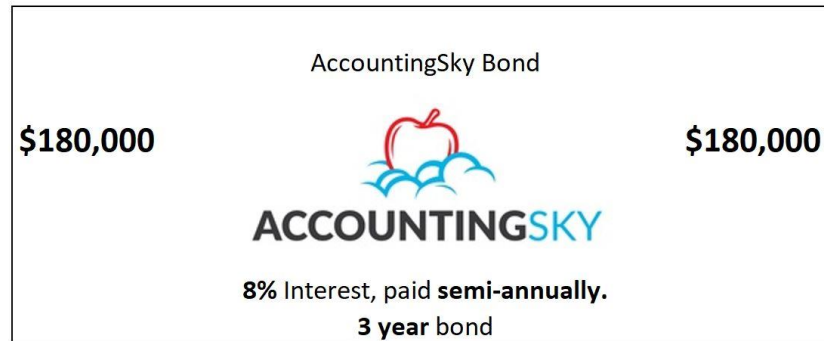
Therefore, at the end of the maturity, we will need to fulfill the promise to pay the face amount at maturity. The journal entry will look like this:

Dec. 31, 2023	Bonds Payable	180,000	
	Cash		180,000

Effective-Interest Method

When we use the effective interest method to record the interest payments, we follow a different procedure to calculate the amount of interest expense. The journal entry will be the same as when we were using the straight-line method. We still

need to record the interest expense (debit Interest Expense), the cash payment (credit Cash), and the amortization of the premium (debit Premium on Bonds Payable). However, some of the amounts will be different.



The cash payment (Credit Cash) will be the same exact amount. however, instead of calculating the amortization of the discount first, in this method, we calculate the amount of the interest expense first. To find the amount of Interest Expense, using the effective interest method, we need to use the following formula:

$$\text{Interest Expense} = \text{Carrying Amount} * \text{Market Rate} * \text{Time Period}$$

Basically, we multiply the carrying amount (in our case \$189,750, but which changes after every interest payment) with the market rate at the date of issuance (in our case 6%). We also need to calculate the interest expense for 6 months out of 12 (6/12) because the interest stated on our bond is paid semi-annually. The journal entry would look like this:

Jun. 30, 2021	Interest Expense	5,693	
	Premium on Bonds Payable	1,508	
	Cash		7,200

We credited Cash for \$7,200 because that is exactly what we promised (8% per year on \$180,000, paid every 6 months). The Interest expense was debited for

\$5,693 ($\$189,750 \times 6\% \times 6/12$). Debits and credits must equal, which means the amount of amortization of the Premium on Bonds Payable account will be \$1,508.

If you remember, the carrying amount was calculated as the balance in Bonds Payable plus the Premium on Bonds Payable. The new balance in the Premium on Bonds Payable account is \$8,242 ($\$9,750 - \$1,508$). That means the new carrying amount of the bond is \$188,242 ($\$180,000 + \$8,242$). This is a very important information because we will need it to calculate the interest expense for the second interest payment. This is how the second interest payment looks like:

Dec. 31, 2021	Interest Expense	5,647	
	Premium on Bonds Payable	1,553	
	Cash		7,200

We still credit cash for \$7,200 because that is what we promised when we issued the bond ($\$180,000 \times 8\% \times 6/12$). The interest expense is \$5,647 ($\$188,242 \times 10\% \times 6/12$), by using the formula for calculating the interest expense using the effective interest method. The difference is the amortization of the premium \$1,553. The new balance in the Premium on Bonds Payable account is \$6,689 ($\$8,242 - \$1,553$). Now, the new carrying amount of the bond is \$186,689 ($\$180,000 + \$6,689$). This new carrying amount will be used to calculate the interest expense for the third interest payment. Here is the schedule of the carrying amount of the bond:

	Carrying Amount	minus Amortized Premium
Jan. 1, 2021	189,750	
Jun. 31, 2021	188,242	1,508
Dec. 31, 2021	186,689	1,553
Jun. 31, 2022	185,090	1,599
Dec. 31, 2022	183,443	1,647
Jun. 31, 2023	181,746	1,697
Dec. 31, 2023	180,000	1,746

For the 4th payment, we will use \$185,090 as carrying amount, and for the 5th payment, \$183,443. For the final interest payment, we will use \$181,746 as the interest payment. (Note that there is some rounding in the calculations above). The amounts of interest expense and amortization of the discount will change in every payment. However, the cash payment will remain the same, \$7,200.

Bond Exercises

Let's solve some problems that involve bonds. Just play the video below.

[WATCH VIDEO](#)

Bond Pricing

Before you even begin learning this section, you will need to learn about the Time Value of Money (TVM). The lesson about Time Value of Money is attached in this Chapter 10 Appendix. After reviewing the appendix, we can continue here.

Bond Pricing is the process we use to determine the selling price of the bond. In our case when we issued the bond at a discount, the face amount of the bond was \$180,000 and we sold it for \$170,862. We knew we had to sell it for less because our contract interest rate (8%) was less than the market interest rate (10%). But how did we determine exactly \$170,862? This section will answer that question.

[WATCH VIDEO](#)

Long-Term Notes Payable

Long-term note payable is very similar to short-term notes payable. The main difference is that the maturity date of the long-term note payable is beyond 1 year. In this section of this book, we will discuss specifically about (1) Installment Notes and (2) Mortgage Notes.

Installment Notes

Installment note is a type of long-term note which requires the borrower to pay multiple payments to the lender. Learning about notes payable is best through practice.

Example

Best Coffee Shop Company borrowed \$50,000 from the bank to purchase furniture for the store. Best Coffee Shop Company signed a 3-year installment note for \$50,000 and 9% interest. Our company is required to make annual payments in the amount of \$19,753.

The entry to record the borrowing is like this:

Jan. 1, 2022	Cash	50,000	
	Notes Payable		50,000

Simply, we borrowed \$50,000 cash, and now we owe the same amount to the bank. When we make the first annual payment, the journal entry will look like this:

Dec. 31, 2022	Interest Expense	4,500	
	Notes Payable	15,253	
	Cash		19,753

We debit Interest Expense for \$4,500 ($\$50,000 \times 9\%$). We credit Cash for the set amount of \$19,753. The amount of principal paid towards the N/P is \$15,253 ($\$19,753 - \$4,500$).

The second payment will be slightly different because the balance in Notes Payable now is \$34,747 ($\$50,000 - \$15,253$). The journal entry for the second payment will look like this:

Dec. 31, 2023	Interest Expense	3,127	
	Notes Payable	16,626	
	Cash		19,753.00

We debit Interest Expense for \$3,127 ($\$34,747 \times 9\%$). We credit Cash for the set amount of \$19,753. The amount of principal paid towards the N/P is \$16,626 ($\$19,753 - \$3,127$).

The third, and final payment, will be different because the balance in Notes Payable now is \$18,121 (\$34,747-\$16,626). The journal entry for the third payment will look like this:

Dec. 31, 2024	Interest Expense	1,632	
	Notes Payable	18,121	
	Cash		19,753.00

Of course, there are some rounding errors. We debit Interest Expense for \$1,632 (\$18,121*9%). We credit Cash for the set amount of \$19,753. The amount of principal paid towards the N/P is \$18,121 (\$19,753-\$1,632).

The balance in Notes Payable now is \$0 (\$18,121-\$18,121). We paid off our obligation within 3 years.

Chapter 10 Appendix 1: Time Value of Money (TVM)

Time value of money is a very important concept in accounting, finance, economics, and business. Time value of money is a concept explaining that \$1 today is not the same as the \$1 one year ago, or the \$1 one year from now.

Simply, if you invest your money today, and you will have more in 5 years, let's say. Your return can be simple interest, or a compounded interest.

Simple Interest (Future Value)

We invested \$100 in 2018, for 5 years with a simple annual interest of 10%. How much will we have in 5 years? In this case we have the Present Value (\$100). Present Value is the value today. We also have the interest rate (10%) per year, and the length of the investment (5 years). Every year, we will earn \$10 on this investment ($10\% \times \100).

	Year 1	Year 2	Year 3	Year 4	Year 5
Interest Earned	\$10	\$10	\$10	\$10	\$10
<i>Calculation</i>	$10\% \times \$100$	$10\% \times \$100$	$10\% \times \$100$	$10\% \times \$100$	$10\% \times \$100$

The total interest earned is \$50. Therefore, in 5 years from now, we will have \$150, the \$100 we invested plus the interest of \$50. Thus, the future value of an investment of \$100, invested now, for a 10% simple interest, is \$150.

Compounded Interest (Future Value)

Compounded interest is different than simple interest. Compounded interest calculates the interest earned on the balance of the investment, or interest on top of the interest earned.

We invested \$100 in 2018, for 5 years with **compounded** annual interest of 10%. How much will we have in 5 years?

	Year 1	Year 2	Year 3	Year 4	Year 5
Interest Earned	\$10	\$11	\$12.1	\$13.31	\$14.641
<i>Calculation</i>	<i>10%*\$100</i>	<i>10%*\$110</i>	<i>10%*\$121</i>	<i>10%*\$133.1</i>	<i>10%*\$146.41</i>

For the Year 1, the balance for the investment is \$100. 10% on that is \$10. For Year 2, the balance in the investment is \$110 (\$100+\$10 interest). 10% on that is \$11. For Year 3, the balance is \$121 (\$100+\$10 from year one +\$11 from year two). 10% on that is \$12.1.

We will earn \$61.051 interest in 5 years. We will have a total of \$161.051. As you can notice, compounded interest brings you more money, because it calculates the interest on top of the interest earned in previous periods.

Present Value

The examples we have seen above are for finding the value of an investment in the future, a future value. But can we do that in the opposite direction? Is it possible to find the present value of an investment?

For example, we want to have \$20,000, 5 years from now, to buy a brand-new car. We have an investment in mind in which we can invest today, and it will have an interest rate of 12%, compounded annually. How much should we invest now in order to have \$20,000, 5 years from now? \$20,000 is the value in the future. The amount we need to invest now is the present value. We need to look for the present value.

We can find the answer following the example above and work backwards. However, that would be very bothersome. There is a simpler way. We have tables designed only for this purpose. In this lesson, we only need to know 2 tables. In the future, as you progress with your accounting, finance, economics, and business education, you will need to know how to use more. The 2 tables we are interested in are Present Value of \$1 and Present Value of Annuity of \$1. They are presented below:

Table 1: Present Value of \$1 Table

Periods	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	20%	25%	30%
1	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.833	0.800	0.769
2	0.925	0.907	0.890	0.873	0.857	0.842	0.826	0.812	0.797	0.783	0.769	0.756	0.694	0.640	0.592
3	0.889	0.864	0.840	0.816	0.794	0.772	0.751	0.731	0.712	0.693	0.675	0.658	0.579	0.512	0.455
4	0.855	0.823	0.792	0.763	0.735	0.708	0.683	0.659	0.636	0.613	0.592	0.572	0.482	0.410	0.350
5	0.822	0.784	0.747	0.713	0.681	0.650	0.621	0.593	0.567	0.543	0.519	0.497	0.402	0.328	0.269
6	0.790	0.746	0.705	0.666	0.630	0.596	0.564	0.535	0.507	0.480	0.456	0.432	0.335	0.262	0.207
7	0.760	0.711	0.665	0.623	0.583	0.547	0.513	0.482	0.452	0.425	0.400	0.376	0.279	0.210	0.159
8	0.731	0.677	0.627	0.582	0.540	0.502	0.467	0.434	0.404	0.376	0.351	0.327	0.233	0.168	0.123
9	0.703	0.645	0.592	0.544	0.500	0.460	0.424	0.391	0.361	0.333	0.308	0.284	0.194	0.134	0.094
10	0.676	0.614	0.558	0.508	0.463	0.422	0.386	0.352	0.322	0.295	0.270	0.247	0.162	0.107	0.073
11	0.650	0.585	0.527	0.475	0.429	0.388	0.350	0.317	0.287	0.261	0.237	0.215	0.135	0.086	0.056
12	0.625	0.557	0.497	0.444	0.397	0.356	0.319	0.286	0.257	0.231	0.208	0.187	0.112	0.069	0.043
13	0.601	0.530	0.469	0.415	0.368	0.326	0.290	0.258	0.229	0.204	0.182	0.163	0.093	0.055	0.033
14	0.577	0.505	0.442	0.388	0.340	0.299	0.263	0.232	0.205	0.181	0.160	0.141	0.078	0.044	0.025
15	0.555	0.481	0.417	0.362	0.315	0.275	0.239	0.209	0.183	0.160	0.140	0.123	0.065	0.035	0.020
16	0.534	0.458	0.394	0.339	0.292	0.252	0.218	0.188	0.163	0.141	0.123	0.107	0.054	0.028	0.015
17	0.513	0.436	0.371	0.317	0.270	0.231	0.198	0.170	0.146	0.125	0.108	0.093	0.045	0.023	0.012
18	0.494	0.416	0.350	0.296	0.250	0.212	0.180	0.153	0.130	0.111	0.095	0.081	0.038	0.018	0.009
19	0.475	0.396	0.331	0.277	0.232	0.194	0.164	0.138	0.116	0.098	0.083	0.070	0.031	0.014	0.007
20	0.456	0.377	0.312	0.258	0.215	0.178	0.149	0.124	0.104	0.087	0.073	0.061	0.026	0.012	0.005
22	0.422	0.342	0.278	0.226	0.184	0.150	0.123	0.101	0.083	0.068	0.056	0.046	0.018	0.007	0.003
24	0.390	0.310	0.247	0.197	0.158	0.126	0.102	0.082	0.066	0.053	0.043	0.035	0.013	0.005	0.002
25	0.375	0.295	0.233	0.184	0.146	0.116	0.092	0.074	0.059	0.047	0.038	0.030	0.010	0.004	0.001
30	0.308	0.231	0.174	0.131	0.099	0.075	0.057	0.044	0.033	0.026	0.020	0.015	0.004	0.001	0.000
35	0.253	0.181	0.130	0.094	0.068	0.049	0.036	0.026	0.019	0.014	0.010	0.008	0.002	0.000	0.000
40	0.208	0.142	0.097	0.067	0.046	0.032	0.022	0.015	0.011	0.008	0.005	0.004	0.001	0.000	0.000

Table 2: Present Value of Annuity of \$1 Table

Periods	4%	5%	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	20%	25%	30%
1	0.962	0.952	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.833	0.800	0.769
2	1.886	1.859	1.833	1.808	1.783	1.759	1.736	1.713	1.690	1.668	1.647	1.626	1.528	1.440	1.361
3	2.775	2.723	2.673	2.624	2.577	2.531	2.487	2.444	2.402	2.361	2.322	2.283	2.106	1.952	1.816
4	3.630	3.546	3.465	3.387	3.312	3.240	3.170	3.102	3.037	2.974	2.914	2.855	2.589	2.362	2.166
5	4.452	4.329	4.212	4.100	3.993	3.890	3.791	3.696	3.605	3.517	3.433	3.352	2.991	2.689	2.436
6	5.242	5.076	4.917	4.767	4.623	4.486	4.355	4.231	4.111	3.998	3.889	3.784	3.326	2.951	2.643
7	6.002	5.786	5.582	5.389	5.206	5.033	4.868	4.712	4.564	4.423	4.288	4.160	3.605	3.161	2.80
8	6.733	6.463	6.210	5.971	5.747	5.535	5.335	5.146	4.968	4.799	4.639	4.487	3.837	3.329	2.925
9	7.435	7.108	6.802	6.515	6.247	5.995	5.759	5.537	5.328	5.132	4.946	4.772	4.031	3.463	3.019
10	8.111	7.722	7.360	7.024	6.710	6.418	6.145	5.889	5.650	5.426	5.216	5.019	4.192	3.571	3.092
11	8.760	8.306	7.887	7.499	7.139	6.805	6.495	6.207	5.938	5.687	5.453	5.234	4.327	3.656	3.147
12	9.385	8.863	8.384	7.943	7.536	7.161	6.814	6.492	6.194	5.918	5.660	5.421	4.439	3.725	3.190
13	9.986	9.394	8.853	8.358	7.904	7.487	7.103	6.750	6.424	6.122	5.842	5.583	4.533	3.780	3.223
14	10.563	9.899	9.295	8.745	8.244	7.786	7.367	6.982	6.628	6.302	6.002	5.724	4.611	3.824	3.249
15	11.118	10.380	9.712	9.108	8.559	8.061	7.606	7.191	6.811	6.462	6.142	5.847	4.675	3.859	3.268
16	11.652	10.838	10.106	9.447	8.851	8.313	7.824	7.379	6.974	6.604	6.265	5.954	4.730	3.887	3.283
17	12.166	11.274	10.477	9.763	9.122	8.544	8.022	7.549	7.120	6.729	6.373	6.047	4.775	3.910	3.295
18	12.659	11.690	10.828	10.059	9.372	8.756	8.201	7.702	7.250	6.840	6.467	6.128	4.812	3.928	3.304
19	13.134	12.085	11.158	10.336	9.604	8.950	8.365	7.839	7.366	6.938	6.550	6.198	4.843	3.942	3.311
20	13.590	12.462	11.470	10.594	9.818	9.129	8.514	7.963	7.469	7.025	6.623	6.259	4.870	3.954	3.316
22	14.451	13.163	12.042	11.061	10.201	9.442	8.772	8.176	7.645	7.170	6.743	6.359	4.909	3.970	3.323
24	15.247	13.799	12.550	11.469	10.529	9.707	8.985	8.348	7.784	7.283	6.835	6.434	4.937	3.981	3.327
25	15.622	14.094	12.783	11.654	10.675	9.823	9.077	8.422	7.843	7.330	6.873	6.464	4.948	3.985	3.329
30	17.292	15.372	13.765	12.409	11.258	10.274	9.427	8.694	8.055	7.496	7.003	6.566	4.979	3.995	3.332
35	18.665	16.374	14.498	12.948	11.655	10.567	9.644	8.855	8.176	7.586	7.070	6.617	4.992	3.998	3.333
40	19.793	17.159	15.046	13.332	11.925	10.757	9.779	8.951	8.244	7.634	7.105	6.642	4.997	3.999	3.333

Let's see the previous example again. For example, we want to have \$20,000, 5 years from now, to buy a brand-new car. We have an investment in mind in which we can invest today, and it will have an interest rate of 12%, compounded annually. How much should we invest now in order to have \$20,000, 5 years from now?

All we have to do is take the future value, \$20,000 and multiply with the appropriate factor from the appropriate table. The investment is happening only once in time, so we will use the Present Value of \$1 Table. We look for the factor where 12% and 5 periods cross, 0.567. Thus, the present value is \$11,340 ($\$20,000 \times 0.567$).

We need to invest \$11,340 today, so that in 5 years we will have \$20,000. (Assuming compounded interest of 12%). To find future values, you need the Future Value tables.

Annuity

As you can notice, we also have a table for annuities. Annuities are series of equal cash flows. For example, we can have a settlement in which we will receive \$1,000 every year, for 10 years. Or maybe, we have an agreement to sell a patent and we will need to receive \$500 every month, for 5 years. Or maybe, we lost a lawsuit, and we were ordered by the court to pay \$5,000 every month, for 2 years. Finally, we might have issued a bond and made many promises.

For example, we borrowed money today, and we agreed to pay equal amounts of yearly payments of \$10,000, for 20 years. The interest, compounded annually, is 7%. How much money did we receive today?

Simply, take the annual payment, \$10,000 and multiply it with the factor from the table. The table we use now is Present Value of Annuity of \$1, because the \$10,000 are series of equal cash payments – an annuity. The factor is 10.594, the point where 7% and 20 periods cross).

Today, we received \$105,940 ($\$10,000 \times 10.594$). How come we received only \$105,940 and we have to repay \$200,000 total ($\$10,000 \times 20$ years)? Well, did you account for the compounded interest? This is exactly how mortgages cost so much at maturity.

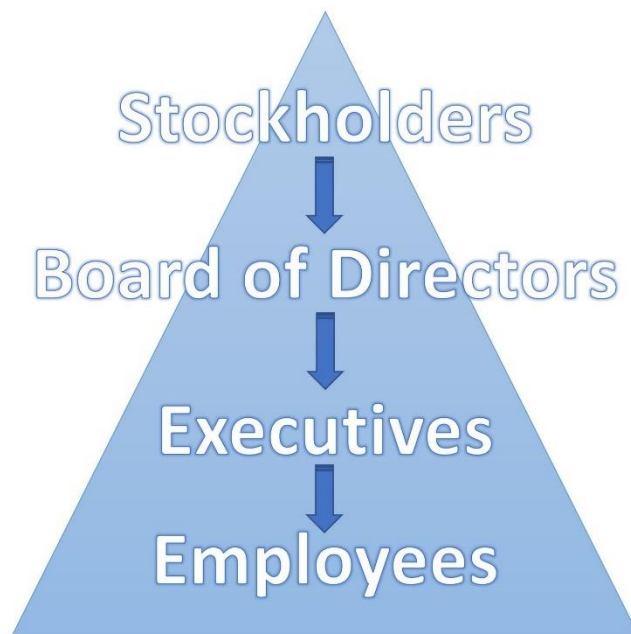
Chapter 11: Stockholders' Equity

Corporate Organization

When discussing about a business, there are 3 general types of businesses. Those are sole proprietorships, partnerships, and corporations. There are also limited liability companies that represent a hybrid between corporations and partnerships and are organized based on the state law in which they are located and operate. In this book, we focus primarily on corporations.

Corporate Characteristics

Corporations are legal entities that are separate from their owners (stockholders). We call the owners stockholders, because they are the ones that are holding the ownership (stock) of the corporation. The stockholders elect the board of directors, and the board of directors hire the company's executives. Finally, the executives hire the employees in the company, as presented below:



The stockholders have limited liability, meaning if Person A sues the corporation for damages, Person A can only get what the corporation has, not the assets

stockholders personally own (such as their house, car, etc.). The board of directors establish company's policies and procedures, besides hiring the company's executives (CEO, CFO, etc.).

When the corporation makes profit, the corporation itself, must pay income taxes. If the corporation decides to distribute dividends (distribution of earnings to owners) to its stockholders, then the stockholders need to pay income tax on those dividends they received. This is called double taxation, and it is presented as the biggest disadvantage of corporations. Let's take a look at the advantages and disadvantages of corporations:

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Paid-In Capital

When we take a look a look at a Balance Sheet, the 2 major sections in the Stockholders' Equity section are Paid-In Capital and Retained Earnings. We learned about retained earnings and the statement of retained earnings in Chapter 1. In this section, we will learn about Paid-In Capital. Paid-In Capital is the amount the stockholders invested in the corporation.

When the corporation is formed, the state will authorize the company to issue a certain number of shares. These shares are called authorized shares. When the company issues these shares of stock, they become issued shares of stock. When the corporation has issued shares of stock, it has owners. Let's see this in an example.

Company A was authorized to issue 100,000 shares of stock. Since none of the shares has been issued yet, nobody owns the company. The company decided to issue 10,000 shares for cash. Now, the only issued and outstanding shares are the 10,000 shares. In order to own 100% of Company A, you need to own 10,000 shares. In order to own 20% of Company A, you need to own 2,000 shares (10,000 shares *20%).



The blue circle represents the 100,000 authorized shares of stock. The yellow circle represents the 10,000 issued shares of stock. The issued shares are both issued and outstanding.

Sometimes, as we will see in the Appendix 1 of this chapter, the company reacquires its own stock (Treasury Stock). That means that the outstanding stock goes into the company's treasury and it is no longer outstanding. It has been issued, but it is no longer outstanding because nobody owns it. Let's say that the company reacquired 4,000 shares of its own stock. Now, 4,000 shares are treasury stock, and only 6,000 shares remain outstanding.



The green circle represents the 6,000 shares that remain outstanding. The yellow circle is the 4,000 shares of treasury stock. The outstanding 6,000 shares are authorized, issued, and outstanding shares.

Issuing **Common** and **Preferred** Stock

As we discussed earlier, a corporation might have both common and preferred stock. It is also possible that a company has only common stock. When a corporation issues common stock, it means that an investor is buying an ownership in the corporation. In this section we will learn about issuing common stock.

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Issuing Par Value Common Stock

Journalize the entry to record Pierce Company's issuance of 30,000 shares of \$10 par value stock for \$300,000 cash on October 30th. This is the journal entry:

Oct. 30, 2022	Cash	300,000	
	Common Stock		300,000

We debit cash because we received cash from the investor. We credit common stock to increase the common stock, since now we have an additional owner.

Issuing Common Stock for more than Par

Journalize the entry to record Pierce Company's issuance of 30,000 shares of \$10 par value stock for \$12 per share on October 30th. This is the journal entry:

Oct. 30, 2022	Cash	360,000	
	Common Stock		300,000
	Paid In Capital in Excess of Par, CS		60,000

We debit cash for \$360,000 because we received that much cash (30,000 shares * \$12 per share). We credit common stock for \$300,000 because the par value of 1 share is \$10 (30,000 shares * \$10 par value per share). We also credit the Paid-In Capital in Excess of Par, Common Stock account for \$60,000. This is the amount that was paid above the par value.

Issuing No-Par Value Common Stock

Journalize the entry to record Pierce Company's issuance of 30,000 shares no-par value stock for \$20 per share on October 30th. This is the journal entry:

Oct. 30, 2022	Cash	600,000	
	Common Stock		600,000

We debit cash for \$600,000 because that is what we received (30,000 shares * \$20 per share). We credit common stock for the same amount because the shares do not have a par value.

Issuing Stated Value Common Stock

Journalize the entry to record Pierce Company's issuance of 30,000 shares with a Stated Value of \$15 for \$20 per share on October 30th. This is the journal entry:

Oct. 30, 2022	Cash	600,000	
	Common Stock		450,000
	Paid In Capital in Excess of Stated Value, CS		150,000

We debit cash for \$600,000 because that is the amount we received. However, we credit common stock for \$450,000 because that is the stated value of the stock (30,000 shares * \$15 stated value per share). We credit the Paid In Capital in Excess of Stated Value, Common Stock account for the amount paid over the stated value, \$150,000.

Issuing Par Value Preferred Stock

Journalize the entry to record Pierce Company's issuance of 50 shares of \$100 par value preferred stock for \$5,000 cash on October 30th. This is the journal entry:

Oct. 30, 2022	Cash	5,000	
	Preferred Stock		5,000

We debit cash for \$5,000 because that is how much cash we received. We credit preferred stock for \$5,000 because that is the par value of the stock (50 shares * \$100 par value per share).

Issuing Preferred Stock for more than Par

Journalize the entry to record Pierce Company's issuance of 50 shares of \$100 par value preferred stock for \$6,000 cash on October 30th. This is the journal entry:

Oct. 30, 2022	Cash	6,000	
	Preferred Stock		5,000
	Paid In Capital in Excess of Par, PS		1,000

We debit cash for \$6,000 because that is how much cash we received. We credit preferred stock for \$5,000 because that is the par value of the stock (50 shares * \$100 par value per share). We credit Paid-In Capital in Excess of Par, Preferred Stock for the amount paid over the par value, \$1,000.

Comparing Common and Preferred Stock

The basic rights of stockholders in a corporation are the right to vote, share earnings, share assets upon liquidation of the company, and preemptive rights. The preemptive right enables the stockholders to retain their ownership percentage in a case where new stocks are issued. Common stockholders are somewhat different than preferred stock. Let's examine some of the differences.

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For example, when dividends are paid, the preferred stockholders get paid first. In the case of a company liquidation, the preferred stockholders have preference over the company's assets.

Participating Vs. Non-Participating Preferred Stock

In the case of dividends, when the preferred stock is classified as non-participating, the company pays dividend to preferred stockholders first, some amount to common stockholders, and whatever is left goes to common stockholders.

However, if the preferred stock is classified as participating, the company pays dividend to preferred stockholders first, some amount to common stockholders, and whatever is left is split between the common and preferred stockholders.

Cumulative Vs. Non-Cumulative Preferred Stock

In the case of dividends, let's see an example of a non-cumulative preferred stock. A company did not declare any dividends for the first 2 years but declared dividends in year 3. Since the preferred stock is non-cumulative, both preferred and common stockholders will get dividends for year 3.

Let's see an example of a cumulative preferred stock. A company did not declare any dividends for the first 2 years but declared dividends in year 3. Since the preferred stock is cumulative, the preferred stockholders will get paid for years 1, 2, and 3, and common for year 3, if there is anything left.

Dividends

Dividends are distributions made by the company to the company's owners (stockholders). If you buy 1 share of McDonald's stock, you become one of the many owners. Can you go to any McDonald's store, tell them you are one of the owners, and get as much money from their cash register as you want? Of course not. The only way for the company to distribute earnings to the owners is through dividends. The company needs to have sufficient cash and retained earnings in order to declare a dividend. We have two types of dividends: (1) Cash Dividends and (2) Stock Dividends.

Cash Dividends

Cash dividends are company's distribution of earnings to stockholders by giving them cash. Three dates are very important when paying cash dividends.

Date of Declaration. The board of directors declares that dividends will be paid on a specified future date. This declaration creates a liability for the company (Dividends Payable).

Date of Record. This is the date at which the company will pull the records to see who the stockholders are. In a publicly traded company, people are buying and selling stock frequently. The date of record is the specific date at which the company will pull the records to see who the stockholders at that specific time are (the ones getting paid).

Date of Payment. The date of payment is the date when the dividends are paid.

One example. On January 10th, 2022 (*date of declaration*) the board of directors declared a cash dividend to the stockholders of record on January 31st, 2022 (*date of record*). The dividend will be paid on February 15th, 2022 (*date of payment*).

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Comprehensive Example

Pierce Company needs more capital to expand, so they incorporated and the charter from the State of California authorized 70,000 shares of 5%, \$100 par value preferred stock, and 110,000 shares of \$2 par value common stock.

1. Pierce Company issued 20,000 shares of common stock for \$250,000.

2. Pierce Company issued 500 shares of preferred stock for \$120 per share.
3. Pierce Company declared a \$10,000 cash dividend for stockholders of record on January 15th.
4. Pierce Company paid the cash dividend.

Solution

We need the journal entries for each of the 4 transactions above.

1. The company issued 20,000 shares of common stock in exchange for cash. That means our company received cash, \$250,000 so we debit cash for that amount. The common stock has a par value of \$2 per share, so we credit common stock for \$40,000 (20,000 shares * \$2 par value). Finally, we credit Paid-In Capital in Excess of Par, Common Stock for the difference, \$210,000. This is the journal entry:

Cash	250,000	
	Common Stock	40,000
	Paid-In Capital in Excess of Par, CS	210,000

2. The company issued 500 shares of preferred stock in exchange for cash. Therefore, we debit cash for \$60,000 (500 shares * \$120 per share). We need to credit preferred stock for \$50,000 (500 shares * \$100 par value). Finally, we credit Paid-In Capital in Excess of Par, Preferred Stock for the difference, \$10,000. This is the journal entry:

Cash	60,000	
	Preferred Stock	50,000
	Paid-In Capital in Excess of Par, PS	10,000

3. When we declare cash dividends, we always debit Dividends and credit Dividends Payable. The declaration tells our promise to pay, thus creating a liability. In our case we pay total of \$10,000 cash dividends. How much of this \$10,000 goes to preferred and how much of it goes to common stockholders? As you can see, the preferred stockholders are 5%, \$100 par value preferred stock. This

means that each preferred share is entitled to \$5 ($5\% \times \100) dividend per year. Since we only pay to outstanding shares, the dividends for preferred stockholders will be \$2,500 ($\5×500 shares). The rest will go to the common stockholders, \$7,500 ($\$10,000 - \$2,500$). This is the journal entry:

Dividends	10,000	
	Dividends Payable, PS	2,500
	Dividends Payable, CS	7,500

4. To pay the cash dividend, we simply debit the Dividends Payable and credit cash. This is the journal entry:

Dividends Payable, PS	2,500	
Dividends Payable, CS	7,500	
	Cash	10,000

Please keep in mind that we only pay dividends on the outstanding shares, not the authorized shares. The authorized shares might not be issued, which means that nobody owns them. Therefore, they are not outstanding, and we have nobody to pay. That is why we pay dividend only to the 500 outstanding preferred stockholders, and 20,000 outstanding common stockholders.

Stock Dividends

Stock dividends are different than cash dividends. While we pay cash when we pay cash dividends, we distribute additional stock with stock dividends. It is possible that the company might not have sufficient cash to declare cash dividends. In some cases, the company might declare Stock dividends, which means the stockholders of the company will receive additional stock. The accounting for Stock dividends is very similar to the accounting for cash dividends. We have small stock dividends (25% or less of outstanding shares) and large stock dividends (more than 25% of outstanding shares). Small stock dividends are recorded by capitalization of retained earnings for an amount equal to the market value of stock. Large stock

dividends are recorded by capitalization of retained earnings for an amount equal to the par value of stock.

Three dates are very important when distributing stock dividends.

Date of Declaration. The board of directors declares that stock dividends will be distributed on a specified future date. This declaration creates a liability for the company (Dividends Distributable).

Date of Record. This is the date at which the company will pull the records to see who the stockholders are. In a publicly traded company, people are buying and selling stock frequently. The date of record is the specific date at which the company will pull the records to see who the stockholders at that specific time are (the ones getting additional shares of stock).

Date of Distribution. The date of distribution is the date when the stock dividends are distributed.

One example. On January 10th, 2022 (*date of declaration*) the board of directors declared a 1% stock dividend to the stockholders of record on January 31st, 2022 (*date of record*). The dividend will be distributed on February 15th, 2022 (*date of distribution*).

Example - Small Stock Dividends

The board of directors of Pierce Company declared a 10% stock dividend on December 31, 2022. Pierce Company has 10,000 outstanding shares. The par value of each share is \$10. The stock dividend will be distributed on January 15th, 2023 to the stockholders of record on January 10th, 2023. The market price of Pierce Company's stock is \$15 per share. Record the journal entries for the stock dividend declaration and distribution.

December 31st, 2022 - Date of Declaration

The amount of the stock dividend is 1,000 shares (10% * 10,000 outstanding shares). This is a small stock dividend because it is 10% of outstanding shares (25% or less is considered a small stock dividend).

Dec. 31, 2022	Stock Dividends	15,000
	Stock Dividends Distributable	10,000
	Paid-In Capital in Excess of Par, CS	5,000

When we declare small stock dividends, we need to debit the stock dividends account for the market price (\$15) of the stock we are distributing as a dividend. The amount is \$15,000 (\$15 market price * 1,000 shares stock dividend). We credit the Stock Dividends Distributable for the par value of the stock (\$10 par value * 1,000 shares stock dividend), because those are the shares we are issuing as a dividend. Finally, we credit Paid-In Capital in Excess of Par, Common Stock for the difference, \$5,000.

January 15th, 2023 - Date of Distribution

On the date of distribution, we need to distribute the additional shares of stock. We do not pay cash because this is not a cash dividend, but stock dividend.

Jan. 15th, 2023	Stock Dividends Distributable	10,000	
	Common Stock		10,000

We debit stock dividend distributable because we have just distributed the additional stock. We credit common stock because we issued additional common stock as a dividend. The amount is \$10,000 because that is the par value of the stock we distributed.

Example - Large Stock Dividends

The board of directors of Pierce Company declared a 30% stock dividend on December 31, 2022. Pierce Company has 10,000 outstanding shares. The par value of each share is \$10. The stock dividend will be distributed on January 15th, 2023 to the stockholders of record on January 10th, 2023. The market price of Pierce Company's stock is \$15 per share. Record the journal entries for the stock dividend declaration and distribution.

December 31st, 2022 - Date of Declaration

The amount of the stock dividend is 3,000 shares (30% * 10,000 outstanding shares). This is a large stock dividend because it is 30% of outstanding shares (more than 25% is considered a large stock dividend).

Dec. 31, 2022	Stock Dividends	30,000	
	Stock Dividends Distributable		30,000

When we declare large stock dividends, we only care about the par value of the stock. Therefore, we need to debit the stock dividends account for the par value (\$10) of the stock we are distributing as a dividend. The amount is \$30,000 (\$10 par value * 3,000 shares stock dividend). We credit the Stock Dividends Distributable for the par value of the stock (\$10 par value * 3,000 shares stock dividend), because those are the shares we are issuing as a dividend.

January 15th, 2023 - Date of Distribution

On the date of distribution, we need to distribute the additional shares of stock. We do not pay cash because this is not a cash dividend, but stock dividend.

Jan. 15th, 2023	Stock Dividends Distributable	30,000	
	Common Stock		30,000

We debit stock dividend distributable because we have just distributed the additional stock. We credit common stock because we issued additional common stock as a dividend. The amount is \$30,000 because that is the par value of the stock we distributed.

Reporting

To report the equity of a company, we need to report both Retained Earnings and Paid-In Capital. We can do that separately, or we can combine them together.

Statement of Retained Earnings

We discussed the statement of retained earnings in chapter one. A sample statement is shown below:

Best Coffee Shop Company
Statement of Retained Earnings
For the year ended, Dec. 31, 2021

Retained Earnings, Beginning	0
Add Net Income	14,500
Less Dividends	<u>(4,000)</u>
Retained Earnings, Ending	<u>10,500</u>

Statement of Stockholders' Equity

We can combine the statement of retained earnings and the paid-in capital into one statement, called Statement of Stockholders' Equity. A sample statement of stockholders' equity is presented below:

Best Coffee Shop
Statement of Stockholders' Equity

	Common Stock (Shares)	Common Stock (\$)	Retained Earnings	Total
Balance, December 31, 2020	100,000 (\$5 par)	\$500,000	0	\$500,000
Net Income			\$14,500	\$14,500
Issued Common Stock	5,000	\$25,000		\$25,000
Cash Dividends			\$4,000	\$4,000
Balance, December 31, 2021	<u>105,000</u>	<u>\$525,000</u>	<u>\$18,500</u>	<u>\$543,500</u>

Chapter 11 Appendix 1: Treasury Stock

Treasury stock is the company's own stock purchased by the company itself. Let's say that Pierce Company purchases 500 shares of its own common stock. This is an example of treasury stock because the company owns itself now. This stock does not have any rights (vote, share earnings, etc.). This stock is simply in the company's treasury. Therefore, it is treasury stock. The treasury stock reduces stockholders' equity because the amount of outstanding common shares has been reduced. Treasury stock is a contra-stockholders' equity account. Its normal balance is debit.

Comprehensive Example

- (1) Pierce Company purchased 1,000 of its own shares for \$12 per share on June 1.
- (2) On June 15th, Pierce company reissued 200 of the treasury shares (which were purchased on June 1) for \$12 per share.
- (3) On June 30th, Pierce company reissued 500 of the treasury shares (which were purchased on June 1) for \$13 per share.
- (4) On July 5th, Pierce company reissued the remaining 300 of the treasury shares (which were purchased on June 1) for \$10 per share. Record the journal entries for each of the 4 transactions.

(1) Purchasing Treasury Stock

Jun. 1, 2023	Treasury Stock	12,000	
	Cash		12,000

When a company purchases its own stock, we debit Treasury Stock and credit cash, for the amount for which the stock was purchased. In our case \$12,000 (1,000 shares * \$12 per share).

(2) Reissuing Treasury Stock (at cost)

Jun. 15, 2023	Cash	2,400	
	Treasury Stock		2,400

When we reissue the treasury stock previously bought, we debit cash for the amount we received \$2,400 (\$12 per share * 200 shares), and credit treasury stock for the cost we purchased it \$2,400 (\$12 cost per share * 200 shares).

(3) Reissuing Treasury Stock (above cost)

Jun. 30, 2023	Cash	6,500	
	Treasury Stock		6,000
	Paid-In Capital, TS		500

In this case we debit cash for \$6,500 (\$13 per share * 500 shares). We credit treasury stock for the cost we purchased it \$6,000 (\$12 cost per share * 500 shares). Finally, we credit Paid-In Capital, Treasury Stock for \$500, the difference.

(4) Reissuing Treasury Stock (lower than cost)

Jul. 5, 2023	Cash	3,000	
	Paid-In Capital, TS	500	
	Retained Earnings	100	
	Treasury Stock		3,600

In this case we debit cash for \$3,000 (\$10 per share * 300 shares). We credit treasury stock for the cost we purchased it \$3,600 (\$12 cost per share * 300 shares). Because we sold the treasury stock below cost, it will be a reduction to retained earnings. However, we first need to eliminate the Paid-In Capital, Treasury Stock account because we do not really have additional Paid-In Capital when we sell lower than cost. Therefore, we debit Paid-In Capital, Treasury Stock for its balance, \$500. Finally, we debit Retained Earnings to reduce it for \$100.

Chapter 11 Appendix 2: Stock Splits

Companies realize that sometime, the par value or the stated value of their stocks is simply too high. In order to reduce the par or stated value of their stock, they can make a stock-split. The ultimate goal of a stock split is to reduce the market price of the stocks, in order for the stock to be more affordable to a larger investor audience. To best understand the stock splits, please watch the lecture video below.

Lecture Video - Stock Splits

[WATCH VIDEO](#)

Chapter 12: Statement of Cash Flows

In the past chapters we learned more extensively about the main financial statements. However, we mostly concentrated on the Income Statement, the Balance Sheet, and the Statement of Retained Earnings. The reason why the Statement of Cash Flows is covered at the end of this book is the fact that in order to understand how to prepare this statement, one needs to understand many of the concepts we covered so far.

What Questions Does the Statement of Cash Flows Answer?

The goals of the Statement of Cash Flows are to explain what happened to company's cash. This statement should answer the change in the cash balance from one period to the next, how cash is received, how cash is spent, etc. When we discuss about cash in a company, we really discuss cash and cash equivalents, as was explained in Chapter 6. Cash equivalents are instruments that can be converted in cash and the maturity date is within 3 months or less.

Statement of Cash Flows - Overview

The Statement of Cash Flows is comprised of 3 sections: (1) Cash Flows from Operating Activities, (2) Cash Flows from **Investing** Activities, and (3) Cash Flows from **Financing** Activities. This is a sample Statement of Cash Flows:

Best Coffee Shop Company
STATEMENT OF CASH FLOWS
For the Year Ended December 31, 2022

Cash Flows From Operating Activities	
Net Income	40,000
Adjustments to reconcile Net Income:	
Add Depreciation Expense	20,000
Deduct A/R Increase	-17,000
Add Inventory Decrease	2,000
Add A/P Increase	40,000
Deduct Accrued Liabilities Decrease	-5,000
Deduct Gain on Sale of Plant Asset	-10,000
Cash Flows Provided by Operating Activities	70,000
Cash Flows from Investing Activities	
Deduct Purchase of Equipment	-300,000
Add Sale of Asset (desks)	40,000
Cash Flows Used for Investing Activities	-260,000
Cash Flows from Financing Activities	
Add Issuing Bonds	87,000
Add Issuing stock	100,000
Deduct Retire Notes Payable	-5,000
Deduct Cash Dividends	-10,000
Deduct Treasury Stock	-2,000
Cash Flows Provided by Financing Activities	170,000
Net Decrease in Cash	-20,000
Cash balance in 2021	40,000
Cash Balance in 2022	20,000

Let's learn what these operating, investing, and financing activities mean.

Operating Activities - Activities that are the main driver of the business. For example, a grocery store would include all activities connected to selling groceries as operating activities. A tutoring business would include all activities with providing tutoring services as operating activities.

Investing Activities - Activities that generally involve the *sale or purchase of long-term assets*. Also, activities that involve *sale or purchase of short-term investments* (except trading securities) and collecting cash from *notes receivable*.

Financing Activities - Activities that involve *long-term liabilities* (i.e. issuing bonds or long-term notes payable and repaying bonds or long-term notes payable) and *equity* (i.e. issuing stock, dividends, treasury stock)

Preparing the Statement of Cash Flows

When we prepare the Statement of Cash Flows, we can choose from 2 methods: (1) the indirect method and (2) the direct method. These methods differ only in the preparation of the Operating Activities section. The indirect method is the most widely used method because it is a much easier method in practice. Regardless of the method used, we need to get the same results. In order to prepare the Statement of Cash Flows, we need to explain how cash was received or disbursed and distinguish between the 3 categories (operating, investing, and financing).

Indirect Method

When we prepare the Operating Activities section of the Statement of Cash Flows, using the indirect method, we begin with Net Income. After all, the Income Statement truly shows most of the operating activities a company had for the past period. However, we also know that Net Income is not Cash. This is the case in accrual accounting.

Non-Cash Transactions

When we make a sale of \$100 on credit, we debit A/R and credit Revenues for \$100. This transaction causes Revenues to increase for \$100, which in turn increased the Net Income for \$100. However, we did not receive \$100 cash. It is only A/R, not cash. In the statement of Cash Flows, we only care about (yes, you guessed it) Cash. Basically, when current assets increase, we need to deduct that amount from Net Income because Net Income was increased by a non-cash transaction.

There are many other non-cash transactions like the one described above. What happens when we owe wages to an employee for \$200? We debit Wages Expense and credit A/P for \$200. Since expenses increased for \$200, Net Income decreased for \$200. However, this was also a non-cash transaction. How about depreciation? When we record depreciation, we debit Depreciation Expense and credit Accumulated Depreciation. Although Net Income ultimately decreases with this

transaction, it is a non-cash transaction. Basically, when current liabilities increase, we need to add that amount to Net Income because Net Income was decreased by a non-cash transaction.

So, should we still use Net Income as our starting point? Of course. Once we begin with Net Income, we need to eliminate all non-cash transactions to arrive at the Cash Flows from Operating Activities amount.

Cash Transactions Not Captured by Income Statement

Besides the non-cash transactions, there are some transactions that are not reflected in the Income Statement, and, therefore, the balance in Net Income does not reflect those transactions. In this case, we have to 'let Net Income know' about these cash transactions not included in the Income Statement.

How do we record the transaction to collect \$500 cash from a customer to whom we previously provided services on account? We debit Cash and credit A/R. In this transaction, even though we received cash, both of the accounts (Cash and A/R) are located on the balance sheet. Therefore, Net Income 'doesn't know' that this cash transaction happened. We need to 'let Net Income know'. Basically, when current assets decrease, we need to add that amount to Net Income, because Net Income is not aware of this transaction.

How do we record the transaction to pay an outstanding A/P balance of \$800? We debit A/P and credit Cash for \$800. In this transaction, even though we paid cash, both of the accounts (Cash and A/P) are located on the balance sheet. Therefore, Net Income 'doesn't know' that this cash transaction happened. We need to 'let Net Income know'. Basically, when current liabilities decrease, we need to deduct that amount from Net Income, because Net Income is not aware of this transaction.

The Actual Preparation of the Statement of Cash Flows

Instead of reading on how to prepare the Statement of Cash Flows, why don't we look at a problem and prepare the Statement of Cash Flows together? In order to prepare the Statement of Cash Flows we need to have the Income Statement, the Comparative Balance Sheet (Balance Sheet that shows the last 2 periods) and selected transactions. Here is the comprehensive example:

Comprehensive Example

Presented are the Income Statement and the Comparative Balance Sheet for Best Coffee Shop Company.

Best Coffee Shop Company		
INCOME STATEMENT		
For Year Ended Dec. 31, 2022		
Sales Revenue		300,000
Cost of Goods Sold		<u>150,000</u>
Gross Profit		150,000
Operating Expenses		
Salaries Expense	57,000	
Depreciation Expense	20,000	
Other Operating Expenses	<u>15,000</u>	
Total Operating Expenses		<u>92,000</u>
Operating Income		58,000
Other Revenues and Expenses		
Interest Revenues	7,000	
Gain on Sale of Assets	10,000	
Interest Expense	<u>-15,000</u>	
Total Other Revenues and Expenses		<u>2,000</u>
Income Before Income Taxes		60,000
Income Tax Expense		<u>20,000</u>
Net Income		<u>40,000</u>

Income Statement for 2022

Best Coffee Shop Company			
COMPARATIVE BALANCE SHEET			
December 31, 2022 and 2021			
	2022	2021	Increase (Decrease)
ASSETS			
<i>Current Assets</i>			
Cash	20,000	40,000	-20,000
Accounts Receivable	70,000	53,000	17,000
Inventory	100,000	102,000	-2,000
<i>Long-Term Assets</i>			
Plant Assets	500,000	245,000	255,000
Accumulated Depreciation	-40,000	-35,000	-5,000
Total Assets	730,000	475,000	255,000
LIABILITIES			
<i>Current Liabilities</i>			
Accounts Payable	60,000	20,000	40,000
Accrued Liabilities	10,000	15,000	-5,000
<i>Long-Term Liabilities</i>			
Notes Payable	150,000	70,000	80,000
Total Liabilities	220,000	105,000	115,000
STOCKHOLDERS' EQUITY			
Common Stock	350,000	230,000	120,000
Retained Earnings	180,000	140,000	40,000
Treasury Stock	-20,000	0	-20,000
Total Stockholders' Equity	510,000	370,000	140,000
Total Liabilities and Stockholders' Equity	730,000	475,000	255,000

Comparative Balance Sheet for 2021 and 2022

The selected transactions are also available and should be useful for preparing the investing and the financing section of the Statement of Cash Flows.

- Purchased equipment for \$300,000 cash.
- Sold old desks with a cost of \$50,000 and accumulated depreciation of \$20,000, yielding a gain of \$10,000.
- Issued bonds for \$87,000 cash.
- Paid \$5,000 cash to retire notes payable.
- Issued common stock for \$100,000 cash.
- Paid cash dividends in the amount of \$10,000.

- Purchased treasury stock for \$2,000.

Required: Prepare the Statement of Cash Flows, using the indirect method.

Solution to Comprehensive Example

Preparing the Operating Activities Section

To prepare the operating activities section, we start with Net Income. Once we have Net Income, we need to follow three steps.

Add and Deduct expenses that are non-cash transactions (i.e. depreciation expense, amortization expense, depletion expense, etc.)

Add and Deduct the changes in current assets and current liabilities.

Add Losses and Deduct Gains from the sale of L-T investing and financing activities.

The Net Income, as shown on the Income Statement above is \$40,000. Following the first step, we take a look at the Income Statement again to find any depreciation, amortization, or depletion expenses. In our case, we have Depreciation Expense in the amount of \$20,000. Since the depreciation expense is a non-cash transaction and decreases Net Income, we need to add it back to eliminate its effect.

Cash Flows From Operating Activities	
Net Income	40,000
Adjustments to reconcile Net Income:	
Add Depreciation Expense	20,000

The second step is to look at the Comparative Balance Sheet. Specifically, we need to look at the current assets and current liabilities changes. **If current assets increased, we need to deduct the amount of increase from Net Income. If current assets decreased, we need to add the amount of decrease to Net Income.** Additionally, **if current liabilities increased, we need to add the amount of increase to Net Income. If current liabilities decreased, we need to deduct the amount of decrease from Net Income.**

In our case we have A/R and Inventory as current assets and A/P and Accrued Liabilities as current liabilities. The accounts receivable increased for \$17,000, and therefore, we need to deduct it from Net Income. The inventory balance decreased for \$2,000, and therefore, we need to add it to Net Income. The accounts payable balance increased for \$40,000, and therefore, we need to add it to Net Income. Finally, the accrued liabilities decreased for \$5,000, and therefore, we need to deduct it from Net Income.

Cash Flows From Operating Activities	
Net Income	<u>40,000</u>
Adjustments to reconcile Net Income:	
Add Depreciation Expense	20,000
Deduct A/R Increase	-17,000
Add Inventory Decrease	2,000
Add A/P Increase	40,000
Deduct Accrued Liabilities Decrease	-5,000

For the last step, we need to look back at the income statement to find if we have any losses or gains. On the Income Statement we can notice a gain of \$10,000. We need to eliminate the gain from net income by subtracting it. We eliminate the gain because this is not an operating activity. It is a sale of plant assets, which is an investing activity.

Cash Flows From Operating Activities	
Net Income	<u>40,000</u>
Adjustments to reconcile Net Income:	
Add Depreciation Expense	20,000
Deduct A/R Increase	-17,000
Add Inventory Decrease	2,000
Add A/P Increase	40,000
Deduct Accrued Liabilities Decrease	-5,000
Deduct Gain on Sale of Plant Asset	<u>-10,000</u>
Cash Flows Provided by Operating Activities	<u>70,000</u>

The result is Cash Flows Provided by Operating Activities of \$70,000. When we have a cash inflow, we use the term 'Provided By'. When we have a cash outflow, we use the term 'Used For'.

Preparing the Investing Activities Section

Investing activities are those activities that involve purchasing and selling long-term assets, short-term investments (except trading securities), and collecting cash from long-term notes receivable. In our example we have 2 such transactions. We purchased equipment for \$300,000 cash and this is a cash outflow because we spent \$300,000. We also sold a long-term asset (desks), but the problem did not provide the information on how much cash we received from the sale. Recalling the knowledge from Chapter 8 about disposal of fixed assets, we can reconstruct the journal entry:

Cash	40,000	
Accumulated Depreciation	20,000	
	Equipment	50,000
	Gain on Sale of Asset	10,000

As you can see, the cash received from this sale was \$40,000. We found this amount by making the debits and credits to equal. The Cash Flows from Investing Activities will look like this:

Cash Flows from Investing Activities

Deduct	Purchase of Equipment	-300,000
Add	Sale of Asset (desks)	<u>40,000</u>

Cash Flows Used for Investing Activities	<u>-260,000</u>
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The result is Cash Flows Used for Investing Activities of \$260,000.

Preparing the Financing Activities Section

Financing activities are those activities that involve issuing and repaying bonds and long-term notes payable, and issuing stock, paying cash dividends, and reacquiring stock treasury stock). In our example, we have multiple transactions like this. Namely, we issued bonds by receiving \$87,000 cash and issued common stock by receiving \$100,000 cash. These are cash inflows. We also paid \$5,000 cash to retire notes payable, paid cash dividends of \$10,000, and purchased treasury stock for

\$2,000. These transactions are all cash outflows. This is how the Financing Activities Section will look:

Cash Flows from Financing Activities

Add Issuing Bonds	87,000
Add Issuing stock	100,000
Deduct Retire Notes Payable	-5,000
Deduct Cash Dividends	-10,000
Deduct Treasury Stock	<u>-2,000</u>
Cash Flows Provided by Financing Activities	<u>170,000</u>

The result is Cash Flows Provided by Financing Activities of \$170,000.

Now that we completed all sections, we can compile the entire Statement of Cash Flows.

Best Coffee Shop Company
STATEMENT OF CASH FLOWS
For the Year Ended December 31, 2022

Cash Flows From Operating Activities	
Net Income	40,000
Adjustments to reconcile Net Income:	
Add Depreciation Expense	20,000
Deduct A/R Increase	-17,000
Add Inventory Decrease	2,000
Add A/P Increase	40,000
Deduct Accrued Liabilities Decrease	-5,000
Deduct Gain on Sale of Plant Asset	-10,000
Cash Flows Provided by Operating Activities	70,000
Cash Flows from Investing Activities	
Deduct Purchase of Equipment	-300,000
Add Sale of Asset (desks)	40,000
Cash Flows Used for Investing Activities	-260,000
Cash Flows from Financing Activities	
Add Issuing Bonds	87,000
Add Issuing stock	100,000
Deduct Retire Notes Payable	-5,000
Deduct Cash Dividends	-10,000
Deduct Treasury Stock	-2,000
Cash Flows Provided by Financing Activities	170,000
Net Decrease in Cash	-20,000
Cash balance in 2021	40,000
Cash Balance in 2022	20,000

When we sum the cash flows provided by and used for operating, investing, and financing activities, we get \$20,000 Cash decrease. This result matches the comparative balance sheet in which the cash balance in 2021 was \$40,000 and decreased by \$20,000 this year, resulting in \$20,000 balance in 2022.

Chapter 13: Financial Statement Analysis

Financial statement analysis is a salient tool in every aspect of business. Employees, managers, executives, investors, creditors, and auditors, all need to know how to perform a financial statement analysis for various reasons. Looking at great amounts of data does not tell us anything. However, when we perform a comprehensive financial statement analysis, everything will start to make sense. In addition to analyzing the past and the present, we might also use that data to predict future performance.

There are three ways to conduct a financial statement analysis: (1) horizontal analysis; (2) vertical analysis; and (3) ratio analysis. Let's learn each of these.

Horizontal Analysis

Horizontal analysis is an analysis of the performance of a company over time. Basically, we analyze any account for multiple periods. For example, we want to see whether Revenues increased or decreased on a year-to-year basis. Maybe we want to perform a horizontal analysis on a quarter-to-quarter basis.

To perform a horizontal analysis, we need comparative financial statements. Comparative financial statements are financial statements that list the balances of the accounts for multiple periods. An example of a comparative balance sheet is presented below:

Best Coffee Shop Company			
COMPARATIVE BALANCE SHEET			
December 31, 2022 and 2021			
	2022	2021	Increase (Decrease)
ASSETS			
<i>Current Assets</i>			
Cash	20,000	40,000	-20,000
Accounts Receivable	70,000	53,000	17,000
Inventory	100,000	102,000	-2,000
<i>Long-Term Assets</i>			
Plant Assets	500,000	245,000	255,000
Accumulated Depreciation	-40,000	-35,000	-5,000
Total Assets	730,000	475,000	255,000
LIABILITIES			
<i>Current Liabilities</i>			
Accounts Payable	60,000	20,000	40,000
Accrued Liabilities	10,000	15,000	-5,000
<i>Long-Term Liabilities</i>			
Notes Payable	150,000	70,000	80,000
Total Liabilities	220,000	105,000	115,000
STOCKHOLDERS' EQUITY			
Common Stock	350,000	230,000	120,000
Retained Earnings	180,000	140,000	40,000
Treasury Stock	-20,000	0	-20,000
Total Stockholders' Equity	510,000	370,000	140,000
Total Liabilities and Stockholders' Equity	730,000	475,000	255,000

As you can notice from the comparative balance sheet, many things changed between 2021 and 2022. Cash decreased for \$20,000, Accounts Receivable increased for \$17,000, etc. These results give us the changes in each of the accounts. Many times, dollar amounts do not paint the picture properly. That is why we need percentages.

For example, Cash decreased for \$20,000 from \$40,000. That means Cash decreased for 50% ($20,000/40,000 \times 100$). Accounts increased for \$17,000 from \$53,000. That means A/R increased for 32% ($17,000/53,000$).

We can do this analysis for each account in every financial statement. We can perform horizontal analysis even for many years in the past. If we perform the horizontal analysis on a quarter-to-quarter basis, we will be able to see specific performance of the company and act accordingly. To sum up, you take the increase/decrease and divide it with the base year amount. That will give you the percentage of change.

Vertical Analysis

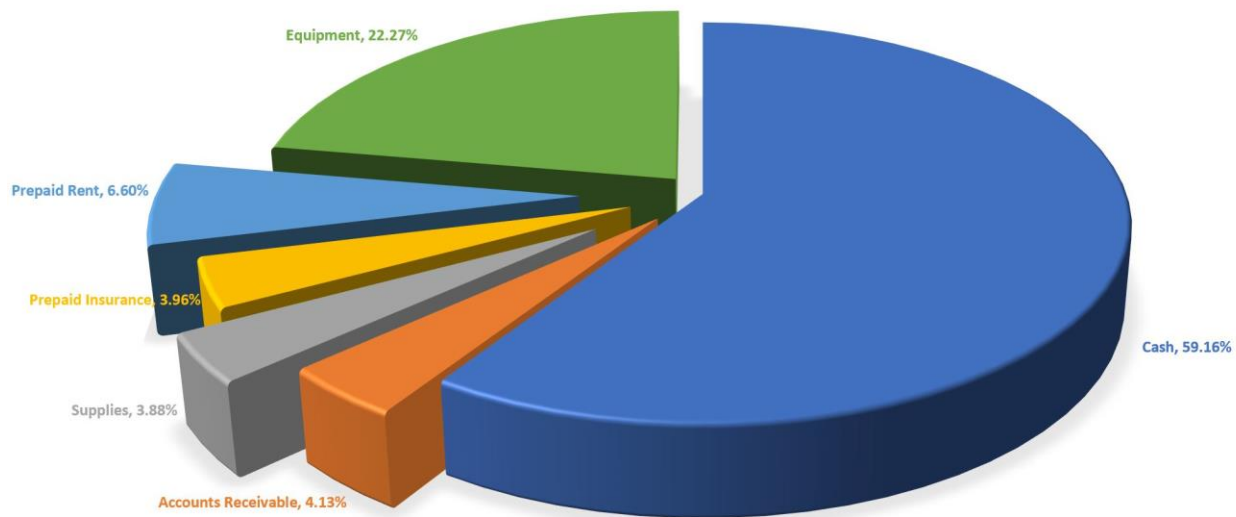
Vertical analysis is an analysis of a company based on a comparison to a base amount. Let's take a look at a balance sheet to understand vertical analysis better.

Best Coffee Shop Company	
BALANCE SHEET	
Dec. 31, 2021	
ASSETS	
Cash	71,700
Accounts Receivable	5,000
Supplies	4,700
Prepaid Insurance	4,800
Prepaid Rent	8,000
Equipment	27,000
Total Assets	<u>121,200</u>
LIABILITIES	
Accounts Payable	700
STOCKHOLDERS' EQUITY	
Common Stock	110,000
Retained Earnings	10,500
Total Liabilities and S. Equity	<u>121,200</u>

Whatever we are analyzing, we need a base amount. For assets, I will choose total assets to be the base amount, 100%. All other assets will be a percentage of the total assets.

For example, Cash is 59.16% of total assets ($71,700/121,200$). Accounts receivable is 4.13% of total assets ($5,000/121,200$). Supplies are 3.88% of total assets. Prepaid Insurance is 3.96% of total assets ($4,800/121,200$). Prepaid rent is 6.6% of total assets ($8,000/121,200$). Finally, equipment is 22.27% of total assets ($27,000/121,200$).

Percentage presented like this do not mean anything. That is why we need a graphical representation, just like the one presented below.



Now, you can see what your main assets are. You can conduct the same analysis for liabilities, revenues, expenses, etc. That is how we get a clearer picture of the company's financials.

Ratio Analysis

Ratio analysis includes specific formulas that measure the relationship between items on the financial statements. This section of the chapter will include many, many formulas, but the interpretation of the results is very useful.

We divide financial ratios in 3 general categories: (1) Liquidity ratios; (2) Solvency ratios; and (3) Profitability ratios.

Liquidity Ratios

Liquidity ratios tell us how fast we can convert our assets in cash. Obviously, cash is the most liquid asset, because it is already cash.

We have 7 ratios in this specific category. Let's look at each of them.

Working Capital

The working capital tells us how whether our company can pay our current liabilities.

$$\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities}$$

If current assets were \$200,000 and current liabilities were \$90,000, then the working capital would be \$110,000. While this is a good result, it is difficult to compare this result to another company of different size. That is why we have the Current Ratio.

Current Ratio

The current ratio serves the same purpose as the working capital, but this measure can be used to compare companies of different sizes.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

In our previous case, the current ratio would be 2.22. Any current ratio larger than 1 is a good sign. However, a huge current ratio may mean that the company is holding too many current assets and is not investing enough. The measure would also depend on the type of industry.

Quick Ratio

The current ratio takes all current assets into account. That is the limitation of the current ratio. That is why we have the Quick Ratio.

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Quick assets are cash, short-term investments, and receivables. Quick assets would never include inventories or prepaid expenses.

A/R Turnover

The A/R Turnover tells us how frequently we can convert accounts receivable to cash dollars.

$$\text{Accounts Receivable Turnover} = \frac{\text{Sales}}{\text{Average Accounts Receivable}}$$

If the A/R at the beginning of the year were \$50,000 and at the end of the year were \$40,000, and sales were \$700,000, then the A/R Turnover would be 15.6.

Inventory Turnover

Inventory turnover ratio tells us the number of times we have sold and replaced our company's inventory.

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

The higher the inventory turnover, the better. Higher inventory turnover means that our company is selling its inventory better than other companies with lower inventory turnover.

Solvency Ratios

Solvency tells us how well the company can pay our long-term liabilities. We will learn 3 solvency ratios: (1) Fixed assets to Long-Term Liabilities; (2) Liabilities to Stockholders' Equity; and (3) Times Interest Earned.

Fixed Assets to Long-Term Liabilities

This ratio helps in analyzing the solvency of our company, because when a company obtains a long-term debt, the debt is often secured (collateralized) by the company's fixed assets.

$$\text{Fixed Assets to Long Term Liabilities} = \frac{\text{Fixed Assets (net)}}{\text{Long - Term Liabilities}}$$

Higher ratio is better in this case because fixed assets will still be sufficient to cover the long-term debt.

Liabilities to Stockholders' Equity

This ratio is also called Debt-to-Equity ratio and it used to assess the financial leverage of our company.

$$\text{Liabilities to Stockholders' Equity} = \frac{\text{Total Liabilities}}{\text{Total Sotckholders' Equity}}$$

When this ratio is high, it means that the company used a lot of debt to finance its operations and growth.

Time Interest Earned

Times interest earned ratio is another ratio that tells us about how well our company can pay its expenses.

$$\text{Times Interest Earned} = \frac{\text{Income Before Income Tax} + \text{Interest Expense}}{\text{Interest Expense}}$$

Higher ratio in this case is better because it means that our company can pay its interest expenses. However, an extremely large ratio would not be a good sign because it means the company is paying its expenses instead of finding the optimal mix of paying debt and investing.

Profitability Ratios

These ratios tell us how well the company can generate profits from its main operations. We will discuss 6 ratios in this section.

Asset Turnover

This ratio tells us how many dollars in sales we make for every dollar we invested in assets.

$$\text{Asset Turnover} = \frac{\text{Sales}}{\text{Average Total Assets}}$$

Obviously, the higher the ratio, the more sales we generate from our assets.

Return on Total Assets

This ratio tells us the earnings dollars of our company generated by every dollar we invested in assets.

$$\text{Return on Total Assets} = \frac{\text{Net Income} + \text{Interest Expense}}{\text{Average Total Assets}}$$

Again, the higher the ratio the better. It would mean that we make more profit as compared to our investment in assets.

Return on Equity

This ratio tells us the amount of dollars we generate for every dollar investors invested in the company.

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Average Total Stockholders' Equity}}$$

This ratio is most commonly used by investors when they are deciding to invest in a company. The higher the ratio, the better. Investors will be more interested to invest in a company with higher return on equity.

Earnings per Share

This ratio tells us the profit of the company that is allocated to its common stockholders.

$$\text{Earnings per Share} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Outstanding Common Stock Shares}}$$

In this case as well, the higher the ratio the better. It means that our company is allocating more of its profit to our stockholders than other companies who have lower earnings per share.

Price-Earnings Ratio

This ratio tells us the market price of a share as compared to the earnings per share. Someone who wants to invest in our company will use this ratio to find out how much money needs to be invested in order to make one dollar of the profit of the company.

$$\text{Price Earnings} = \frac{\text{Market Price per Share of Common Stock}}{\text{Earnings per Share on Common Stock}}$$

When comparing two companies, a lower price earnings ratio is better. For example, the P/E ratio of Company A is 8 and the P/E ratio of company B is 2. It means that an investor will have to invest \$8 in company A to receive \$1 of the company's earnings, or only \$2 in Company B to receive \$1 of the company's earnings.